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Ministry of Health Malaysia

Maternal DIETARY GUIDELINES FOR MALAYSIA

National Coordinating Committee on Food and Nutrition (NCCFN)

Ministry of Health Malaysia

2023

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Ministry of Health Malaysia

Maternal DIETARY GUIDELINES FOR MALAYSIA

National Coordinating Committee on Food and Nutrition (NCCFN)
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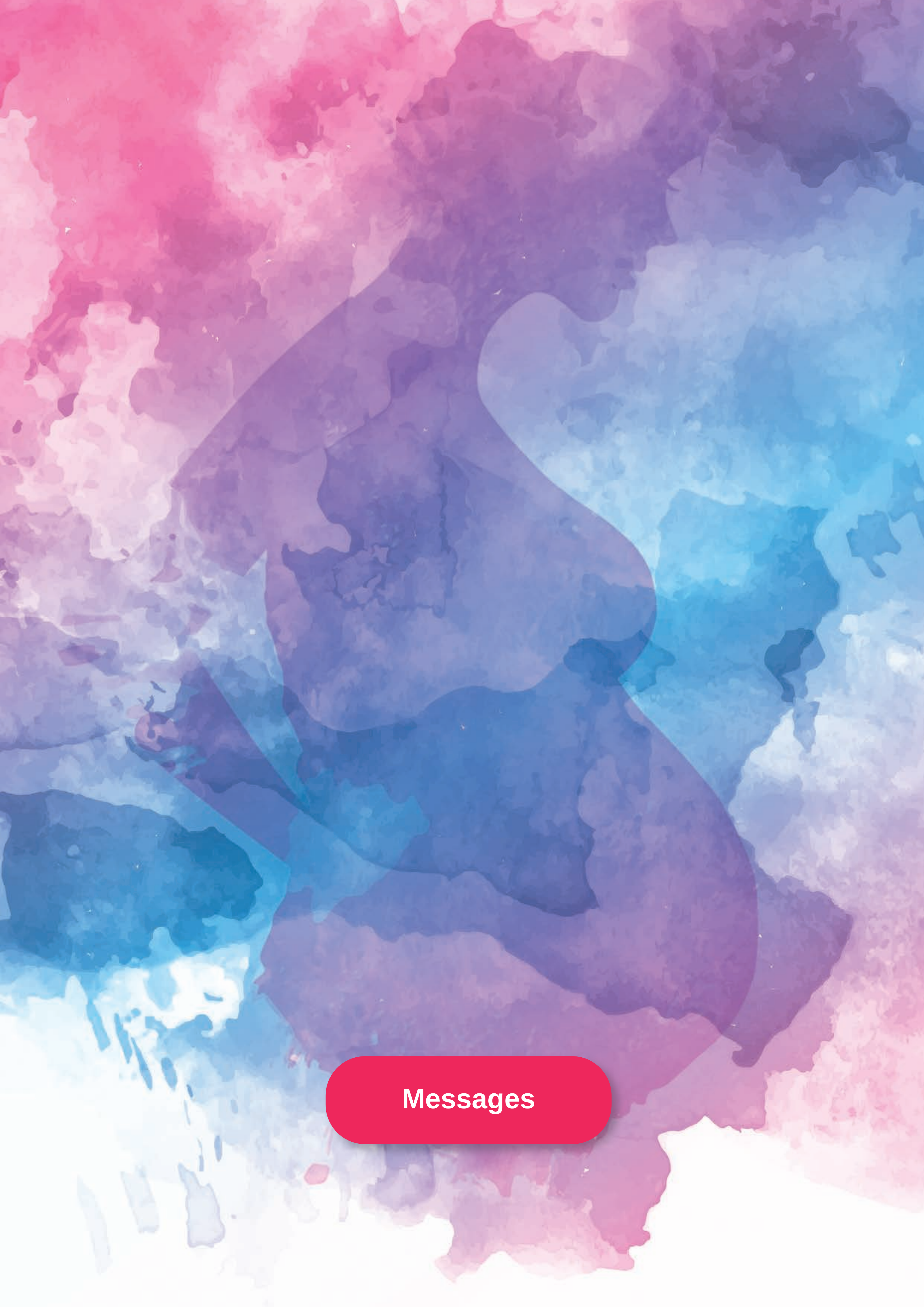


Messages

Message by Minister Ministry of Health Malaysia

Appropriate maternal nutrition is an important determinant of a child's nutritional status, growth, health and development outcomes particularly during the first 1,000-days of life from conception to the age of two years. Ensuring access to nutritious, safe, affordable and sustainable diets, along with essential nutrition and health services are fundamental to women's survival, health and well-being. In the country, nutritional status of women at reproductive age is one of the primary components of national nutrition policies and plans, aiming to prevent malnutrition in women before, during pregnancy and while breastfeeding. In addition, appropriate maternal nutrition is essential to prevent hidden hunger for their young children.

Maternal nutrition knowledge, attitudes and practices (KAP) play a significant role in ensuring better nutritional status of children. Mothers who have better KAP would have better nutritional status of their children. Thus, the publication of Maternal Dietary Guidelines for Malaysia (MDGM) is timely and important in achieving a healthy and good nutritional status of women and their children.

Therefore, it is my fervent hope that all the relevant ministries, agencies, academic institutions, hospitals, the healthcare industries, professional organisations, and the food industries would adopt and apply the MDGM 2023 to meet the demands of the population looking for such dietary guidance. The great efforts of the Technical Working Group on Nutritional Guidelines, under the auspices of the National Coordinating Committee on Food and Nutrition (NCCFN), in establishing this guideline for women at reproductive age in the country, is commendable.

YB Dr. Zaliha Mustafa
Minister
Ministry of Health Malaysia

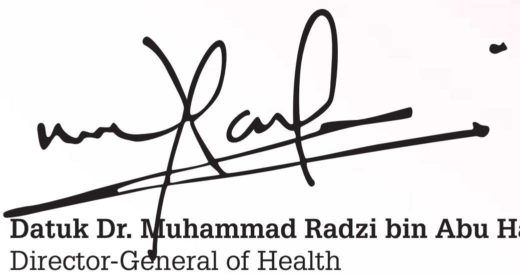
Foreword By
Director-General of Health
Ministry of Health Malaysia

Nutrition care and support on healthy diets and appropriate weight gain are essential particularly for women at the reproductive age (WRA) to ensure optimal pregnancy outcomes for the mother and child. The rapid rise in overweight and obesity among WRA in the country also means that more attention must be given to the distinct needs of these groups before and during pregnancy. Maternal undernutrition and inappropriate gestational weight gain (GWG) are the major contributors to the increased incidence of preterm birth (PTB), low birth weight (LBW) and poor foetal growth.

Studies have consistently shown that maternal nutrition has a definite and significant effect on nutritional status during the child-bearing year. Strengthening maternal nutrition is therefore one of the pivotal strategies to align with the government goals to achieve the nutritional wellbeing for Malaysians. This will facilitate our efforts to address the challenge of maternal nutrition in ensuring optimal pregnancy outcomes to combat double burden of malnutrition. Considering that Malaysia is a unique country with its multiracial and multicultural eating practices, it is important to understand the relationship between maternal nutrition and birth outcomes in this country.

This Maternal Dietary Guidelines for Malaysia (MDGM) is an important educational reference that translates scientific information on nutritional recommendation into a simple and practical application. The aim of this guideline is to be the main reliable healthy eating references for the health professionals and related stakeholders in delivering nutrition messages to the intended groups.

Thus, I believe that the publication of the Maternal Dietary Guidelines for Malaysia is an important effort to assist women at reproductive age in adopting healthy eating habits. It is my hope that all healthcare professionals will make use of this MDGM to ensure that vital nutrition information is effectively communicated. I would like to extend my congratulations to the Technical Working Group on Nutritional Guidelines for their efforts in creating this significant document.



Datuk Dr. Muhammad Radzi bin Abu Hassan
Director-General of Health
Ministry of Health Malaysia

Preface by Deputy Director-General of Health (Public Health) Ministry of Health Malaysia

Good nutrition in reproductive age is pivotal in preparation for becoming pregnant and giving birth. Pregnancy is associated with increased iron demand, and therefore, increases the risk of iron deficiency anaemia. Throughout pregnancy, iron deficiency anaemia adversely affects the maternal and foetal well-being, and is linked to increased morbidity and foetal death. In Malaysia one in three women of reproductive age (15-49 years old) is having an anaemia problem. Despite a significant reduction in the prevalence of anaemia among pregnant mothers from 29.3% in 2016 to 19.3% in 2012, maternal anaemia remains a major public health concern in Malaysia. This issue warrants serious attention and needs to be intervened. Adequate nutrient intake during the life stages of pregnancy and lactation is important for the optimal foetal growth and production of breastmilk. Poor nutrition during pregnancy and lactation periods can cause mothers and growing children at a greater risk of disease and death.

The publication of this Maternal Dietary Guidelines for Malaysia (MDGM) is to provide adequate access to reliable nutrition information to facilitate women in achieving an optimal nutrition before, and during pregnancy as well as during lactation. The MDGM also adopts the International of Medicine (IOM) recommendation for the maternal weight gain.

Special appreciation goes to the Technical Working Group on Nutritional Guidelines and the sub-committee on Maternal Dietary Guidelines for Malaysia (MDGM) for their tremendous efforts in successfully publishing this guidelines. Moreover, heartfelt gratitude goes out to all individuals who have contributed in various ways towards the development of the Maternal Dietary Guidelines for Malaysia (MDGM) 2023.

Datuk Dr. Norhayati binti Rusli
Deputy Director-General (Public Health)
Ministry of Health Malaysia

Preface by Chairman & Co-Chairman Maternal Dietary Guidelines for Malaysia

Women of reproductive age have distinct nutritional requirements, especially before and during pregnancy and while breastfeeding, when nutritional vulnerability is greatest. Ensuring women have access to nutritious diets and adequate care and health services is fundamental for the survival and well-being of mothers and their off- spring. Worldwide, maternal nutrition education and food-based intervention strategies have documented positive impacts on maternal and child health outcomes.

The Maternal Dietary Guidelines for Malaysia (MDGM) 2023 is the first edition of healthy eating practices guidelines developed for healthy women at reproductive age (15 - 49 years old) in the country. The publication of the MDGM is under the coordination of the Technical Working Group for Nutritional Guidelines, National Coordinating Committee on Food and Nutrition (NCCFN). The main focus of the guidelines includes promoting healthy eating practices, meeting caloric and essential nutrient intakes, being physically active, achieving ideal gestational weight gain, and ensuring food safety and hygiene during pregnancy. The guidelines are conveyed through Key Messages (KM), Key Recommendations (KR) and How to Achieve (HTA). The MDGM is primarily intended for health professionals for use in nutrition education and promotion for the intended population, especially at the clinic and community settings.

The Technical Working Group is hopeful that the MDGM will be widely used as the main reference for healthy lifestyle behaviours, primarily dietary and physical activity, during pregnancy and postpartum. We look forward for feedback from stakeholders and end-users to help us update and improve future editions of the guidelines as to ensure the relevance and effectiveness of the guidelines over time.

It is also worth noting the various contributors to the MDG, including writers, members of the Technical Working Group, members of the Focus Group Discussion, the editorial team, Consensus Workshop participants, and the TWG Secretariat. Recognizing the collaborative effort and expertise of these individuals and groups is essential for fostering a sense of ownership and engagement in the implementation of the guidelines.

Ms. Zaiton Haji Daud
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Maternal Dietary Guidelines for Malaysia

Prof. Dr. Zalilah Mohd Shariff
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Acknowledgement

Individuals from the various Departments and Institutes, the Ministry of Health Malaysia. Academicians from local universities, nutritionists, dietitians, representatives from related professional bodies, representatives from the food manufacturing and trading industry, and consumer bodies are all acknowledged by the Technical Working Group on Nutritional Guidelines. Their invaluable contributions and dedication to complete this document successfully are sincerely appreciated.

A word of gratitude is also conveyed to the:

- Director, Food Safety and Quality Division, MOH
- Directors, State Department of Health, MOH (all over Malaysia)
- Director, Hospital Sultan Idris Shah, Serdang
- Dean, Faculty of Health Sciences, Universiti Kebangsaan Malaysia
- Dean, Faculty of Science and Technology, Universiti Kebangsaan Malaysia
- Dean, Faculty of Medicine and Health Sciences, Universiti Putra Malaysia
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- Dean, School of Health Sciences, Universiti Sains Malaysia
- Dean, School of Medical Sciences, Universiti Sains Malaysia
- Dean, Faculty of Hospitality, Tourism and Wellness, Universiti Malaysia Kelantan
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for their generous support and cooperation.

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Coordination and Documentation

The documentation of the Maternal Dietary Guidelines for Malaysia 2023 was coordinated by the Technical Working Group (TWG) on Nutritional Guidelines, which is under the purview of the National Coordinating Committee on Food and Nutrition (NCCFN), Ministry of Health Malaysia. The Nutrition Division, Ministry of Health Malaysia served as the secretariat for the Maternal Dietary Guidelines for Malaysia.

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List of Tables, Figures and Appendices

EXECUTIVE SUMMARY

A woman's nutritional status is a robust indicator of her overall health and well-being. A healthy woman has a strong immune system and adequate nutrient reserves that protect her against infections and diseases and provide support for physiological changes, especially during pregnancy and lactation. The National Health and Morbidity Survey in 2016 and 2022 reported that the proportion of women with hyperglycaemia and hypertension during pregnancy increased from 13.5% to 27.1%, and 5.8% to 6.5%, respectively. Meanwhile, anaemia in pregnancy has slightly reduced from 29.3% to 19.3%. Poor maternal nutrition increases the risk of adverse pregnancy outcomes that could have a lifelong impact on the health and nutrition of the woman and offspring.

Nutritional requirements during pregnancy and lactation differ considerably from those of non-pregnant and non-lactating women. Hence, a specific approach to dietary recommendations is needed to guide these women in meeting their nutritional requirements. The Maternal Dietary Guidelines for Malaysia (MDGM) is a government-endorsed document intended to be the main dietary guideline for relevant stakeholders, including public and private health professionals, in delivering health and nutrition education. The current MDGM aims to provide scientific but practical recommendations on healthy eating and active lifestyle for women of reproductive age (15 - 49 years). In line with the Sustainable Development Goals (SDGs), the MDGM incorporates the element of a sustainable food system that promotes the consumption of foods that are available, accessible, affordable, healthy, and culturally appropriate for the population. In addition, the MDGM addresses energy and nutrient requirements, weight management, food safety and physical activity during pregnancy and postpartum through Key Messages (KM), Key Recommendations (KR) and How to Achieve (HTA).

The Technical Working Group for Nutritional Guidelines, under the auspice of the National Coordinating Committee on Food and Nutrition (NCCFN) invited experts from the academia, Ministry of Health Malaysia and related professional bodies who have extensive knowledge on nutrition and health to undertake the preparation of the MDGM. The process, initiated on 24th January 2021, has produced several drafts that were presented, reviewed, validated, and approved by the TWG Nutritional Guidelines. The KM, KR and HTA were then vetted by health professionals and other related end-users to assess their relevance, clarity, and comprehension through focus group discussions on 21st February 2023. The final draft of KM, KR and HTA were presented in a consensus meeting with various stakeholders on 8th February 2023.

The MDGM features the following key components:

- The MDGM uses the Recommended Nutrient Intakes (RNI) 2017 for Malaysia as the main reference for energy and nutrient recommendations. In addition, the Malaysian Food Pyramid 2020 is referred for food-based recommendations that emphasize on balance, moderation, and variety.
- The MDGM is primarily designed for women of reproductive age (15 – 49 years).
- Recommendations favour wholesome foods over processed and ultra-processed foods.

Key Message 1: Choose a variety of foods to meet energy requirements for healthy pregnancy outcomes and during breastfeeding

Key Message 2 : Choose variety and micronutrient-dense foods

There are five **Key Messages** in the MDGM as follows:

Key Message 3 : Achieve optimal weight for healthy pregnancy

Key Message 4 : Be physically active everyday

Key Message 5 : Choose clean and safe food

Key Message 1

Choose a variety of foods to meet energy requirements for healthy pregnancy outcomes and during breastfeeding

Key Message 1

Choose a variety of foods to meet energy requirements for healthy pregnancy outcomes and during breastfeeding

Prof. Dr. Barakatun Nisak Mohd Yusof, Dr. Norsyamlina Che Abdul Rahim, Ms. Junidah Raib, Dr. Noor Baizura Md Yusop, Dr. Nurzalinda Zalbahar@Zabaha, Dr. Nor. Aini Jamil @ A. Wahab, Ms. Nur Wahidda Azmi and Ms. Nur Amalina Muhamad

1.1 Terminology

Basal metabolic rate

The basal metabolic rate is the minimum energy consumption rate for life needs. It is measured through a supine lying position in a state of rest, fasting, immobility and a resting mind. The basal metabolic rate is usually expressed as energy per minute, hour, or day.

Calorie

A calorie is a unit used to measure the energy produced by food. One gram of calories (cal) is the amount of heat energy required to increase the temperature for one gram of water by 1°C. Kilocalorie (kcal) is the amount of heat energy necessary to raise the temperature of a kilogram of water. Therefore 1 kcal = 1000 cal.

Energy balance

Energy balance is achieved when calorie-intake (food intake) is balanced with calorie expenditure (physical activity). Individuals gain weight when calorie or energy intake exceeded energy expenditure while, individuals lose weight when the energy expenditure exceeded the energy intake over a period of time.

Energy requirements
Energy requirements are the amount of energy from food necessary for energy balance. Energy intake is used to maintain the body size, body composition and physical activity needs of pregnant women. Sufficient energy is also necessary for the optimal growth and development of the foetus, the accumulation of female fat tissue throughout pregnancy and milk production. This is important for the long-term retention of women and infant health.

Exclusive breastfeeding
Exclusive breastfeeding is defined as feeding an infant only breast milk from his or her mother or a wet nurse or with expressed breast milk. The infant does not receive other fluids, including water or solids, except for drops or syrups consisting of vitamins, mineral supplements or medicine.

Food groups
A food group puts together foods of similar nutrient content and function. There are five food groups which are vegetables; fruits; rice, other cereals, whole grain cereal-based products and tubers; fish, poultry/ eggs/ meat, and legumes; milk and milk products. These food groups contain foods that are similar in calories, carbohydrate, protein and fat contents.

Food taboo
Food taboo is abstaining from food and/or beverage consumption due to religious and cultural reasons. It can be permanent or temporal. Permanent food taboos are avoidance of food and/or drinks throughout their life, while some foods are avoided for specific periods. These restrictions often apply to women and are related to the reproduction cycle (during pregnancy, birth, and lactation periods).

Healthy snack
Any nutrient-dense food eaten between main meals in small portions and should not replace main meals.

Malaysian Food Pyramid 2020
A food pyramid is a visual tool that is used as a guide to your DAILY food intake in achieving a healthy diet. It is developed to provide a guide for the types and amounts of food that can be eaten in combination to provide a balanced diet. A food pyramid consists of four levels that represent five food groups. The recommended number of servings per day for each food group is indicated next to it. From the bottom to the top of the food pyramid, the number of servings of each food group becomes smaller indicating that an individual should eat more of the foods at the base of the pyramid and less of the foods at the top of the pyramid.

Malaysian Healthy Plate
Malaysian Healthy Plate is a visual guide to show the total food in each food group that needs to be consumed in a meal to achieve a healthy and balanced diet based on the principle of quarter-quarter-half. It is used to translate the recommendations of the Malaysian Dietary Guidelines and Malaysian Food Pyramid to help Malaysians practice healthy eating habits by planning their daily meals.

Postpartum period
The Postpartum period begins immediately after the baby's birth and extends up to six weeks (42 days) after birth.

Planetary health diet
Planetary Health Diet is a healthy and sustainable model diet, published by EAT-Lancet 2019, that aims to provide health to the population and the planet. The recommendations are based on the predominant consumption of vegetables, greens, fruits, and whole grains and reduced consumption of meat, fish, eggs, refined cereals, and tubers.

Serving size
In the Malaysian Food Pyramid, serving size is the recommended amount of foods consumed daily in household measures used for foods and drinks, for example cup, plate, bowl, tablespoon, teaspoon and glass. However, serving size defined in the Malaysian Food Pyramid may not be equal to a serving size defined in a food label.

Total energy consumption
Total energy consumption is the average energy consumed within 24 hours by an individual or a group. It is an example of the average energy consumption consumed in one day but not the actual amount of energy consumed daily.

Trimester
The trimester is the phase throughout pregnancy, which is divided into the first (1-12 weeks), second (13-27 weeks) and third trimesters (28-40 weeks). Each trimester coincides with a period of about three months.

Ultra-Processed Food (UPFs)

NOVA characterised ultra-processed foods as industrial formulations generated through compounds extracted, derived or synthesized from food or food substrates. Ultra-processed foods also commonly contain artificial substances such as colours, sweeteners, flavours, preservatives, thickeners, emulsifiers and other additives that promote aesthetics, enhance palatability and increase shelf life. Ready-to-eat food and beverage, spreads, packaged snacks and pastries, cakes, instant noodles, and pre-prepared ready-to-heat products are some examples of ultra-processed foods high in sugar, salt, fat and artificial substances.

Vegetarian diets

The term vegetarian is often used to describe a range of diets practised by a particular group of populations with varying degrees of food restrictions by omitting animal products.

1.2 Introduction

A healthy and varied diet is essential at all times in life, particularly during pregnancy. The dietary intake of the women before, during and after pregnancy affects their nutritional status. It has long-lasting impacts on the child's growth, development and health across the first 1000 days and beyond. The fundamental principles of healthy eating remain the same for the general population. A balanced diet with various foods and moderate macronutrient intake provides the best chance for a healthy pregnancy and optimal perinatal outcomes (Jasper et al., 2019; Marshall et al., 2022). Thus, pregnancy is a critical period during which the mother requires different nutrients for healthy gestation.

Nutrition plays a critical role during this developmental period to meet the requirements of energy and other nutrients that are increased during pregnancy and breastfeeding (Jouanne et al., 2021). It promotes optimal foetal development and reduces the risk of non-communicable diseases (NCDs) in adulthood. The progression brings out a concept known as the "developmental origins of health and disease" (DOHaD) or "developmental programming" which predisposes the infant to lifelong NCDs (Abdul-Hussein et al., 2021).

Adequate consumption of energy and nutrients during lactation determines the proper nutrition status of a woman for recovery, facilitates weight control and, to some extent, affects the production of milk, which is essential for the proper development of the infant (Jouanne et al., 2021). This is because the energy requirement is the primary determinant of gestational weight gain (GWG) and is consistent with the clinical and public health intervention designed to improve GWG, which is directed at energy intake or expenditure (Ehrhardt et al., 2022). The RNI 2017 (NCCFN, 2017) recommends that energy requirements for individuals should be based on their actual physical activity level (PAL), and for population groups, a PAL of 1.6 can be selected. The RNI 2017 also recommend additional energy and protein for each trimester (NCFFN, 2017).

Suboptimal energy intake has been associated with abnormal foetal growth patterns, including low birth weight and large gestational age, which are associated with increased risks of developing childhood and adult chronic disease (Marshall et al., 2022). Previous studies have established that undernourished pregnant women suffer from a combination of chronic energy deficiency (CED) that leads to low birth weight (LBW), and preterm and unsuccessful birth outcomes. Similarly, a high incidence of LBW babies has been reported among mothers with low pre-pregnancy body mass index (BMI). This is pertinent as exposure to undernutrition in utero is associated with LBW and stunting in childhood. Therefore, improving maternal health and nutrition in pregnancy and early life could be critical in preventing wasting and stunting (Verma & Prasad, 2021).

It is clear that following a healthy dietary pattern before, during pregnancy and breastfeeding can positively affect health outcomes for both the mother and child in subsequent life stages. Therefore, the chapter aims to provide recommendations for meeting energy needs and requirements by choosing various foods to achieve healthy pregnancy outcomes and during breastfeeding.

1.3 Scientific Basis

Additional dietary energy is required during pregnancy to support the energy deposited in maternal and foetal tissues. Besides, pregnancy also causes a rise in energy expenditure due to increased basal metabolism and changes in the energy cost of physical activity. The same pattern is also observed during breastfeeding. Therefore, meeting optimal energy and macronutrient needs are critical during pregnancy and breastfeeding, which can be achieved by consuming various foods from each food group. Optimal maternal and foetal pregnancy outcomes depend on sufficient nutrient intakes to fulfil the maternal and foetal requirements. As previously mentioned, energy intake is the primary nutrient determinant of GWG and the process of pregnancy (IOM, 2009). Women with low or high pre-pregnancy BMI, such as being underweight, overweight and obese, and other special requirements, such as women who practice a vegetarian diet, multiple pregnancies and teen pregnancy, should be given extra attention to reduce the risk of excessive or insufficient GWG throughout pregnancy.

1.3.1 Energy needs throughout pregnancy

The main recommendation for women of reproductive age (WRA) is to eat a healthy, balanced diet as described in the Malaysian Dietary Guidelines 2020 (NCCFN, 2021) with the proportion of 90% macronutrients from total energy intake of 50-65% carbohydrate, 10-20% protein and 25-35% fat. Whilst, the energy requirement throughout pregnancy depends on factors such as maternal pre-pregnancy weight status, physical activity and the physiological demands for each trimester. The energy must match the demands of maternal metabolism, physical activity, foetal growth and other tissue growth such as fat mass, breast tissue, uterus and placenta. The energy requirement aims to achieve appropriate weight gain to minimize the risks of adverse maternal and foetal outcomes.

Maternal weight gain during pregnancy is the main basis of energy requirement during pregnancy. This is because the energy accumulated due to increasing basal metabolic rate and the total energy expenditure (TEE). Energy expenditure for pregnant women can be calculated using two different approaches. The first approach is based on serial measurements of resting energy expenditure (REE), assuming that energy expenditure of physical activity (EEPA) and the thermic effect of food (TEF) are not affected by pregnancy (Prentice et al., 1996a). The second approach is based on the serial measurements of TEE using the doubly labelled water (DLW) method. This method includes the energy expenditure for tissue deposition and any changes in TEF and EEPA (EFSA, 2013). Nonetheless, the measurements of both methods should be started before conception, which is a challenge in the practice setting.

Table 1.1 shows the recommendation of additional energy and protein intake for pregnant and breastfeeding women. The total energy requirement for women aged 18 – 29 is 1610 kcal/day. The energy needs increase following the increase of gestational age and week of pregnancy. Pregnant women are recommended an additional energy intake of 80 kcal/day in the first trimester, 280 kcal/day in the second trimester, and 470 kcal/day in the third trimester (NCCFN, 2017). The additional energy required can be achieved by increasing food intake to ensure pregnant women achieve the recommended GWG and undergo safe pregnancy. Additionally, 500 kcal/day is recommended for breastfeeding mothers.

The estimation of extra energy required during pregnancy is calculated with a mean GWG of 12 kg using factorial approaches (FAO/WHO/UNU, 2004), assuming that pre-pregnancy BMI is within the healthy range. Based on these data, the average additional energy requirement for pregnancy is 320

Table 1.1: Recommendations of additional energy and protein needs for pregnant women in Malaysia

	Age	Pregnancy	1st Trimester	2nd Trimester	3rd Trimester	Breastfeeding
Energy (kcal/day)	18 – 29 years 30 – 59 years	1610 1660	+ 80 + 80	+ 280 + 280	+ 470 + 470	+ 500 + 500
Protein (g/day)	18 – 29 years 30 – 59 years	53 52	+ 0.5 + 0.5	+ 8 + 8	+ 25 + 25	+ 19 + 19

Source: NCCFN (2017)

KM1

Choose a variety of foods to meet energy requirements for healthy pregnancy outcomes and during breastfeeding

MJ (76,530 kcal) which equates to approximately 0.29 MJ/day (70 kcal/day), 1.1 MJ/day (260 kcal/day) and 2.1 MJ/day (500 kcal/day) during the first, second and third trimesters, respectively (EFSA, 2013). The recommendations are also consistent with a recent systematic review by Savard et al. (2021) that documented the increase in TEE ranging from 4.0% to 17.7% (84–363 kcal) between early and mid-pregnancy, from 0.2% to 30.2% (5–694 kcal) between mid- and late pregnancy, and from 7.9% to 33.2% (179–682 kcal) between early and late pregnancy. The median increases in TEE were 6.2% (144 kcal), 7.1% (170 kcal), and 12.0% (290 kcal) between early and mid-, mid- and late, and early and late pregnancy, respectively (Savard et al., 2021).

1.3.2 Metabolism during pregnancy

Extra energy is required during pregnancy to grow and maintain the foetus, placenta, and maternal tissues (Jasper et al., 2019; Tran et al., 2019). The increase in energy needs is due to the increased mass of active tissue metabolism, such as maternal cardiovascular, renal, and respiratory work and tissue synthesis. The average increase in metabolic rate throughout pregnancy is 4.5% for the first trimester, 10.8% for the second trimester and 24.0% in the third trimester (Butte & King, 2005).

Energy is the primary nutrient determinant of gestational weight gain and the process of pregnancy. Energy consumption during whole pregnancy is associated with increased women's metabolism and physical activity, estimated at 77,000 kcal (FAO/ WHO/ UNU, 2004). The pregnancy process causes an increase in metabolic levels at 10.7 ± 5.4 kcal/week of pregnancy, while the amount of energy used increases by 5.2 ± 12.8 kcal/week (Most et al., 2019). The total metabolism increases can be estimated according to the pregnant woman's REE (FAO/ WHO/ UNU, 2004). An average cumulative increment of 147.8 MJ (35,330 kcal) for a gain in body mass of 12 kg was estimated from studies of well-nourished women who gave birth with adequate body masses (EFSA Panel, 2013). The increase in energy requirement during pregnancy is needed to provide energy storage for postpartum and breastfeeding.

1.3.3 Carbohydrates requirements during pregnancy

Carbohydrates are essential nutrients providing energy for both mother and the growing foetus. The recommended carbohydrate intake for pregnant women is similar to healthy non-pregnant women. The optimal proportion of carbohydrates (including starches and sugars) is essential in reducing chronic disease risk. However, there are no conclusive guidelines regarding carbohydrate consumption during pregnancy. Thus, the RNI (2017) would be used to support that the carbohydrate intake of 50-65% TEI (NCCFN, 2017) for the generally healthy Malaysian adult with the optimal proportion should be between 50 and 55% TEI (Seidelman et al., 2018). Sources of carbohydrates include starches and fibres, preferably from wholegrain sources. Adequate fiber consumption of 20-30 g/day is highlighted for preventing and treating constipation, which is common during pregnancy. Sugars are also part of the carbohydrate but are not recommended due to the impacts of dietary sugars on unhealthy dietary patterns. High simple sugar intake during gestation may contribute to an excessive GWG and develop other pregnancy complications such as GDM, preeclampsia and preterm birth (Casas et al., 2020).

1.3.4 Protein and fats requirement during pregnancy

Protein is deposited unequally across pregnancy, predominantly in late pregnancy. The deposition of protein occurs primarily in the second (20%) and third trimesters (80%) (FAO/WHO/UNU, 2004). Approximately 925 g of protein is required to support a pregnant woman's weight gain of 12.5 kg. The average protein accumulation needed is 611 g, with no difference between the weight status of pregnant women either underweight, normal weight or overweight (Butte & King, 2004).

The protein requirements during the first trimester remain the same for the general population of at least 1 g protein/kg body weight per day for adults to achieve optimal health (NCCFN, 2017). However, dietary protein must increase to 1.1 g/kg daily after the first trimester (IOM, 2009). Thus, the daily protein intake requirement is 71 g throughout pregnancy. The focus should be given to undernourished women to reduce the risk of LBW neonates. Protein consumption during pregnancy should come from foods with high biological value, such as lean meats and poultry, fish, eggs, tofu, nuts, seeds, legumes and beans.

The rate of fat deposition follows the same pattern as the gestational body mass gain; 11% of fat is deposited in the first, 47% in the second and 42% in the third trimester (FAO/WHO/UNU, 2004). The total fat stores are according to the weight status of pregnant women; underweight (5.3 kg), normal weight (4.6 kg) and overweight (8.4 kg) (Butte & King, 2004). The average rate of fat gain is 8 g per day in the first trimester, 26 g per day in the second trimester, and there is variation in fat accumulation during the third trimester, ranging between 7 g and 23 g per day (Butte & King, 2004). A study by Branco et al. (2016) showed that body fat increased by 2.9 kg in the first and 3.7 kg in the second. However, it reduced to 2.1 kg at the end of the pregnancy until the postpartum phase.

Dietary fat is an essential energy source; however, an excess of high-fat diets leads to maternal obesity. The dietary intake for total fat should be between 25 – 35% of daily energy consumption. Achieving daily fat intake recommendations should reduce the consumption of saturated fat (SFA) by choosing lean meat and low-fat milk and modifying cooking methods (Tsakiridis et al., 2020). Intake of adequate amounts of long-chain polyunsaturated fatty acids (PUFA) such as α-linolenic acid, docosahexaenoic acid (DHA) and eicosapentaenoic acid (EPA) (omega 3) and omega 6 are essential. Omega-3 PUFA has been linked with improved foetal brain development and reduced risk of preterm delivery. However, excessive intake has been associated with adverse events such as prolonged gestation and high birth weight (Akerele & Cheema, 2016). A combination of EPA and DHA at 1750 mg/week is suggested in maternal dietary fat intake to achieve the beneficial effects of PUFA (McGuire, 2011). Thus, The omega-3 recommendation is 1.2% (1.4 g/d), and omega-6 is 5 – 10% (13 g/d) of total energy intake.

1.3.5 Breastfeeding

Breastfeeding and breast milk are the global standards for infant feeding. Exclusive breastfeeding is recommended for the first six (6) months and continues for up to two (2) years of age with the appropriate complementary feeding, starting at 6 months (WHO/UNICEF, 2006). Like pregnancy, energy and dietary requirements also differ during breastfeeding. The energy requirement of a breastfeeding woman is defined as the level of energy intake from food that will balance the energy expenditure needed to maintain a body size and composition, a level of physical activity, and breast milk production, which are consistent with good health for the woman and her child. Sufficient intake allows women to perform economically necessary and socially desirable activities.

Milk production would not be affected even if the woman's diet is less nutritious, as fat deposits during pregnancy are used for milk production. Breast milk production will only decrease when women suffer severe malnutrition (WHO/UNICEF, 2006). Most women store an extra 2 to 5 kg (19,000 to 48,000 kcal) during pregnancy, mainly fat stored in tissue, in preparation for breastfeeding. The energy stores will be used to maintain breastfeeding if additional energy requirements are unmet. It is common for breastfeeding women to lose 0.5-1.0 kg/month after the first 6-8 weeks postpartum (Kominiarek & Rajan, 2016). Nonetheless, weight loss during breastfeeding does not affect the quantity or quality of breast milk. Successful breastfeeding is indicated when the breastfed infant is gaining appropriate weight. Thus, monitoring an infant's weight is critical.

The recommended energy intake during exclusive breastfeeding ranges from 2100 to 2400 kcal/L after the additional 500 kcal/day, depending on PAL (Table 1.2, see Appendices). This additional energy requirement is due to energy expenditure of 675 kcal/day for women who breastfed exclusively during the first 6 months (FAO/WHO/ UNU, 2004). As for protein, an additional 19 g for the first 6 months and 13 g for the second 6 months is recommended daily during breastfeeding (Table 1.2).

1.3.6 Benefits of meeting the energy and macronutrients needs

The benefits of meeting energy needs can be translated into desirable GWG and optimal pregnancy outcomes (Most et al., 2019). However, in a systematic review and meta-analyses of observational studies, no significant relationship was observed between the increment in energy intake from early to late pregnancy and the GWG (Jebeile et al., 2016). Underreporting food intake may explain the findings, which warrant high-quality longitudinal studies.

While there is no direct association between energy intake and GWG, meeting energy requirements with nutrient-dense food is critical as it helps women meet the vital micronutrients. Women will unlikely meet the micronutrient demand if their energy intake is not increased. They may consume essential nutrients at levels similar to pre-pregnancy, jeopardizing pregnancy outcomes. Besides, additional energy is needed during pregnancy to support foetal growth, the incremental increases in the sizes of the maternal tissues and changes in maternal energy metabolism.

This is supported by a recent prospective birth cohort study in Japan involving 89,817 mother-child pairs of live-born non-anomalous singletons investigating how energy intake during pregnancy and GWG influenced birth weight (Minami et al., 2022). They found a positive association between energy intake during pregnancy and birth weight mediated by GWG. In this study, excessive energy intake (too less and too much) during pregnancy also increased the risk of abnormal birth size (Minami et al., 2022), which strongly supports the need to meet optimal energy needs during pregnancy for appropriate GWG and foetal growth.

Meeting the energy and macronutrient needs during breastfeeding is also crucial for mothers' and infants' health and well-being. Proper nutrition supports breast milk production, ensures optimal growth and development for the infant, and helps maintain the mother's nutritional status. A recent meta-regression analysis suggested a positive association between maternal BMI and human milk fat in term-born infants between 1 and 6 months postpartum (Daniel et al., 2021). Specifically, the analysis found that for every 1 kg/m^2 increase in maternal BMI, there was a corresponding 0.56 g/L increase in milk-fat concentration. However, no association was observed between maternal BMI and human milk energy or total protein levels (Daniel et al., 2021). Furthermore, a study conducted among 367 pregnant women aged 21-45 years in Minneapolis and Oklahoma found that higher maternal diet quality at one and three months postpartum was associated with lower infant adiposity (Tahir et al., 2019). These findings highlight the importance of maternal nutrition and diet quality during breastfeeding in influencing both the composition of breast milk and the long-term health outcomes of infants.

1.3.7 Strategies to achieve the recommended energy and macronutrient needs

Although it is commonly said that pregnant women are "eating for two," the additional energy needs are modest for most women. Thus, the vital message to pregnant women is to "eat better, not more." Also, there is growing evidence that diet, at preconception, seems equally or even more critical than during pregnancy. Good nutrition affects fertility and the early development of the placenta and foetus, which occur well before a woman recognizes she is pregnant (Stephenson et al., 2018).

It is strongly recommended that women eat a variety of nutrient-dense, whole foods, including fish, fruits, vegetables, omega-3 fatty acids, and whole grains, in place of processed and ultra-processed foods and beverages to enhance nutritional quality without excessive energy intake (Marshall et al., 2022). A greater ultra-processed food intake was associated with lower diet quality (Nansel et al., 2022). An intervention study promoting the intake of minimally processed food in Brazilian pregnant women has shown a significant reduction of ultra-processed food leading to a risk reduction of excess GWG (Sartorelli et al., 2023). Thus, a recommendation to reduce ultra-processed food may help to improve adherence to dietary guidelines in pregnant and postpartum women.

Besides the variety of nutrient-dense foods, eating patterns during pregnancy and postpartum can also influence maternal energy intake and diet quality. In the Pregnancy Eating Attributes Study (PEAS), eating frequency was associated with higher daily energy intake and diet quality (Schwedhelm et al., 2022). Meanwhile, skipping meals was associated with lower diet quality and overeating at the subsequent meals. Overeating can result in weight gain, which increases the risk of GDM and other pregnancy complications such as pre-eclampsia. Furthermore, skipping meals may cause insufficient intake of essential nutrients such as proteins, vitamins, and minerals required for foetal growth and development (Chen, Loy & Chen, 2023). The finding suggests modifying eating patterns and meal regularity to support optimal energy intake and diet quality during pregnancy and postpartum.

Intake of a healthy diet is critical because a recent systematic review of observational studies on 167,507 women suggested that an unhealthy diet (e.g., refined grains, processed meat, and sources of saturated fat and sugar) during pregnancy could cause inflammation in physiological systems (Chia et al., 2019). Such inflammation further decreases placental foetal blood flow, which is associated with lower birth weight and increased risk of prematurity.

1.3.8 Special Considerations

a. Maternal obesity and underweight

The ideal condition is to have an optimal BMI and good nutritional status before pregnancy, but an increasing proportion of women entering pregnancy at a weight exceeding the healthy range. This is critical as obesity in pregnancy is a significant contributor to maternal and perinatal morbidity, with a constantly rising global prevalence among reproductive-aged women (Giouleka et al., 2023). Women with obesity are likely to have pregnancies complicated by excess GWG (>60% prevalence) and to have an increased incidence of GDM, hypertensive disorders, nonelective cesarean section, and macrosomic infants (Kim & Ayabe, 2023). Nonetheless, weight loss attempt is not recommended during pregnancy, primarily through restrictive diets, as it may lead to inadequate nutrient intake, which can negatively impact foetal growth and development.

The IOM recommendations for energy intake were mainly based on studies in pregnant women without obesity (IOM, 2009). Women with overweight and obesity are twice as likely to exceed the IOM guidelines, and the prevalence of excess GWG is forecasted to continue to rise. This results in only one-third of pregnant women achieving the recommended weight gain that is believed to lead to healthy maternal, neonate and infant outcomes.

Recommending the appropriate energy and macronutrient needs are crucial for women with obesity during pregnancy. In their analyses, Most et al. (2019b) recommended that the daily energy intake throughout the second and third trimesters not exceed energy expenditure to achieve the recommended rate of weight gain. In other words, pregnant women with obesity should not consume additional energy intake. The extra energy for foetal development would be compensated by maternal fat mass with no adverse effects on maternal and foetal outcomes (Most et al., 2019a). Therefore, the appropriate energy needs for women with obesity must be personalized and given together with physical activity and improvement in dietary quality. Besides, excessive energy intake rather than decreased energy expenditure after pregnancy would also lead to postpartum weight gain (Most et al., 2020). Thus, women should be advised not to overeat after pregnancy, and appropriate dietary intake recommendations must be given to women with obesity during the postpartum period.

Similarly, women with underweight pre-pregnancy BMI had a significantly higher risk of delivering an infant with LBW. A cross-sectional survey in Taiwan observed that processed food consumption in the third trimester was associated with LBW. Nutritious food during pregnancy is a critical factor in LBW, and the findings are even significant among older and pre-pregnant underweight women. Thus, prenatal nutrition advice should emphasize individualized optimal nutrition (Chen, Loy & Chen, 2023). The LBW babies are more likely not to survive during the first month of life and tend to suffer from long-term consequences of a higher risk of stunting, low IQ and high risk of adult-onset non-communicable disease (NCDs) such as obesity and type 2 diabetes (UNICEF, 2023).

b. Multiple pregnancy

The maternal metabolic rate in twins is approximately 10% greater than in singletons. The physiological changes in multiple gestations are highly demanding than the singleton pregnancy. This includes increased plasma volume and decreasing haemoglobin, albumin, and water-soluble vitamins. Generally, there is a consensus in the literature that the energy demand is higher than in a singleton pregnancy. Still, more position statements from scientific societies must focus on the required energy intake. Due to a lack of studies and the actual energy needs of women pregnant with twins remaining unknown, Wierzejska (2022) theoretically assumed that 150 kcal/day might need to be additionally consumed compared to the recommendations for singleton pregnancies, taking the target GWG of 20 kg. It is also estimated that a 40% higher-calorie diet may maintain the woman's nutritional state during a twin pregnancy. Evidence for nutritional management of other multiple pregnancies (triplets, quadruplets) is lacking, but they can be managed similarly to twin gestations (Kominiarek & Rajan, 2016).

c. Adolescent pregnancy

The physiology of pregnancy is different in teenagers compared to adult pregnant women. Young pregnant teenage girl (particularly <16 years) is a significant risk factor for stillbirth, preterm delivery, low-birth weight, and neonatal mortality (Marvin Dowle & Soltani, 2020) due to pregnancy coincides with continuing and incomplete growth of pregnant teenagers (Wallace, 2019). As in adult pregnant women, the nutritional needs of teen pregnancy change during pregnancy, with requirements for energy and several micronutrients increasing as pregnancy progresses. Thus, optimal dietary intake is critically important during continued growth and development. However, the dietary patterns of pregnant teens are generally less healthy than those of pregnant adult women. The diets of adolescents were characterized by a higher intakes of snacks and processed foods, fewer fruits and vegetables and fewer nutritional supplements than adult women (Marvin-Dowle et al., 2018; Samano et al., 2022).

Systematic reviews have reported that intakes of energy and other micronutrients in adolescents pregnancy were below the recommended level. Blood markers showed that anaemia and low iron status might be prevalent in this population (Marvin-dowle, Burley & Soltani, 2016). Teen pregnancy is also a risk factor of excessive (37%) and insufficient (41%) GWG (Samano et al., 2022). The diet of Mexican teen pregnant girls associated with excessive GWG included cereals, grains and animal-sourced foods along with inadequate intake of legumes. However, energy, macronutrient intake, and eating habits were not associated with GWG (Samano et al., 2022). Thus, the recommended energy intake for the adolescent pregnant girl must be consistent with a recommended intake that suits their age, considering pregnancy trimesters.

d. Vegetarians

According to the American Dietetic Association, vegan diets are healthful, nutritionally adequate, and safe for all age groups and physiological conditions, including childhood, adolescence, pregnancy and Breastfeeding (Vesanto, Craig & Levin, 2016). The Physicians Committee for Responsible Medicine also stated that a plant-based diet is a healthful choice at every stage of life, including pregnancy and breastfeeding (Hanson et al., 2015).

However, it is crucial to include all food groups in the daily diet, considering the nutritional values of each food group. Insufficient or excessive intake of these foods should be avoided as both situations have significant implications on an individual's health. Balanced vegetarian diets meet the nutritional needs of all life cycle stages, including pregnancy, breastfeeding, infancy, childhood, adolescence, and older adults (Vesanto, Craig & Levin, 2016). Well-planned vegetarians tend to consume less overall energy; a lower dietary fat (particularly saturated fat) and dietary cholesterol; and higher intakes of fruits, vegetables, whole grains, nuts, soy products, fibre, and phytochemicals than non-vegetarians. Because of the variability in vegetarian diets, individuals need to become familiar with their nutritional needs and potential dietary deficiencies.

The proportions of macronutrients required do not differ during breastfeeding from those of nonlactating women. Lactating women can acquire all essential amino acids from high-quality protein sources such as meat, fish, eggs, and milk. For vegetarian women, plant protein sources, including legumes, nuts, fruits, starchy root vegetables, and cereals, should be combined to ensure all essential amino acids are consumed. Minimally processed carbohydrates and fibre should be consumed in similar proportions as in the pregnancy diet (Hanson et al., 2015). It is advised that pregnant (including those planning to become pregnant) and breastfeeding women on a vegan diet should give special attention to dietary intake and get advice from healthcare professionals.

1.3.9 Risk of suboptimal intake during pregnancy and breastfeeding

Various populations are at risk of suboptimal energy and macronutrient intake during pregnancy. This population must receive individualized nutrition consultation and appropriate coordination of care, and work as multidisciplinary teams. The risks are prone to pregnant women with morning sickness, food taboos, fasting, substance usage and other factors.

a. Morning sickness

It is important to differentiate between morning sickness and hyperemesis gravidarum. Morning sickness is mild nausea and vomiting commonly occurring in early pregnancy, affecting as many as 70-85% of pregnant women. It usually goes away by the second trimester. Treatment is primarily symptomatic and typically involves avoiding certain foods and eating small but frequent meals. There is typically no long-term effect of morning sickness on pregnant women and their infants (IOM, 2009).

About 0.5 to 2.0 per cent of pregnant women will experience hyperemesis gravidarum (ACOG, 2004). Hyperemesis gravidarum is persistent vomiting unrelated to other medical conditions and weight loss of 5 per cent or greater of pre-pregnancy weight at < 16 weeks' gestation (IOM, 2009). Pregnant women who experience morning sickness or hyperemesis gravidarum should see a healthcare provider to discuss their conditions and develop a strategy for ensuring adequate nutrition.

b. Foods taboos

Food taboos may limit the quantity and quality of food that pregnant and breastfeeding women choose to consume, making the woman more vulnerable to energy, macronutrients and micronutrients deficiencies (Kohler et al., 2018). Avoiding certain protein foods without a proper replacement may put them at risk for malnutrition. It is important to note that pregnant and breastfeeding women require a variety of energy-rich and nutrient-rich foods for the health of both the mother and offspring. A systematic review has identified that animal protein is among the most common practice of food taboos (Kohler, Lambert & Biesalski, 2019). Animal products have better protein quality and the highest protein digestibility (EFSA, 2013). Protein intake is critical during pregnancy to support protein biosynthesis for maternal tissue requirement and foetus development and growth.

c. Fasting during pregnancy

Fasting is practised in several religions, such as Muslim, Christian and Hinduism. While religions exempt pregnant women from fasting, most still choose to fast. For example, a systematic review reported that Ramadhan fasting did not affect the health of healthy pregnant women and their offspring. A cohort of Indonesian family life surveys involving a 2000 database found that adult Muslims who were conceived during Ramadhan were slightly thinner than Muslims who did not. However, another study has shown that the overall energy intake was not dramatically reduce but may affect the overall dietary quality, which warrants further investigation. A pregnant woman who decides to fast must discuss with their doctor and maintain a healthier dietary intake and pattern and good hydration status (Oosterwijk et al., 2021).

d. Other factors

Maternal factors such as stress, anxiety, and smoking can decrease milk production, but breastmilk's quantitative and caloric value does not change with dieting and exercise. Moreover, body weight, BMI, body fat percentage, and weight gain during pregnancy do not influence milk production. Also, the volume of milk is not affected by the nutritional status of mothers except in severe dehydration (vomiting and diarrhea with loss of over 10% of the body fluids). During severe famines and malnutrition crises, the nutritional status of pregnant women and the foetus can be affected, not the breastfeeding mother and her baby. Research shows that "energy sparing adaptations" were associated with normal lactation when energy intake is limited. These will decrease basal metabolic rate, thermogenesis, and physical activity (Savard et al., 2021).

1.3.10 Planetary diet during pregnancy and postpartum

The planetary diet was proposed by EAT-Lancet Commission in 2019 as a healthy and sustainable model diet that aims to provide health to the population and the planet (Willet et al., 2019). The diet includes a range of possible intakes by food group and substantially restricts the intake of highly processed and animal-source foods. The diet is similar to a plant-based diet, and its benefits to human health are clear. Nonetheless, no direct study has assessed planetary diet and energy intake during pregnancy and breastfeeding. Concerns have also been raised about how the diet provides adequate essential micronutrients, particularly those generally found in higher quantities and more bioavailable forms in animal-source foods (Beal, Ortenzi & Fanzo, 2023). Deficiencies in these micronutrients would contribute to substantial public health burdens, especially among reproductive-age women.

1.4 Current Status

1.4.1 Energy and dietary intake during pregnancy

The RNI recommends women of reproductive age consume around 1610 - 1660 kcal per day (NCCFN, 2017). Studies among pregnant women in Malaysia reported that energy consumption did not meet the Malaysian RNI. A cross-sectional study conducted at selected health facilities in Hulu Langat District, Selangor, found that only 1.7% of pregnant women aged 18 to 45 met the energy recommendation (Samiun et al., 2019). Another study among 236 pregnant Malay women aged 20 to 45 years was conducted in two private hospitals in Klang Valley and found that the mean (SD) daily energy intake was 1,748 (526) kcal/day, which achieved only 76.0% of the Malaysian RNI (Zahara et al., 2014).

Pregnant women in Thailand and Indonesia experienced the same issue of inadequate energy intake. According to a study in Narathiwat, Thailand, the average calorie consumption among pregnant women was 1,204 kcal/day, compared to the Thai RDA of 2,300 kcal/day (Sukchan et al., 2010). Meanwhile, in Indonesia, the average calorie intake of pregnant women was 1,670 kcal/day, compared to the Indonesian dietary recommendation of 2,200 kcal/day (Yayuk et al., 2012). This is of concern because women's diet and nutritional status during pregnancy have a significant impact on delivery outcomes (Mohamed et al., 2022). Indeed, about 70% of Malay pregnant women had avoided at least one food item due to food taboos, and those practising food taboos had inadequate GWG (Mohamad & Ling, 2016).

In Malaysia, a nationwide survey reported about 1 out of 10 live birth babies (10%) were born LBW (IPH, 2016). This is comparable to the neighbouring countries Thailand (9.2%) but lower than Indonesia (20.2%) (Sukchan et al., 2010; Yayuk et al., 2012). While no data was reported for the adequacy of energy intake in these studies, data from a Japanese cohort, found that the group with low energy intake had a higher risk of small-gestational for age (SGA) birth than the moderate energy intake group (adjusted RR: 1.12; 95% CI 1.04-1.19) (Minani et al., 2022).

With regards to specific food groups, a study among pregnant women in Kelantan, Malaysia showed that higher fruits intake was associated with greater birth weight, while intakes of confectioneries and condiments were both associated with lower birth weight (Loy et al., 2013). Assessing a single food item may not provide a clear link between food groups and pregnancy outcomes. Thus, the assessment of dietary patterns in the GUSTO (Growing Up in Singapore Towards Healthy Outcomes) birth cohort created a better understanding of the whole diet and pregnancy outcomes (Chia et al., 2016). In this GUSTO cohort, the maternal dietary pattern characterized as high in vegetables, fruits and rice was associated with a lower risk of preterm births but an increased risk of an LGA birth. High intake of rice, in particular, white rice, could be related to high GI diets. Diet that high in GI has been associated with a large birth size and lower preterm birth risk (Chia et al., 2016). However, there is no elaboration on about energy or individual nutrients.

The dietary pattern of Malaysian pregnant women is distinct from what observed in Western countries (Yong et al., 2019). Pregnant women with a higher waist circumference at the first prenatal visit were likely to portray low adherence to certain dietary patterns characterized by high intake of condiments, spices, sugar, spread and creamer, which could explain their prior exposure to nutrition education to reduce unhealthy dietary component in preventing excess GWG. Achieving appropriate weight gain during pregnancy is critical for the offspring. The NHMS for the maternal and child health (IPH, 2023) reveals the double burden of malnutrition among Malaysian children, in which 6.0% are overweight and nearly 20% are stunted with equally distribution in rural and urban area (IPH, 2023).

While certain dietary patterns are protective against metabolic obesity, a SECOST cohort study of pregnant women in Negeri Sembilan found relatively low healthy eating index (HEI) scores, and the total HEI score varied throughout pregnancy (Yong et al., 2019). For pregnant women without obesity, higher total HEI scores (better dietary quality) in the second and third trimesters were significantly associated with a lower risk of inadequate GWG. Thus, achieving appropriate energy intake with high dietary quality is important for optimal GWG. Nearly 15% of pregnant women in Malaysia have obesity (IPH, 2016). In a study conducted at a health clinic in Batu Pahat district, women who were overweight/ obese at pre-pregnancy were at-risk of having excessive GWG rates (Chee et al., 2019). The findings highlight the need for preconception counselling on GWG via appropriate energy and dietary intake.

1.4.2 Dietary intake during breastfeeding

In exclusive breastfeeding women, despite the increased requirements, intakes of total energy, protein, total fat, dietary fibre, and water did not meet the recommended values throughout the exclusive breastfeeding period (Basir et al., 2019). The inadequate intakes suggest that traditional postpartum dietary practices may influence mothers' ability to meet their nutritional requirements during lactation. A qualitative study explored the perceptions towards dietary practices that would increase breastmilk production among working mothers in Selangor, Malay (Suhaimi & Teng, 2022). Mothers believed that a fluid intake and a healthy and balanced diet facilitate breastmilk production. They also thought that eating a lot would promote breast milk production. A higher maternal energy intake as per recommended intake was associated with a higher success rate of exclusive breastfeeding at 6 months than those with lower energy intake (Fikawati & Shafiq, 2017).

While optimal dietary intake of the mother is critical during lactation, it may not influence the quality of human milk. Basir et al. (2019) determined the relationship between maternal dietary intake with nutritional composition of human milk among Malay mothers during the postpartum period of exclusive breastfeeding. While the composition of the maternal diet and nutritional content of human milk differed between confinement and post-confinement periods, no associations was observed between nutritional contents of human milk and maternal dietary intake. Besides, maternal energy intake was not associated with the energy value of human milk, which is consistent with a systematic review that reported maternal total calorie intake did not have any influence on the calorie concentration of human milk (Bravi et al., 2016).

1.5 Key Recommendations

Key Recommendation 1

Choose a variety of foods to meet your energy needs during pregnancy according to the trimester.

How to achieve:

1. Increase energy and protein needs according to trimesters by having a small and frequent meal containing a variety of foods, including a healthy snack (Table 1.2, see Appendices).

   i. For the first trimester; +80 kcal/day and +0.5 g/day of protein (increase slightly from the current protein intake).

   ii. For the second trimester; +280 kcal/day and +8.0 g/day of protein or about ½ additional serving of protein.

   iii. For the third trimester; +470 kcal/day and +25.0 g/day of protein or about 2 additional servings of protein.

2. Ensure to practise healthy eating habits throughout the pregnancy, as pregnancy cravings tend to favor unhealthy food choices, by practicing the following strategies. The strategies are similar across trimesters.

   i. Choose a combination of all food groups in the Malaysian Food Pyramid 2020 (Figure 1.1) to ensure the body gets all the nutrients needed within the recommended number of servings for each food groups based on daily energy needs (Table 1.3, see Appendices).

   ii. Consume protein-based foods of high biological value, such as fish, lean meats and poultry, eggs, tempeh, tofu, nuts, seeds and legumes (combine bean, lentil and soy) in the appropriate amount (Table 1.3, see Appendices).

   iii. Choose cereal-based products from whole grains such as brown rice, oats, corn, wholemeal bread, chapati, putu mayam, ragi and barley, which are also rich in fibre.

   iv. Choose foods rich in omega 3 (e.g., *siakap, cencaru, kembung,* nuts and seeds) and omega 6 (corn oil, soybean oil, and sunflower oil) fats.

   v. Reduce intake of foods high in saturated fats (e.g., butter, fatty meat, full cream milk, coconut milk and coconut oil).

   vi. Consume 2-3 servings of milk and milk products daily. Replace sweetened condensed milk, sweetened condensed filled milk, and sweetened creamer with milk and milk products.

   vii. Limit intake of ultra-processed foods such as soft drinks, sweetened breakfast cereals, salty fatty packaged snacks and instant noodles, which are nutritionally imbalanced.

Key Recommendation 2

Eat the main meals (breakfast, lunch and dinner) as recommended by the Malaysian Healthy Plate (Figure 1.2).

How to achieve:

1. Ensure regular mealtimes and do not skip the main meal.

2. Use the Malaysian Healthy Plate (Figure 1.2) for your main meal, i.e., breakfast, lunch and dinner.

   i. Fill half of the plate with non-starchy vegetables and fruits.

   ii. Fill in the first quarter of the plate with a part of fish, poultry, lean meat, eggs, legumes (e.g., dhal, tempeh, soybean curd), or dairy products.

   iii. Fill in the second quarter of the plate with a scoop of rice or other cereals, preferably from wholegrain cereal-based products or tubers (e.g., sweet potato).

3. Drink water or unsweetened beverages with the meal.

4. Choose nutritious food at home more often and healthier options when eating out, such as:

    i. Food with less salt, sugar, and fat, sauces, and flavor enhancer.

    ii. Healthier cooking methods such as stir-frying, steaming, simmering, stewing, grilling, baking, or boiling and reducing deep frying, or adding thick *santan* to the dishes.

    iii. Use fresh ingredients for cooking such as fish, tauhu, tempeh instead of using ingredients made from commercially prepared processed or ultra-processed foods such as fish balls, meatballs, salami, sausage, etc.

Key Recommendation 3

Choose a healthy snack to meet additional energy and protein needs.

How to achieve:

1. Keep your snack times scheduled in between your main meals.

2. Add 1-3 healthy snacks according to trimester stage in a recommended serving size (Table 1.4 and Table 1.5, see Appendices).

3. Choose nutrient-dense snacks (high nutrient contents) such as milk and milk products (e.g., milkshakes, yoghurt, low salt cheese), legumes (e.g., boiled or steamed chickpeas, green peas porridge, baked beans, soybeans), nuts and seeds (e.g., peanuts, almonds, walnuts).

4. Limit intake of ultra-processed foods such as soft drinks, sweetened breakfast cereals, salty fatty packaged snacks and instant noodles, which are nutritionally imbalanced.

5. Reduce the frequency of eating fast food.

Key Recommendation 4

Choose a variety of food to meet energy and protein needs during exclusive breastfeeding.

How to achieve:

1. Consume an additional energy and protein of 500 kcal/day and 19.0g/day (1 servings), respectively for the first 6 months during exclusive breastfeeding (Table 1.2, see Appendices).

2. Consume an additional 13.0g/day (~1 serving) of protein for the second 6 months (Table 1.2, see Appendices) by having a small and frequent meal containing a variety of foods. No additional energy needs at this stage.

3. For all breastfeeding mothers, choose a variety of food groups to ensure that the body obtains all of the nutrients it need in the recommended amount (Table 1.3, see Appendices), as in the Malaysian Food Pyramid 2020 (Figure 1.1).

4. Use the Malaysian Healthy Plate (Figure 1.2) for the main meal, i.e., breakfast, lunch, and dinner.

    i. Fill half of the plate with non-starchy vegetables and fruits.

    ii. Fill in the first quarter of the plate with a part of fish, poultry, lean meat, eggs, legumes (e.g., dhal, tempeh, soybean curd) or dairy products.

    iii. Fill in the second quarter of the plate with a scoop of rice or other cereals, preferably from wholegrain cereal-based products or tubers (e.g., sweet potato).

5. Consume main meal that includes:

    i. Protein-based food with high biological value, such as lean meats and poultry, fish, eggs, tempeh, tofu, nuts, seeds, legumes and beans in the appropriate amount (Table 1.3, see Appendices).

ii. Omega-6 polyunsaturated fatty acid-rich oil (i.e., corn oil, soybean oil & sunflower oil) and omega-3 polyunsaturated fatty acid-rich oil (i.e., *siakap, cencaru, kembung,* nuts, and seeds) and reduce intake of high-fat food, especially from saturated fat (i.e., butter, fatty meat, full cream milk, coconut milk and coconut oil).

iii. Cereal-based products from whole grains such as brown rice, oats, corn, wholemeal bread, chapati, putu mayam, ragi and barley are also foods rich in fibre.

6. Consume 2-3 servings of milk and milk products daily. Replace sweetened condensed milk, sweetened condensed filled milk and sweetened creamer with milk and milk products.

7. Choose nutrient-dense snacks (high nutrient contents) such as milk and milk products (e.g., milkshakes, yogurt and low salt cheese), legumes (e.g., boiled or steamed chickpeas, mung beans porridge, baked beans and soybeans), nuts and seeds (e.g., peanuts, almonds and walnuts).

8. Limit intake of ultra-processed foods such as soft drinks, sweetened breakfast cereals, salty fatty packaged snacks and instant noodles, which are nutritionally imbalanced.

9. Reduce the frequency of eating fast food.

Key Recommendation 5

Drink plenty of water daily during pregnancy and breastfeeding.

How to achieve

1. Drink plenty of water throughout the day; six to eight glasses (1.5 – 2.0 liter; 1 glass= 250ml) of plain water daily, even if not feeling thirsty.

2. Women who are breastfeeding should drink the amount needed to satisfy their thirst. Look for signs and symptoms of dehydration, such as excessive thirst, dark yellow urine colour, dry mouth, lips and eyes, and less frequent urination.

3. Reduce intake of drinks that high sugar especially for women that gains too much weight. Water, milk, and milk products are the best choices.

4. Limit the caffeinated beverage intake not more than 200 - 300 mg/day (about 1 cup) in order not to interfere too much with iron absorption. This includes brewed coffee, instant coffee, tea or chocolate drink.

5. Avoid alcoholic drinks.

1.5.1 Additional Recommendations for Pregnant Women with Special Considerations

1. Women with special considerations must receive individualized nutrition consultation:

   i. Underweight pregnant women must achieve the recommended energy and protein needs according to the trimester (Table 1.2, see Appendices).

   ii. Pregnant women with obesity are not recommended to reduce weight during pregnancy. The energy and protein requirements must be individualized according to the gestational weight gain, physical activity, health status, and lifestyle practices.

   iii. Pregnant adolescent must receive individualized energy and macronutrient prescriptions to cater to the higher requirements for their age and pregnancy.

   iv. Women with multiple pregnancies must achieve the recommended energy and protein needs per pregnancy stage. Any additional energy and protein requirements must be individualized according to gestational weight gain, physical activity, health status, and lifestyle.

2. Pregnant and breastfeeding women who are practicing vegetarianism must get medical and nutritional advices. They must include all food groups in daily diet, as they are at higher risk of protein and certain nutrients deficiencies. Vegetarian women must achieve the recommended energy and macronutrient needs according to the pregnancy stage and during breastfeeding by having small and frequent meals containing a variety of foods, including a healthy snack.

    i. Choose a combination of all food groups to ensure the body gets all the nutrients wihin the recommended amount (Table 1.6, see Appendices).

    ii. Consume plant protein-based with high biological value, such as tempeh, tofu, nuts, seeds, legumes and beans, in the appropriate amount.

    iii. Reduce intake of foods high in saturated fats (butter, fatty meat, full cream milk, coconut milk, and coconut oil) and consume food rich in fatty acids omega 3 (e.g. nuts and seeds) and omega 6 (corn oil, soybean oil and sunflower oil).

    iv. Increase consumption of whole grains and fibre as nutritional recommendation for the normal pregnancy.

3. Food taboos during pregnancy

    i. Food taboos can also influence pregnant women during pregnancy and postpartum based on cultural perceptions and beliefs. Most food taboos are based on the thought that consuming a certain food can cause danger to pregnant women and their babies. The food taboos usually are not supported by scientific evidence.

    ii. Pregnant mothers are strongly recommended to practice healthy eating and a balanced diet during pregnancy and postpartum to ensure optimum baby growth for the babies, preparation for lactation and mothers' speedy recovery after giving birth.

    iii. Women who avoid certain food during pregnancy and the postpartum phase are advised to replace the food from the same food groups to ensure it will not affect healthy eating.

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Appendices

Table 1.2: Recommended energy intake, additional energy and total estimated energy requirements based on the pregnancy stage and during exclusive breastfeeding

	Recommended Energy Intake				
	Energy intake (kcal/day)*	Additional Energy Requirement (kcal/day)	Total Estimated Energy Requirement (kcal/day)	Protein intake (g/day)**	Additional Protein Requirement (g/day)
During Pregnancy					
First Trimester	1600 – 1900	+ 80	1680 – 1980	50 – 53	+ 0.5
Second Trimester	1600 – 1900	+ 280	1880 – 2180		+ 8.0 (~ ½ serving)
Third Trimester	1600 – 1900	+ 470	2070 – 2370		+ 25.0 (~ 2 servings)
For Breastfeeding					
First 6 months of exclusive breastfeeding	1600 – 1900	+ 500	2100 – 2400	50 – 53	+ 19.0
Second 6 months	1600 – 1900	No additional energy needs			+ 13.0 (~1 serving)

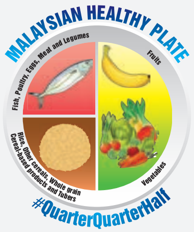
*Estimated Energy Requirement for women of reproductive age (WRA, 15–49) was based on the reference weight between 50–53 kg and PAL between low active (PAL 1.4) and moderate active (PAL 1.6). Choose the lower range of energy for those at higher range of weight and lower range of PAL.
**Estimated Protein Intake for women of reproductive age (WRA, 15-49) was based on the reference weight between 50–53 kg
Source: Recommended Nutrient Intake (RNI), 2017

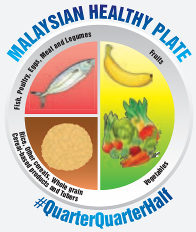
Table 1.3: Recommended number of servings for food groups based on daily energy needs

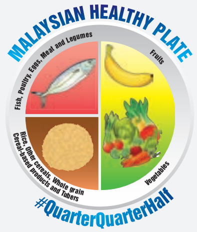
Food Group	~ 1700 kcal/day	~ 1900 kcal/day	~ 2100 kcal/day	~ 2300 kcal/day
Vegetables[1]	3	3	3	3
Fruits[2]	2	2	2	2
Rice, other cereals, wholegrain cereal-based products and tubers[3]	3.5	4.5	5	6
Meat/ Poultry[4]	1	1	1.5	1.5
Egg[4]	1	1	1	1
Fish/ Shellfish[5]	1.5	2	2	2
Legumes (combine bean, lentil and soya)[6]	1	1	1	1
Milk and milk products[7]	3	3	3	3
Fats/ oil (including 1 serving from nuts and seeds)[8]	5	5	6	7
Sugar[9]	1	1	1	1

Notes:
Tips to remember, the more physically active you are, the more calories are required per day. However, fewer calories are needed per day if you are very sedentary.
[1] Calorie free
[2] Based on 15 g carbohydrate and 60 kcal per serving,
[3] Based on 30 g carbohydrate, 4 g protein, 1 g fat and 150 kcal per serving,
[4] Based on 14 g protein, 8 g fat and 130 kcal per serving,
[5] Based on 14 g protein, 2 g fat and 70 kcal per serving,
[6] Based on 40 g carbohydrate, 14 g protein, 0.5 g fat and 220 kcal per serving,
[7] Based on 15 g carbohydrate, 8 g protein, 1 g fat and 90 kcal per serving,
[8] Based on 5 g fat and 45 kcal (including 1 serving of nuts & seeds = 5 g of fat and 65 kcal),
[9] Based on 15 g CHO and 60 kcal. 1 serving of sugar = 3 teaspoons; 1 teaspoon = 5 g of carbohydrate and 20 kcal.
Sources: Suzana et al. (2015); *Recommended Nutrient Intake (RNI), 2017.

Table 1.4: Timing and amount of snacks according to the stage of pregnancy and during exclusive breastfeeding







Main Mealtime : 1500 kcal	Breakfast	Lunch	Dinner
Additional Healthy Snacks	**Morning Tea**	**Afternoon Tea**	**Supper**
Add 1 healthy snack during the first trimester (+ 80 kcal/day)	√		
Add 2 healthy snacks during the second trimester (+ 280 kcal/day)	√	√	
Add 3 healthy snacks during the third (+ 470 kcal/day)	√	√	√
Add 3 healthy snacks during exclusive breastfeeding (+ 500 kcal/day)	√	√	√

The healthy snack is characterized as a nutrient-densed meal and must include about 1 serving of protein

Table 1.5: Example of one day menu and serving size for 1500 kcal/day and snacks according to the stage of pregnancy and during exclusive breastfeeding

Main Meal			
Breakfast	• 2 slices (60 g) of wholegrain bread • 2 tablespoon tuna • Salad and tomatoes • 1 glass of milk	**Dinner**	• 2 scoops of brown rice • 1 scoop (60 g) of stir fried *kangkung* • 1 bowl (250 g) of chicken soup with carrot • 1 slice (110 g) of medium size papaya • 1 glass of plain water
Lunch	• 2 scoops of brown rice • 1 scoop (60 g) of stir fried spinach with *tauhu* • 1 scoop (30 g) of string beans and brinjal • 1 piece of fried fish with chilli (80 g) • 1 whole medium apple • 1 glass of plain water		
Additional Healthy Snack	**Take 1 snack per day as morning tea or afternoon tea**		
First Trimester *(+ 80 kcal/day and + 0.5 g/day of protein)*	Example: • 3 pieces cream-crackers (~89 kcal and 1.8 g protein) OR • 1 whole orange (~71 kcal and 1.0 g protein) OR • 1 handful (15 g) of unsalted almond (~ 86 kcal and 3.0 g protein)		
Additional Healthy Snack	**Take 2 snacks per day as morning tea and afternoon tea**		
Second Trimester *(+280 kcal/day and +8.0 g/day of protein)*	**Morning Tea** Example: • 3 pieces cream-crackers (~89 kcal and 1.8 g protein) OR • 1 whole orange (~71 kcal and 1.0 g protein) OR • 1 handful (15 g) of unsalted almond (~ 86 kcal and 3.0 g protein)	**Afternoon Tea** Example: • 1 glass (250 g) of low-fat milk (~131 kcal and 10 g protein) OR • 1 tub (184 g) of plain yoghurt (~115 kcal and 9 g protein) OR • 1 glass of unsweetened soya milk (~112 kcal and 7 g protein) OR • ½ cup (80 g) of boiled chickpeas (~151 kcal and 7 g protein)	

Additional Healthy Snack	Take 3 snacks per day as morning tea, afternoon tea and supper		
Third Trimester *(+ 470 kcal/day and + 25.0 g/day of protein)*	**Morning Tea** Example: • 1 glass of unsweetened soya milk (~112 kcal and 7 g protein) OR • ½ cup (80 g) of boiled chickpeas (~151 kcal and 7 g protein)	**Afternoon Tea** Example: • 1 piece (93 g) of pau ayam (~199 kcal and 10.1 g protein) OR • 1 bowl (230 g) of bubur kacang hijau (~244 kcal and 9 g protein)	**Supper** Example: • 1 glass (250 g) of low-fat milk (~131 kcal and 10 g protein) OR • 1 tub (184 g) of plain yoghurt (~115 kcal and 9 g protein)

Table 1.6: Number of servings for food groups based on daily energy needs for vegetarians

Food Group	~ 1700 kcal/day	~ 1900 kcal/day	~ 2100 kcal/day	~ 2300 kcal/day
Vegetables	3	3	3	3
Fruits	2	2	2	2
Rice, other cereals, wholegrain cereal-based products and tubers*	3.5	4	4.5	5.5
Eggs/ meat substitute**	3	3.5	3.5	4
Legumes	1–2	1–2	2	2
Milk and milk products	2–3	2–3	3	3
Fats/ oil	4	4	4	3
Sugar	1	1	1	1

*based on 30 g carbohydrates per serving.
**based on 14 g protein per serving.

Figure 1.1: Malaysian Food Pyramid 2020

Notes:
- The number of servings is calculated based on 1500 to 2300 kcal.
- This pyramid is meant for children aged 7 years and older; for younger children, refer to the Malaysian Dietary Guidelines (MDG) for Children and Adolescents.
- For adolescents aged 13 to 15 years, the recommendation for fruits is 2-3 servings and for milk and milk products 2-3 servings.
- For adolescents aged 16 to < 18 years, the recommendation for fruits is 2-3 servings, milk and milk products 2-3 servings and for rice, other cereals, whole grain cereal-based products and tubers 3-6 servings.

* This includes ultra-processed foods which contain artificial substances such as colours, sweeteners, flavours, preservatives, and other additives.

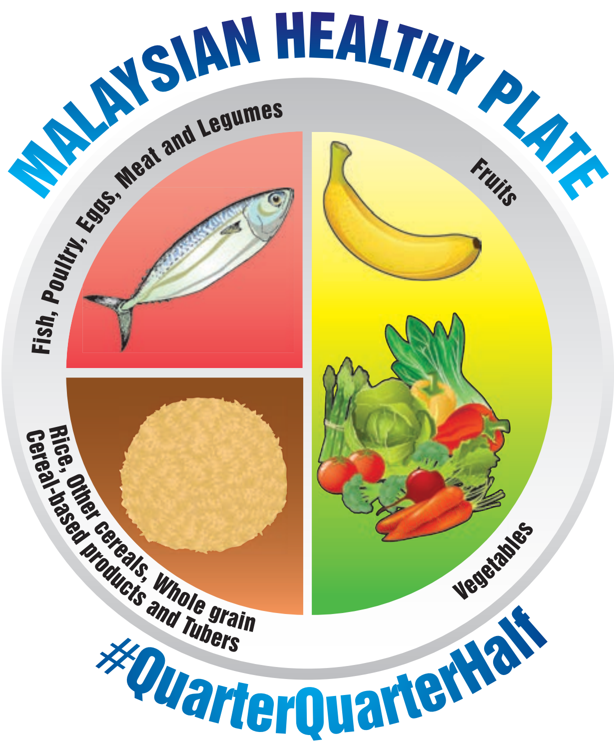
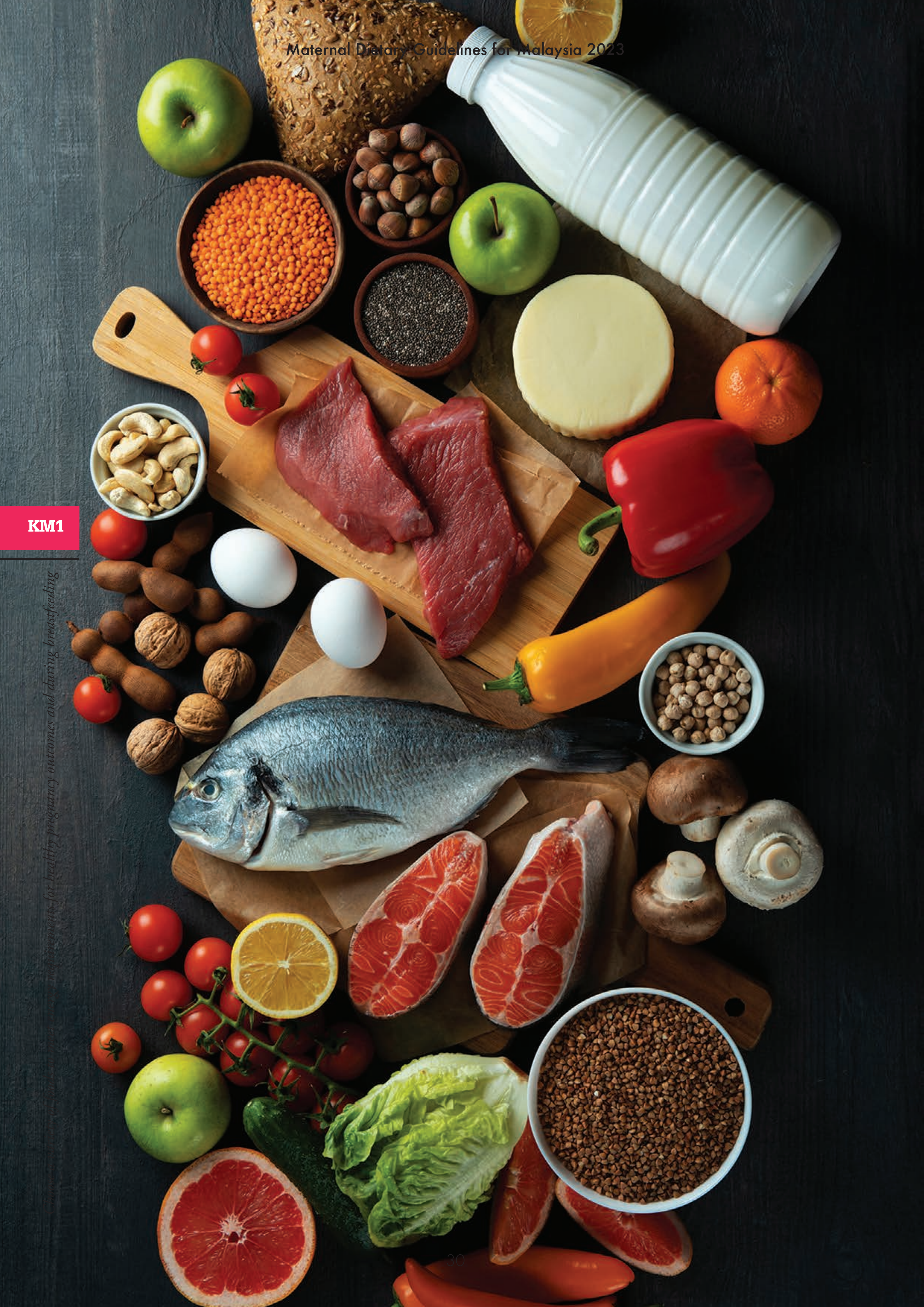


Figure 1.2: Malaysian Healthy Plate 2016

KM1

Choose a variety of foods to meet energy requirements for healthy pregnancy outcomes and during breastfeeding



Choose variety and micronutrient-dense foods

Key Message 2

Choose variety and micronutrient-dense foods

Dr. Tan Sue Yee, Asst. Prof. Dr. Serene Tung En Hui, Dr. Wirdah Mohamed, Prof. Dr. Ruzita Abd Talib, Ms. Siti Mariam Ali, Mr. Suhaidi Sudin and Ms. Wong Hui Juan

2.1 Terminology

Fortified foods
Foods that have undergone fortification deliberately increase the content of one or more micronutrients in a food or condiment to improve its nutritional quality (WHO, n.d.). Examples include wheat flour fortified with folic acid and iron, breakfast cereals fortified with iron, and milk and milk products fortified with calcium and vitamin D.

Heme iron
Dietary iron is found in two forms - heme and non-heme iron. More than 95% of functional iron in the human body is heme (Beutler et al., 2006). Heme iron is well absorbed and presents mainly in meat, poultry and fish (Hooda et al., 2014).

Micronutrient
Micronutrients are vitamins and minerals needed by the body in very small amounts. However, their impact on a body's health are critical, and deficiency in any of them can cause severe and even life-threatening conditions (WHO, n.d.).

Mineral
Minerals are inorganic substances that must be ingested and absorbed in adequate amounts to satisfy a wide variety of essential metabolic and/or structural functions in the body. Macrominerals (calcium, phosphorus, magnesium, sodium, potassium, chloride) are required by humans in amounts of hundreds of milligrams to several grams per day; while microminerals or trace elements (iron, zinc, copper, manganese, selenium, iodine, chromium, molybdenum) are required in amounts of a few milligrams or less per day (Cockell, 2011).

Nutrient
A chemical compound in foods that is used by the body to function and grow (National Cancer Institute, n.d.). Nutrients consist of macronutrients such as proteins, fats and carbohydrates, and micronutrients such as vitamins and minerals.

Nutrient-dense
Nutrient-dense is defined as foods that supply relatively more nutrients than calories (Drewnowski & Fulgoni, 2014).

Non-heme iron
Non-heme iron is a form of dietary iron found mainly in plants and is less well absorbed (Hooda et al., 2014).

Vitamin
Vitamins are vital micronutrients that cannot be synthesised endogenously or in insufficient amounts, thus they can only be obtained through diet. Vitamins can be classified as water-soluble (vitamins B and C) or fat-soluble (vitamins A, D, E, and K) (Reddy & Jialal, 2022).

2.2 Introduction

Nutrient requirements for women, particularly micronutrients, increase more than dietary energy requirements during pregnancy and lactation (Marangoni et al., 2016). Macro- and micronutrients in the maternal diet are crucial to supply adequate nutrients to the foetus and baby, ensuring its optimal development and long-term health (Rees, 2019; Tuncalp et al., 2020). However, dietary intake of fruits, vegetables, meat and dairy products is often insufficient during pregnancy and lactation, and this may lead to micronutrient deficiencies (WHO, 2020). Evidence has shown that inadequate maternal micronutrient status has significant potential to adversely affect many developmental processes in the foetus and in breastfed infants, with both immediate and long-term consequences (Cetin et al., 2019). Therefore, adequate maternal nutrition is essential not only to improve the nutritional status of mothers, but also to reduce the risk of poor birth outcomes (Kavle & Landry, 2018).

Strategies to prevent micronutrient deficiencies take place at the individual, population, and target group levels. Thus, in an effort to prevent micronutrient deficiencies in a susceptible/ vulnerable population such as pregnant and lactating women, there are several micronutrient interventions that can be carried out to address and improve their micronutrient status; these include sustainable food-based strategies such as dietary diversification and food fortification, which promote the consumption of foods that are naturally rich in micronutrients or are enriched through fortification (FAO & ILSI, 1997). Besides that, short term strategies like micronutrient supplementation is also an important approach in supporting the control of micronutrient deficiencies (FAO & ILSI, 1997).

Pregnant and lactating women require additional intakes of iron, folic acid, iodine, cobalamin (vitamin B12), vitamin C, vitamin A and other nutrients (NCCFN, 2017). Therefore, a varied and balanced diet, besides intakes of several vitamin and mineral supplements, such as iron, folic acid, and vitamin C will help mothers to optimally achieve their nutritional needs before, during and after pregnancy (WHO, 2016). Diets with diverse sources of foods are most likely to provide sufficient micronutrients required for the mother's health and the development of the foetus (Gernand et al., 2016).

While consuming foods of diverse sources may benefit micronutrient sufficiency, the choice of foods is pertinent to maternal diet quality and its influence on birth outcomes (Chia et al., 2019). Maternal adherence to healthy dietary patterns, which are characterised with high intakes of vegetables, fruits, wholegrains, low-fat dairy, and lean protein foods, are associated with positive pregnancy outcomes such as lower risk of preterm birth and small-for-gestational-age babies (Amati, Hassounah & Swaka, 2019; Chia et al., 2019; Strohmaier et al., 2020). On the other hand, unhealthy dietary patterns with high intakes of refined grains, processed meat, and high intakes of saturated fat or sugar, are associated with unfavourable birth outcomes such as lower birth weight, greater foetal size and adiposity, and lower Psychomotor Development Index (PDI) score at 12 months of age (Chia et al., 2019).

Although a healthy and varied diet remains the preferred means for pregnant and lactating women in meeting their nutritional requirements, some nutritional needs during this period are undeniably challenging to meet with diet alone (Mousa, Naqash & Lim, 2019). As such, supplements may be prescribed and the use of fortified foods should be recommended in supporting women to meet their increased nutritional demands during this period (Gernand et al., 2016). However, it is important to note that not all micronutrients are necessary or recommended as supplements during the pregnancy and lactation stages. The WHO's recommendation on antenatal care for a positive pregnancy experience summarised that micronutrients that are recommended to be supplemented to pregnant women are iron and folic acid, while the use of calcium and vitamin A supplements are context-specific, specifically recommended to population of women with low intake of the mentioned nutrients (Table 2.1, see Appendices).

2.3 Scientific Basis

2.3.1 Vitamins

a. Vitamin A

Vitamin A is a fat-soluble vitamin that is essential for many bodily functions including vision, skin structure, the immune system, and embryo formation (WHO/FAO, 2004). Vitamin A is important for pregnant women and their foetus for the maintenance of maternal night vision and foetal ocular health, besides the development of other organs and foetal skeleton, as well as maintenance of the foetal immune system (Sommer & Vyas, 2012). Maternal and infant concentrations of vitamin A compounds have been associated with neonatal outcomes (Hanson et al,. 2018). Vitamin A intake of pregnant women directly influences the circulating retinol available to be transferred through the maternal-foetal interface and is important for the survival, growth, and establishment of the foetus's long-term health (Gannon, Jones & Metha, 2020).

The recommended vitamin A requirement for pregnant women is 800µg RE/day, while for breastfeeding mothers it is as much as 850µg RE/day (NCCFN, 2017). Mothers are encouraged to consume foods rich in vitamin A such as liver, red meat, nuts and legumes, chicken, eggs, seafood (examples: oysters, crabs, lobster, mussels, sardines, shrimp), whole grains, and fortified grains (NCCFN, 2017). Study has shown that the content of vitamin A in breast milk increases when the intake of vitamin A through the mother's diet increases (Kodentsova & Vrzhesinskaya, 2006). Thus, breastfeeding mothers need to ensure adequate intake of vitamin A from their diet to replace its loss during breastfeeding (Humphrey, West & Sommer, 1992). Premature birth is associated with lower levels of retinol in breast milk (Souza et al., 2015).

Excessive intake of vitamin A can have teratogenic effects. It is vitamin A in the form of retinol (from animal sources) that is associated with such effects and not carotenoids found in plant sources such as carrots (Humphrey, West & Sommer, 1992; NCCFN, 2017). Excessive doses of vitamin A (>10,000IU/day) have been linked to cranial-face problems and congenital heart defects (Humphrey et al., 1992). Therefore, vitamin A supplementation is not required as it can cause toxicity if taken excessively.

b. Vitamin B1

Thiamine (vitamin B1) is an essential water-soluble micronutrient required for energy metabolism. It is an important cofactor for critical enzymes in glucose metabolism (Huang et al., 2018) and nerve impulse conduction. Thiamine, also known as vitamin B1, helps the body metabolise food for energy and plays an important role in maintaining a healthy cardiovascular and nervous system (Wang et al., 2018).

Thiamine is important during pregnancy to reduce the risk of low birth weight, anencephaly, and maternal gestational diabetes (Adams et al., 2022). Thiamine deficiency results in the impairment of the production and secretion of insulin, resulting in a reduction in glucose utilisation (Adams et al., 2022). The RNI Malaysia 2017 recommends 1.4 mg thiamine for pregnancy and 1.5 mg for lactation (NCCFN, 2017). Thiamine exists in a variety of food sources including wheatgerm, sunflower seeds, dhal, peanuts, soybeans, pork, milk, mung beans, lotus seeds, legumes, oats, peas, and fortified grains (NCCFN, 2017).

Thiamin decreases substantially during pregnancy unless supplemented. It was reported that in the United States, half of the women developed thiamin deficiency after birth, unless supplemented (Adams et al., 2022). Therefore, prenatal supplementation of 6mg of thiamine and more may be needed especially for women with intrauterine growth restrictions, pending further research (Adams et al., 2022). The United Nations International Multiple Micronutrient Antenatal Preparation (UNIMMAP), an established multiple micronutrient formulation recommends 1.4 mg of thiamine supplementation during pregnancy (WHO, 2020).

c. Vitamin B2

Riboflavin (vitamin B2) is important to produce thyroid hormones, immune cells, red blood cells, and for improving photoreceptor functioning. It also works in synergy with other vitamins and minerals to reduce anaemia and night blindness (Adams et al., 2022). Riboflavin deficiency may be associated with low birth weight and a higher risk of serious birth defects (loss of limb and heart defect) (Adams et al., 2022).

The RNI Malaysia 2017 recommends 1.4 mg/day for pregnant women and 1.6 mg/day for lactation (NCCFN, 2017). Good sources of riboflavin include dairy products especially milk, chicken liver, egg, leafy vegetables, legumes, kidneys, and mushrooms (NCCFN, 2017). Riboflavin is safe even at high doses, and there is no Tolerable Upper Limit that has been established for this micronutrient (Adams et al., 2022).

In pregnant women, riboflavin supplementation alone may prevent hypertension and severe pre-eclampsia (Adams et al., 2022). The United Nations International Multiple Micronutrient Antenatal Preparation (UNIMMAP) recommends the supplementation of 1.4 mg riboflavin (WHO, 2020).

d. Vitamin B3

Niacin (vitamin B3) is important for energy production and development of the nervous system, digestive system, and skin (Adams et al., 2022). Low niacin is associated with higher risk of birth defects (spina bifida, serious heart defect) (Adams et al., 2022). Inadequate maternal niacin intake is also associated with a higher risk of congenital anomalies (Palawaththa et al., 2022). The RNI Malaysia 2017 recommends 18 mg/day niacin for pregnant women and 17 mg/day niacin for lactation (NCCFN, 2017). Niacin is found naturally in tuna, anchovies, peanuts, liver, pork, wholemeal wheat flour, beef, seeds, mushrooms, and nuts (NCCFN, 2017).

Several local studies have shown that niacin inadequacy is prevalent among pregnant women (Abdul Manaf et al., 2014; Hamid et al., 2019). This is because niacin level decreases substantially during pregnancy unless supplemented (Adams et al., 2022). Thus, prenatal supplements containing approximately 35 mg/day may reduce the risk of spina bifida and heart defects (Adams et al., 2022).

e. Vitamin B6

Pyridoxine (vitamin B6) is a water-soluble vitamin with its metabolically active form pyridoxal 5'-phosphate (PLP) (NCCFN, 2017), which acts as a coenzyme for numerous enzymes involved in amino acid metabolism; haem, nucleic acid, hormone, neurotransmitter, carbohydrate and lipid biosynthesis; trans-sulfuration of homocysteine to cysteine, and release of glucose from stored glycogen (Food and Nutrition Board, 1998). Classic symptoms of vitamin B6 deficiency are microcytic anaemia, seborrheic dermatitis, epileptiform convulsions, depression and confusion, most of which are related to the role of vitamin B6 as a coenzyme in haemoglobin and neurotransmitter biosynthesis (Dror & Allen, 2012).

Vitamin B6 status (indicated by plasma PLP) declines during pregnancy more by increase in blood volume, with the most significant drop during the third trimester when blood volume plateaus (Dror & Allen, 2012). Hence, the Malaysian RNI recommends an increased intake of vitamin B6 during pregnancy at 1.9 mg/day and during lactation at 2.0 mg/day (NCCFN, 2017). Vitamin B is widely available in a variety of foods, particularly in meat, organ meats, fish, poultry, beans, lentils, and bananas (NCCFN, 2017).

In a Cochrane review by Salam et al. (2015), the authors noted that oral pyridoxine supplementation was associated with a statistically significant decrease in the risk of dental decay in pregnant women, suggesting protection against this condition during pregnancy. However, they concluded that there was not enough evidence to detect clinical benefits of vitamin B6 supplementation in pregnancy and/or labour, as the data assessed were not sufficient to detect any statistically significant differences in the risk of eclampsia, pre-eclampsia, breastmilk production or low Apgar scores among women receiving pyridoxine supplementation or control groups. Aside from that, most studies did not include important neonatal outcomes such as orofacial clefts, cardiovascular malformations, neurological development, and preterm birth.

f. Folate (Vitamin B9)

During pregnancy, increased folate intake is required for rapid cell proliferation and tissue growth of the uterus and the placenta, growth of the foetus, as well as for the expansion of maternal blood volume (Rondo & Tomkins, 2000). The increase in folate requirement among breastfeeding mothers is to replace losses during exclusive breastfeeding. Folate is recognised to reduce the risks of spinal cord defects/neural tube defects (NTDs) and low birth weight/small-for-gestational age (SGA) among babies, which are associated with increased risks of neonatal morbidity and mortality (McIntire et al., 1999).

Due to the increasing need and the low bioavailability of folate, which is easily lost during the process of transportation and food preparation, the requirement during pregnancy is as much as 600 µg/day and 500µg/day for mothers who are breastfeeding (NCCFN, 2017). Sources of folate can also be obtained from cereals and cereal products enriched with folate such as breakfast cereals, brown rice, and bread. For example, two scoops of brown rice can provide 45 µmg/100g of folate, three slices of white bread can provide 18 µmg/100g of folate (NCCFN, 2017), one slice of papaya can provide 31 µmg/100g of folate, two ciku (120 g) provide 30 µmg/100g of folate, one orange (124 g) provides 24 µmg/100g (Tee et al., 1997).

Since folate requirements are 5- to 10-fold higher during pregnancy, pregnant women are recommended to take folic acid supplements to ensure an adequate intake. According to a systematic review and meta-analysis, folate supplementation has a positive effect on birth weight whereby it reduces the risk of low birth weight babies (Hodgetts et al., 2015). More importantly, a Cochrane review in 2015 concluded that folic acid supplementation, alone or in combination with vitamins and minerals, prevents NTDs, although it had no clear effect on other birth defects such as cleft palate, cleft lip, and congenital cardiovascular defects (De-Regil et al., 2015). Because of its preventive effect on NTDs, guidelines have recommended that women take folic acid supplementation even during the peri-conceptional period (Toivonen et al., 2018), which was found to abolish folate deficiency and reduced folate insufficiency (Iglesias-Vázquez et al., 2022).

g. Cobalamin (Vitamin B12)

Cobalamin is a water-soluble vitamin that is essential for red blood cell formation, normal brain and nerve functions, and DNA formation. Cobalamin requirements increase during pregnancy and lactation to meet the demands of the mother, the foetus, and the infant. Cobalamin is secreted into the breast milk to meet the needs of the baby (IOM, 1998).

The recommended intakes of cobalamin for pregnant women is 4.5 µg/day and 5.0 µg/day for lactating mothers (NCCFN, 2017). For example, ½ cup of cooked lala (113 g) can supply 112 µg of cobalamin, ½ cup of fortified cereals (113 g) can supply 22.6 g of cobalamin, and 2 pieces of tempeh (88g) can supply 0.08 µg of cobalamin (USDA, 2012; Suzana et al., 2015).

Cobalamin is a vitamin that can only be obtained from animal sources. Hence, women of childbearing age from low-income settings and those with low intake of animal products are at risk of cobalamin deficiency (Rosenthal, Lopez-Pazos & Dowling, 2015). In view of that, vegetarian mothers are encouraged to increase their intake of fermented foods such as tempeh and foods enriched with vitamin B12 (bread and wheat flour) to meet cobalamin requirements (USDA, 2012). Aside from that, cobalamin supplementation during pregnancy and lactation can also prevent maternal depletion and deficiency in infants (Duggan et al,. 2014).

h. Vitamin C

Vitamin C, or also known as ascorbic acid, is an essential water-soluble vitamin that prevents oxidative stress and enhances dietary iron absorption (Rumbold et al., 2015). Vitamin C is needed for the formation of the foetus in pregnant mothers. Poor vitamin C status, especially during late pregnancy, is associated with increased risk of complications, particularly for pregnant mothers with diabetes (Juhl, Lauszus & Lykkesfeldt, 2017).

In Malaysia, the recommended nutrient intake (RNI) of vitamin C for pregnant mothers is 80 mg per day (NCCFN, 2017). This is to meet the needs of the growing foetus during the third trimester and for the prevention of membrane rupture. As for lactating mothers, the RNI is 95 mg per day, mainly due to loss through breast milk (NCCFN, 2017). Many fruits and vegetables are rich in vitamin C including guava, longan, papaya, lime, tangerine/ mandarin, vegetables such as green leafy, tapioca shoots, *cekur manis, kale(kai-lan),* mustard leaves *(sawi).* For example, the consumption of half a guava (150g) is able to provide a total of 228mg of vitamin C (ascorbic acid), while 1 cup of mustard leaves (sawi) (100g) provides 89mg vitamin C (ascorbic acid) (Suzana et al., 2015; Tee et al., 1997).

Routine vitamin C supplementation or in combination with other supplements is not recommended during pregnancy as there is no convincing evidence that supplementation results in additional benefits (WHO, 2016a). This was further confirmed by a Cochrane systematic review of 29 trials involving 24,300 pregnant mothers where no significant difference was found between women supplemented with vitamin C alone or in combination with other supplements compared to placebo or no control in the risks of stillbirth, neonatal death, perinatal death, birth weight, intrauterine growth restriction, preterm birth, preterm and term PROM (pre-labour rupture of membranes), and clinical pre-eclampsia (Rumbold et al. 2015). Nevertheless, the role of vitamin C in relation to PROM requires further investigation (Mousa, Naqash & Lim, 2019).

i. Vitamin D

Vitamins D2 (ergocalciferol) and D3 (Cholecalciferol) are isoforms of vitamin D (calciferol), a fat-soluble vitamin. Vitamin D2 is mostly found in foods and added to foods (fortified), while vitamin D3 is synthesised in the skin, activated by the exposure of bare skin to sunlight. The function of vitamin D is to maintain normal blood levels of calcium and phosphorus, which are important for the normal mineralisation process of our bones. Vitamin D deficiency during pregnancy leads to lower vitamin D concentrations in neonatal cord blood, which subsequently leads to vitamin D deficiency in neonates at birth (Mustapa et al., 2020; Mustapa et al., 2021). Besides, vitamin D deficiency also causes multiple adverse obstetric and child health outcomes, and has been reported to affect foetal growth and development (Mustapa et al., 2020).

The requirements of vitamin D for pregnant and lactating women are the same as adults, which is 15 µg/day (NCCFN, 2017). Dietary vitamin D sources are from natural foods, fortified foods, and supplements. Foods such as salmon, mackerel, sardine, whole egg, beef liver, catfish, beef, chicken, lamb are good sources of vitamin D, while oyster mushroom is a good plant source for vitamin D. Vitamin D is usually fortified in cereals, biscuits, butter, fat spread, yoghurt, milk, and soy milk; it is carried out voluntarily by the manufactures (NCCFN, 2017). The source of sunlight is important to activate the formation of vitamin D in our body. The amount of time to be exposed in sunlight depends on the amount of skin exposed, the time of day, latitude and season, skin pigmentation (darker skin pigments synthesise less vitamin D than lighter pigments), and sunscreen use (WHO, 2016b).

Presently, there is no recommendation of vitamin D supplement for pregnant women. For pregnant women with suspected vitamin D deficiency, vitamin D supplement may be given at the current recommended nutrient intake of 200 IU (5 µg) per day. Like the general population, pregnant women are encouraged to expose themselves to sunlight, which is the most important source of vitamin D. A study by Norliyana et al. (2021) showed that exposure to sunlight for 15 minutes biweekly on face, lower arms, hands and feet significantly increased serum vitamin D.

g. Vitamin E

Vitamin E is an important micronutrient in the human body that maintains various bodily functions. Vitamin E can be in two compounds, tocopherol and tocotrienol. It plays a very important role in maternal health and child development. Vitamin E is an antioxidant that ensures the formation and development of healthy cells in the foetus and protects pregnant women from toxins. Vitamin E enters the foetal circulation from maternal blood during the twelfth week of pregnancy (Aftab Siddiqui et al., 2021). A lack of vitamin E can lead to female infertility, miscarriage, premature delivery, eclampsia, foetal intrauterine growth restriction, and other diseases associated with pregnancy (Wahid, Khan & Feroz, 2014).

The RNI for vitamin E during pregnancy and lactation is 7.5 mg/day (NCCFN, 2017). The richest food sources of vitamin E are soybean oil, palm olein, sunflower oil, sunflower seeds, almond and peanuts. For example, 2 tablespoons (15 g) of palm olein provide 6.5 mg of tocopherol, half cup of peanuts (150 g) provides 6.1 mg tocopherol (Tee et al., 1997).

Although vitamin E has antioxidant properties that may support pregnancy, and its supplementation may help reduce the risk of pregnancy complications involving oxidative stress, there is still a need to evaluate the efficacy and safety of vitamin E supplementation in pregnancy (Rumbold et al. 2015).

2.3.2 Minerals

a. Iron

Iron is very important among pregnant women for blood formation that supports foetal growth. Blood volume increases significantly within the first few weeks of gestation and continues to increase progressively throughout pregnancy. This causes pregnant women to likely have relative anaemia or physiological anaemia that is caused by increment of blood volume and lower concentration of blood (Kassa et al., 2017).

Iron deficiency affects and reduces the formation of blood. The most common type of anaemia is iron deficiency anaemia among women, especially pregnant women, due to inadequate diet and poor prenatal iron and folic acid intakes (Kassa et al., 2017). Several studies have shown that anaemia during pregnancy has several adverse effects such as stillbirth, prematurity, intrauterine foetal death, low birth weight, and perinatal mortality (WHO, 2014).

Iron requirement increases dramatically throughout pregnancy, and is usually unable to be met through daily food intake. Therefore, FAO/WHO (2004) and NCCFN (2017) recommend the consumption of daily iron supplements (about 100 mg/day) to meet such needs. There are two types of iron from foods, which are heme and non-heme iron. Sources of heme iron are flesh foods such as meat (2 match boxes of beef (62 g) can provide 1.4 mg of iron), fish (1 medium cincaru (265 g) can provide 2.8 mg of iron), and poultry (1 medium thigh (175 g) can provide 1.4 mg of iron); while non-heme sources are from plants such as grains (one cup of *kuetiau* (100 g) can provide 3.7 mg of iron), legumes (one cup of chickpea (186 g) can provide 12.8 mg of iron), and green leafy vegetables (one cup of *kangkung* (80 g) can provide 4.0 mg of iron) (EFSA, 2015; Tee et al., 1997). The absorption of heme iron is better than non-heme iron. However, foods containing caffeine, phytate, polyphenols, and calcium can inhibit the absorption of iron in the duodenum and small intestines. On the other hand, ascorbic acid (vitamin C) increases the absorption of iron (EFSA, 2015).

Daily oral iron and folic acid supplementation with 30 mg to 60 mg of elemental iron and 400 µg (0.4 mg) of folic acid is recommended for pregnant women to prevent maternal anaemia, puerperal sepsis, low birth weight, and preterm birth. Oral iron supplementation's most common side effects are gastrointestinal (stomach and bowel) problems like constipation, nausea, vomiting and diarrhoea. If daily iron is not acceptable due to side effects and in population with an anaemia prevalence of less than 20% among pregnant women, the suggested dose is 120 mg elemental iron and 2800 µg (2.8 mg) folic acid provided weekly throughout the pregnancy, beginning as early as possible after conception (WHO, 2016).

b. Iodine

During pregnancy, iodine needs increase by at least 50% of the normal needs of women. This dramatic increase in demand is due to, i) increased production of thyroxine (T4) in pregnant women to maintain normal thyroid function and for the transfer of thyroid hormones to the foetus in the first trimester; ii) iodine transfer to the foetus, especially in the final phase of gestation; iii) increased removal of iodine in the kidneys (Zimmermann, 2012).

Iodine deficiency during pregnancy can cause pregnant women and their foetus to experience hypothyroidism (underactive thyroid), which can damage the neurological development of the foetus (Zimmermann, 2012). Zimmermann (2012) meta-analysis estimated that the population with iodine deficiency disorders experience a decrease of 12-13.5 points in mean IQ (Zimmermann, 2012).

Therefore, the iodine requirement for pregnant women and nursing mothers is 200 µg/day (NCCFN, 2017). Food sources rich in iodine are iodised salt, seafood, milk, eggs and seaweed (Tee at al., 1997). However, consuming foods that contain goitrogens such as tapioca and soybeans, and vegetables such as cabbage, cauliflower, broccoli, and sweet potatoes, can interfere with iodine absorption (Rohner et al., 2014). For example, 1 teaspoon of salt provides 100-200 µg iodine (Malaysian Food Regulations 1985). Half cup of seaweed (15g) provides 220 µg iodine (Tee et al., 2017).

Harding et al. (2017) in a Cochrane review that included 14 randomised and quasi-randomised controlled trials on iodine supplementation concluded that there was insufficient data to advocate for routine iodine supplementation in women before, during or after pregnancy. The evidence assessed were found to be of low or very low quality, with no clear effects on important maternal or child outcomes. The authors also added that almost all of the evidence came from settings with mild or moderate iodine deficiency, thus may not be applicable to settings with severe deficiency.

c. Zinc

Adequate zinc stores in the body are extremely important during periods of accelerated growth, such as during infant growth and development. Zinc deficiency is common in developing countries and low maternal circulating zinc concentrations have previously been associated with pregnancy complications (Wilson et al., 2016). Zinc deficiency during the first and third trimesters of pregnancy can lead to growth and foetal development retardation, which in turn increases the risk of giving birth to an underweight baby (Wang et al., 2015). Besides that, its deficiency is also a key factor in increased susceptibility to infection found in states of nutritional deficiency.

Therefore, zinc requirement during pregnancy increases according to gestational age, which is 5.5 mg/day during the first trimester, 7.0mg/day in the second trimester, and 10.0mg/day in the third trimester. Zinc requirements during lactation are also higher at 9.5mg/day at 0 to 3 months, 8.8 mg/day at 4-6 months, and 7.2 mg/day at 7-12 months of lactation period (NCCFN, 2017). Breastfeeding is crucial in the early neonatal period and adequate sources of zinc can be obtained by the baby from breast milk (Ackland & Michalczyk, 2016).

Thus, to meet the increasing need for zinc, mothers are encouraged to consume food sources high in zinc, namely liver (half cup of ox liver (125 g) can provide 5.3-7.6 mg of zinc), chicken (1 medium thigh (175 g) can provide 3.2-5.3 mg of zinc), beef (3 match box-sized pieces of meat (100 g) can supply between 2.9 to 4.7 mg of zinc), eggs (one whole egg (55 g) can provide 0.6-0.8 mg of zinc), vegetables (one cup of lady's finger (97 g) can provide 1.3 mg zinc), one cup of spinach (56 g) can provide 1.0 mg zinc), as well as milk and dairy products (one cup of milk (250 g) can provide 1.0-7.8 mg zinc (NCCFN, 2017).

Despite the importance of zinc during the gestational period, WHO (2021) does not recommend zinc supplementation as part of routine antenatal care. Research is still needed particularly on how zinc status is impacted by other nutritional supplementation (e.g., iron and/or calcium) that are given as part of routine antenatal, as well as its efficacy, provided either alone or with other nutritional supplements, on maternal and neonatal outcomes. Multiple doses of zinc, iron and/or calcium may need to be evaluated based on the current national standard of care. Research on the effectiveness or the implementation of zinc supplementation is not identified as a priority at this time.

d. Calcium

Calcium is a key nutrient for bone mineralisation. During pregnancy and lactation, calcium is needed to meet the high demands and needs for bone mineralisation and growth of the foetus. To meet such needs, there is an increased intestinal calcium absorption and during lactation, an increase in calcium resorption of the bone (Winter et al., 2020). Hence, it is expected that there are whole-bone changes during pregnancy and lactation.

Some of the negative consequences of low intake of calcium during pregnancy include poor bone mineralisation and density in children during the early phases of adolescence. Low levels of calcium could also lead to low-birth-weight infants and high blood pressure in pregnant mothers. Although pregnancy and lactation-associated osteoporosis is a rare condition, a small number of women may suffer fragility fractures due to the demands caused by changes in calcium metabolism and physiology of the skeletal following pregnancy and lactation (Hardcastle, 2022).

Lactation is associated with temporary bone loss and this is especially during exclusive breastfeeding period (Grizzo et al., 2020), where an increase in volume or duration of feeding is expected (Winter et al., 2020). Hence, dietary insufficiency of calcium during breastfeeding period can contribute to bone changes. Nevertheless, following lactation, full recovery and even improvements in bone mass is expected, although not the same as in adolescents. This was further confirmed by a systematic review and meta-analysis of 32 papers where lactation-associated bone loss was reported to be temporal and tendency of recovery was found depending on the return of menstruation and weaning from breastfeeding (Grizzo et al., 2020).

In Malaysia, the recommended nutrient intake (RNI) for calcium throughout pregnancy and lactation is 1000 mg per day and 1300 mg per day for pregnant and lactating adolescents aged 13 to 19 years (NCCFN, 2017). Dairy products such as milk, yoghurt and cheese are some of the best calcium-rich sources. Other good sources of calcium are some green leafy vegetables, nuts or fortified foods including flour and dairy alternatives such as soy products (Cormick & Belizán, 2019). Consumption of 1 cup of fresh cow's milk (250 ml) provides 265 mg of calcium, ¾ cup of non-flavoured yoghurt provides 287 mg of calcium, 1 slice of tau-kua (extra firm tofu) (64 g) provides 105 mg of calcium, and 1 slice of tempeh (fermented soybean) (44 g) provides 30 mg of calcium (Suzana et al., 2015; Tee et al., 1997).

In general, calcium supplementation during pregnancy is not recommended for mothers with sufficient dietary calcium intake. However, according to the WHO guidelines for calcium supplementation in pregnant women, supplementation of calcium as part of the antenatal care is recommended only for populations where calcium intake is low for the prevention of pre-eclampsia in pregnant mothers. The WHO recommends supplementation of 1.5-2.0 g per day of calcium (divided into three doses and preferably to be taken during mealtimes) during pregnancy for women at high risk of developing hypertension or women with low dietary calcium intake (WHO, 2018). Such measures may help to reduce the risk of pre-eclampsia and preterm birth, especially among pregnant mothers with low-calcium diets, in accordance to a Cochrane review of 27 studies of 18,064 women (Hofmeyr et al., 2018). However, these findings are to be interpreted with caution due to the low evidence for outcomes other than hypertensive disorders (Hofmeyr et al., 2018).

In addition, calcium supplementation is also recommended by WHO to relieve cramps during pregnancy as a small study found that women who were supplemented with calcium were less likely to have leg cramps compared to the non-treatment group (Marie et al., 2021). It was suggested that even among pregnant women who do not have insufficient calcium intake, they are encouraged to consume dairy products for prevention of such condition. As for the lactation period, a systematic review and meta-analysis of five randomised controlled trials involving 567 lactating women found that calcium supplementation did not have additional clinical benefits to the bone mineral density of lactating women (Cai et al., 2020).

e. Copper

Copper is an essential micronutrient. It has been estimated that the amount of this element in an adult human body ranges from 50 to 120 mg (Collins, 2014). The highest concentrations of copper are found in organ meat and shellfish, especially liver and oyster (NCCFN, 2017).

Copper involves in formation and metabolism of bone tissue and participates in oxidation–reduction reactions as a coenzyme, a regulator of iron metabolism and transport, as well as collagen metabolism (Lanocha-Arendarczyk & Kosik-Bogacka, 2019). It also participates in the metabolism of fatty acids, in RNA synthesis, supports the absorption of iron in the gastrointestinal tract, and participates in the synthesis of myelin (Cetin et al., 2009). Besides that, copper also participates in the synthesis of melanin and acts as a component of tyrosinase, involved in the conversion of tyrosine to melanin. During pregnancy, copper requirement increases due to synthesis of pregnancy products and increased foetal needs.

Copper deficiency is rare due to the high availability of this element in foods. However, a deficiency of this micronutrient during pregnancy may lead to oxidative stress, which often results in reduced foetal growth. As copper has an important role in the production of collagen and elastin, an insufficient amount of this element can lead to a reduction in the tensile strength of the foetal membrane, resulting in its interruption and premature birth. Hypocupremia is therefore known to lead to hypochromic anaemia, intrauterine growth restriction (IUGR), spontaneous delivery, and spontaneous abortion. On the contrary, hypercupremia may lead to Wilson's disease and Fenton's reagent, and may contribute to premature birth, low birth weight, and gestational diabetes (Collins & Klevay, 2011).

The RNI for copper during pregnancy is 1000 µg/day and 1300 µg/day during lactation. Copper is found in a wide variety of foods. The greatest source of this element is beef liver (14587 µg/100g), oysters (5707 µg/100g), and cocoa powder (3788 µg/100g). Other good sources include soybean curd, beans, nuts, lentils, and tempeh (NCCFN, 2017).

A meta-analysis by Lewandowska et al. (2019) showed that reduced serum copper levels at the beginning of pregnancy were associated with a higher risk of pregnancy-induced hypertension. Kashanian et al. (2018) studied the effects of copper on pregnancy and they showed that women who had been supplemented with 1000 mg copper daily from the 17th week of pregnancy experienced reduced depression symptoms in the 2nd and 3rd trimesters of pregnancy compared to the control group. On the other hand, elevated copper levels in the first trimester of pregnancy may lead to later pregnancy complications, including spontaneous miscarriages (Hao et al., 2019). Li et al. (2019) concluded that too much copper can increase the incidence of gestational diabetes, which, if untreated, can lead to macrosomia, intrauterine hypotrophy, birth defects, and miscarriages. Therefore, since copper can be found in a wide variety of foods and its deficiency is rare, while an excess of copper is harmful, copper supplementation is deemed unnecessary during pregnancy (Grzeszczak, Kwaitkowski & Kosik, 2020).

f. Selenium

Selenium is a component of several selenoproteins that are required for the activity of glutathione peroxidase, a vital intracellular antioxidant that prevents oxidative cellular damage (Rayman, 2000). It also plays an important role in thyroid function, as most enzymes involved in thyroid hormone synthesis contain selenium (Turanov et al., 2011). Aside from that, selenium is also known to play a crucial role in immune response regulation (Hubalewska, Duntas & Gilis, 2020).

Selenium deficiency is rare, but because the literature has reported that increased oxidative stress during pregnancy is associated with the occurrence of foetal/ neonatal diseases and adverse pregnancy outcomes (Biswas, McLay & Campbell, 2022), and the fact that thyroid diseases (such as hypothyroidism, thyroid autoimmunity, and hyperthyroidism) are common in pregnancy (Hubalewska, Duntas & Gilis, 2020), selenium could therefore be an important micronutrient during the gestational period. Studies have shown that maternal selenium deficiency is linked with a higher risk of preterm delivery and the delivery of low-birth-weight infants (Okunade et al., 2018), as well as exacerbating mild iodine deficiency (Wilson et al., 2018).

Selenium requirement increases during pregnancy to allow for growth of the foetus and increased selenoprotein synthesis and tissue accumulation. The RNI for selenium according to trimesters is 25 µg/ day during the first trimester, 27 µg/day in the second trimester, and 29 µg/day in the third trimester. During lactation, 12 µg of selenium is reportedly secreted into breast milk per day (NHMRC, 2006); thus the recommended requirements for selenium are even higher during breastfeeding at 34 µg/day for 0 to 6 months, and 4 µg/day for 7-12 months (NCCFN, 2017).

To ensure sufficient intake of selenium, pregnant and lactating mothers are recommended to consume foods rich in selenium such as fish, meat (especially organ meats), eggs, milk, and shellfish. In general, the higher the protein content of a food, the more selenium it contains. For plant food sources (grains, cereals, nuts, lentils) especially, selenium content will depend on environmental conditions and agricultural practices, such as the selenium content in soil (NCCFN, 2017).

Although the relationship between selenium status and pregnancy outcomes are conflicting (Okunade et al., 2018), there appear to be areas in which intervention with selenium could potentially help to prevent pregnancy complications such as intrauterine growth restriction, miscarriage, pre-term labour, preeclampsia, gestational diabetes, postpartum thyroiditis, and exacerbation of autoimmune thyroid diseases (AITDs) during the first postpartum year; while in foetus, the prevention of inter alia, decreased psychomotor development, neural tube defect, and deficits in the cognitive, intelligence, and behavioural development of the child (Hubalewska, Duntas & Gilis, 2020). However, a recent systematic review, which only included 22 randomised controlled trials on human subjects, concluded that selenium supplementation on maternal or newborn outcomes is understudied and currently unknown. Thus, there is insufficient information to recommend the safe use of selenium supplementation during pregnancy and the postnatal period (Biswas, McLay & Campbell, 2022).

2.4 Current Status

According to a cross-sectional study by Mirsanjari et al. (2012), intakes of vitamin B1, vitamin B2, niacin, folate, vitamin B12, phosphorous, and zinc were reportedly above the recommended dietary allowances (RDA); while magnesium and potassium were lower than RDA among pregnant mothers (Mirsanjari et al., 2012). In a more recent study, maternal micronutrient intakes including calcium, iron, vitamin D, folic acid, and niacin were reportedly below the national recommendations (Jan Mohamed et al., 2022). Specifically for niacin, inadequacy was found among pregnant mothers in Selangor, whereby 71.2% reportedly did not meet the requirement (Abdul Manaf et al., 2014), with only 11.5% achieving the requirement (Hamid et al., 2019).

Iron deficiency is also common among pregnant women, with approximately one in four third-trimester pregnant women aged 18-40 years old reportedly experiencing this form of micronutrient deficiency (Tan et al., 2020). The recent NHMS 2022 found that the prevalence of anaemia among pregnant mothers is 19.3% (IPH, 2023), which is lower than the previous national prevalence of 29.3% (IPH, 2016). In terms of iodine status, Lim and colleagues reported that pregnant women from the rural divisions in Sabah exhibited iodine deficiency disorder (Lim et al., 2017), while pregnant women in Sarawak showed inadequate iodine status (Lim et al., 2021) despite the mandatory universal salt iodisation programme.

Lee et al. (2017) reported that more than 70% of third-trimester pregnant women in the capital of Malaysia, Kuala Lumpur had vitamin D deficiency [25(OH)D <50 nmol/L]. The more recent Mother and Infant Cohort Study (MICOS) showed that the prevalence of vitamin D deficiency was 42.6% among third-trimester pregnant mothers in Selangor and Kuala Lumpur (Woon et al., 2019). The same study also reported that food sources contributed 80.4% of vitamin D, while dietary supplements contributed 19.6%; fish and fish products were the highest contributors towards vitamin D intake (35.8%) (Woon et al., 2019). According to Lee et al. (2021), more than half of pregnant women (67.7%) had inadequate vitamin D intake of <15 µg/day. Thus, pregnant women are considered as a population at risk of vitamin D deficiency because of poor dietary vitamin D intake and poor sun exposure.

Micronutrient deficiency is also common and widespread among pregnant women and lactating mothers around other South East Asian countries. Suboptimal micronutrient status have been reported among pregnant women in Indonesia, Vietnam and Thailand, notably for iron (Madanijah et al., 2016), zinc (Nguyen et al., 2013) and vitamin D (Aji et al., 2020; Pratumvinit et al., 2015). Studies among lactating mothers have shown that iron, zinc and selenium deficiencies were prevalent among these women in Indonesia (Madanijah et al., 2016), Vietnam (Nakamori et al., 2009) and Thailand (Dumrongwongsiri et al., 2021). Iodine is reportedly low among mothers in the Philippines and Vietnam, while vitamin A deficiency is prevalent in Cambodia, Laos and Myanmar (Chaparro, Oot & Sethuraman, 2014).

In Malaysia, while maternal fruit intake was associated with increased birth weight; low pre-pregnancy BMI, inadequate GWG, intake of confectioneries and condiments, and hypertension were associated with low birth weight (Jan Mohamed et al., 2022). Lactating women in Bogor, Indonesia, reportedly doubled their vegetable intake across all studied economic quintiles as 71% of them indicated a belief that this dietary practice enhances lactation performance (Madanijah et al., 2016).

Different dietary patterns were identified at pre- and during pregnancy among pregnant Malaysian women, reflecting clear changes in eating and drinking habits throughout their course of pregnancy (Yong et al., 2019). There were reportedly some healthy dietary adaptations, but unfortunately not sustained to the end of pregnancy (Yong et al., 2019). These dietary patterns were substantially different from those reported in the Western populations (Yong et al., 2019), highlighting that diet is essentially unique to a specific geographical area and culture.

2.5 Key Recommendations

Key Recommendation 1

Eat a variety of foods to obtain the necessary micronutrients to support pregnancy and lactation.

How to achieve

1. Eat a variety of foods from each of the 5 food groups in the food pyramid according to the serving sizes required to meet your daily vitamin and mineral needs.

2. Practise the #QuarterQuarterHalf healthy plate concept in every main meal and add a nutrient-dense snack such as egg/tuna sandwich and yoghurt with fruits in between meals.

Key Recommendation 2

Eat a variety of foods from plant sources everyday to meet your needs for vitamins A, C and E.

How to achieve

1. Eat at least 3 servings of vegetables and 2 servings of fruits everyday.

2. Eat a variety of different coloured leafy vegetables such as spinach and mustard green and fruit vegetables such as tomato and brinjal during main meals and as snacks everyday.

3. Eat a variety of different coloured fruits such as papaya, guava and orange during main meals or as snacks everyday.

4. Eat a variety of nuts and seeds as snacks everyday.

5. Use vegetable oil (such as palm oil, sunflower oil, soybean oil, olive oil) in food preparation.

Key Recommendation 3

Eat a variety of foods rich in vitamin Bs to meet your daily needs.

How to achieve

1. Eat vitamin B-rich foods, especially whole grain based products and food from animal sources (meat, egg, milk) everyday.

2. Eat food rich in folate such as green leafy vegetables, legumes, and fortified grains everyday.

3. Eat foods rich in vitamin B12 (milk and milk products, eggs, fermented foods such as tempeh) and foods fortified with vitamin B12 (fortified grains, dairy and soy products) everyday, especially for women who are practising a vegetarian or vegan diet.

Key Recommendation 4

Meet your vitamin D requirement everyday through vitamin D-rich foods and sunlight exposure.

How to achieve

1. Eat a variety of foods high in vitamin D such as salmon, mackerel, sardine, whole egg, beef liver, catfish, beef, chicken, lamb and oyster mushroom in your weekly diet.

2. Choose foods fortified with vitamin D such as cereals, butter, yoghurt, milk and soy milk.

3. Go outdoors for recreation or exercise, especially in the morning, for at least 15 minutes everyday, to get sunlight exposure.

Key Recommendation 5

Eat foods rich in iron everyday together with food that enhances iron absorption.

How to achieve

1. Eat dark leafy green vegetables (such as *kangkung*, spinach, *pegaga)*, legumes and nuts rich in non-heme iron everyday.

2. Eat dried fruits (prunes, raisin) as snack in your weekly diet.

3. Eat protein sources from lean red meat, poultry, fish, seafood, and egg everyday as the main source of heme iron.

4. Eat a variety of legumes and nuts in main meals and as a snack everyday.

5. To enhance the absorption of iron from plant sources, take them with fruits high in vitamin C such as guava, *longan,* papaya, lime, tangerine/ mandarin. Avoid consuming tea, coffee, milk, soy milk, chocolate or carbonated drinks during main meals.

6. For women who are practising a vegetarian or vegan diet, choose legumes and nuts as sources of protein and iron in main meals and as snacks everyday.

Key Recommendation 6

Eat foods rich in iodine in daily meals.

How to achieve

1. Eat foods rich in iodine such as fish, seafood, egg, milk and milk products, in main meals and as snacks everyday.

2. Eat seaweed or food that contains seaweed in your diet.

3. Eat cooked tapioca shoots, bamboo shoots and cruciferous vegetables (broccoli, cabbage, cauliflower) to reduce goitrogen.

4. Use iodised salt during food preparation at the end of the cooking to preserve the iodine content.

Key Recommendation 7

Consume adequate amounts of milk and milk products daily to meet your calcium needs.

How to achieve

1. Consume milk and milk products such as cheese and yoghurt everyday.

2. Use milk and milk products as part of ingredients to dishes and beverages.

3. Women who do not consume milk or milk products, eat foods high in calcium such as soy, tofu, spinach, cabbage, kailan, broccoli, dried fruits, and grains. Other dairy alternatives include calcium-fortified soy beverages and yoghurt.

4. Women who are lactose intolerant, choose products low in lactose and lactose-free dairy products to meet calcium needs.

Key Recommendation 8

Consume micronutrient supplement(s) according to the instructions prescribed by healthcare professionals.

How to achieve

1. Consume folic acid supplement as prescribed by healthcare professionals.

2. Consume iron supplement everyday as prescribed by healthcare professionals.

   i. Take iron supplements with fruits such as guava, longan, papaya, lime, tangerine/ mandarin rich in vitamin C to enhance its absorption.

   ii. Consume foods high in calcium or calcium-fortified, such as milk, at least two hours before or after taking iron supplement.

   iii. Avoid taking iron supplements with tea, coffee, milk, soy milk, chocolate or carbonated drinks.

3. Women who are practising a vegetarian or vegan diet, please consult with a healthcare professional on the possible need of vitamin B12 supplementation.

4. Consult a healthcare professional before you consume any other micronutrient supplement(s).

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KM2

Choose variety and micronutrient-dense foods

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Appendices

Table 2.1: Summary list of WHO recommendations on antenatal care (ANC) for a positive pregnancy experience

These recommendations apply to pregnant women and adolescent girls within the context of routine ANC

A. Nutritional interventions		
	Recommendation	**Type of recommendation**
Dietary interventions	**A.1.1:** Counseling about healthy eating and keeping physical active during pregnancy is recommended for pregnant women to stay healthy and to prevent excessive weight gain during pregnancy[a].	Recommended
	A.1.2: In undernourished populations, nutrition education on increasing daily energy and protein intake is recommended for pregnant women to reduce the risk of low-birth-weight neonates.	Context-specific recommendation
	A.1.3: In undernourished populations, balanced energy and protein dietary supplementation is recommended for pregnant women to reduce the risk of stillbirths and small-for-gestational-age neonates.	Context-specific recommendation
	A.1.4: In undernourished populations, high-protein supplementation is not recommended for pregnant women to improve maternal and perinatal outcomes.	Not recommended
Iron and folic acid supplements	**A.2.1:** Daily oral iron and folic acid supplementation with 30 mg to 60 mg of elemental iron[b] and 400 µg (0.4 mg) of folic acid[c] is recommended for pregnant women to prevent maternal anaemia, puerperal sepsis, low birth weight, and preterm birth[d].	Recommended
	A.2.2: Intermittent oral iron and folic acid supplementation with 120 mg of elemental iron[e] and 2800 µg (2.8 mg) of folic acid once weekly is recommended for pregnant women to improve maternal and neonatal outcomes if daily iron is not acceptable due to side-effects, and in populations with an anaemia prevalence among pregnant women of less than 20%[f].	Context-specific recommendation
Calcium supplements	**A.3:** In populations with low dietary calcium intake, daily calcium supplementation (1.5 - 2.0 g oral elemental calcium) is recommended for pregnant women to reduce the risk of pre-eclampsia.[g]	Context-specific recommendation
Vitamin A supplements	**A.4:** Vitamin A supplementation is only recommended for pregnant women in areas where vitamin A deficiency is a severe public health problem[h] to prevent night blindness.[i]	Context-specific recommendation
Zinc supplements	**A.5:** Zinc supplementation for pregnant women is only recommended in the context of rigorous research	Context-specific recommendation (research)

A. Nutritional interventions (cont.)		
	Recommendation	**Type of recommendation**
Multiple micronutrient supplements	**A.6:** Multiple micronutrient supplementation is not recommended for pregnant women to improve maternal and perinatal outcomes.	Not recommended
Vitamn B6 (pyridoxine) supplements	**A.7:** Vitamin B6 (pyridoxine) supplementation is not recommended for pregnant women to improve maternal and perinatal outcomes.	Not recommended
Vitamin E and C supplements	**A.8:** Vitamin E and C supplementation is not recommended for pregnant women to improve maternal and perinatal outcomes.	Not recommended
Vitamin D supplements	**A.9:** Vitamin D supplementation not recommended for pregnant women to improve maternal and perinatal outcomes.	Not recommended
Restricting caffeine intake	**A.10:** For pregnant women with high daily caffeine intake (more than 300 mg per day), lowering daily caffeine intake during pregnancy is recommended to reduce the risk of pregnancy loss and low-birth-weight neonates.	Context-specific recommendation

Source: WHO (2016)

a. A healthy diet contains adequate energy, protein, vitamins and minerals, obtained through the consumption of a variety of foods, including green and orange vegetables, meat, fish, beans, nuts, whole grains and fruits.
b. The equivalent of 60 mg of elemental iron is 300 mg of ferrous sulfate hepahydrate, 180 mg of ferrous fumarate or 500 mg of ferrous gluconate.
c. Folic acid should be commenced as early as possible (ideally before conception) to prevent neural tube defects.
d. This recommendation supersedes the previous recommendations found in the WHO publication *Guideline: daily iron and folic acid supplementation in pregnant women* (2012).
e. The equivalent of 120 mg of elemental iron equals 600 mg of ferrous sulfate heptahydrate, 360 mg of ferrous fumarate or 1000 mg of ferrous gluconate.
f. This recommendation supersedes the previous recommendation in the WHO publication *Guideline: intermittent iron and folic acid supplementation in non-anaemic pregnant women* (2012).
g. This recommendation is consistent with the *WHO recommendations for prevention and treatment of pre-eclampsia and eclampsia* (2011) and supersedes the previous recommendation found in the WHO publication Guideline: calcium supplementation in pregnant women (2013).
h. Vitamin A deficiency is a severe public health problem if $\geq$5% of women in a population have a history of night blindness in their most recent pregnancy in the previous 3–5 years that ended in a live birth, or if $\geq$20% of pregnant women have a serum retinol level <0.70 wmol/L. Determination of vitamin A deficiency as a public health problem involves estimating the prevalence of deficiency in a population by using specific biochemical and clinical indicators of vitamin A status.
i. This recommendation supersedes the previous recommendation found in the WHO publication *Guideline: vitamin A supplementation in pregnant women* (2011).

Key Message 3

Achieve optimal weight for healthy pregnancy

Key Message 3

Achieve optimal weight for healthy pregnancy

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3.1 Terminology

Chronotype
A person's tendency to sleep or be awake at a particular time of day.

Postpartum
A period that begins after childbirth and typically lasts about 6 weeks.

Postpartum weight retention
A weight difference between weight at a specific time after childbirth (e.g. 6 months, 12 months, > 1 year) and weight prior to pregnancy.

Pregnancy
A period in which a foetus develops within the womb or uterus of a woman. A pregnant woman undergoes pregnancy that lasts about 40 weeks, starting from the last menstrual period to delivery.

Pre-pregnancy body mass index (BMI)
A measure of weight (kg) divided by square of height (m) taken at pre-conception or in early pregnancy. Pre-pregnancy body mass index (BMI) is then categorized based on the categories of the World Health Organization (2010) as underweight (< 18.5 kg/m^2); normal weight (18.5–24.9 kg/m^2); overweight (25.0–29.9 kg/m^2); and obese (≥30.0 kg/m^2).

Psychosocial health
Dimensions of health that include mental, emotional, social and spiritual.

Rate of gestational weight gain
The rate is calculated by dividing the difference of weight between the last and first prenatal care visit in the second or third trimester by the corresponding difference of the respective gestational weeks.

Sleep quality
A measurement of how well one's sleeping, whether the sleep is restful and restorative and includes dimensions such as sleep latency, sleep waking, sleep efficiency and wakefulness.

Total gestational weight gain
The amount of weight gain during pregnancy that is calculated as the weight difference between pre-pregnancy weight and weight prior to delivery and is categorized as inadequate, adequate and excessive weight gain.

Trimesters of pregnancy
A pregnancy comprises three semesters or 40 weeks with each semester is between 12-14 weeks.

3.2 Introduction

Globally, overweight and obesity are increasing steadily across all age groups, particularly in low- and middle-income countries (Global Nutrition Report, 2021). Of children under 5 years of age, despite the persistence of stunting (22%) and wasting (6.7%), 5.7% (38.9 million) of young children are overweight. Among adults, 2.2 billion are overweight or obese (40.8% of women and 40.4% of men) of whom 772 million are afflicted by obesity with a higher prevalence reported among women (16.2%) than men (12.3%). There is also an increasing trend of overweight and obesity among women of childbearing age in both developed (WHO, 2022a; Stierman et al., 2021) and developing countries (He et al., 2016; Oddo, Maehara & Rah, 2019; Khanam et al., 2021; Islam et al., 2022; Kumar, Manala & Kundu, 2022; Owobi et al., 2022) that has contributed to its growing public health significance as women of childbearing age are more likely to enter pregnancy with higher BMI.

Pre-pregnancy BMI and gestational weight gain (GWG) are important determinants of short- and long-term maternal and child health outcomes (Goldstein et al., 2017; Farpour-Lambert et al., 2018; Haque et al., 2021). While underweight women entering pregnancy are at increased risk of pre-term births and delivering small-for-gestational age newborns, being overweight or obese pre-conceptionally is a risk factor for gestational diabetes mellitus, pre-eclampsia, caesarean delivery, macrosomia newborns and long-term child obesity. In addition, inappropriate (inadequate or excessive) GWG is associated with adverse pregnancy outcomes that include micronutrient deficiency (e.g. iron), low birthweight, small-for-gestational age, premature birth, gestational diabetes mellitus, hypertensive disorders, postpartum haemorrhage, stillbirth and macrosomia. Although pre-pregnancy BMI and GWG are found to have independent effect on obstetrical and neonatal outcomes, the combination of both (e.g. obese women with excessive gestational weight gain) could further increase the risk of adverse maternal and child health outcomes (Voerman et al., 2019; Sun et al., 2020; Choi et al., 2022).

While the relationship between pre-pregnancy BMI and postpartum weight retention (PPWR) is unclear, GWG is known to be a major determinant of PPWR (Mannan, Doi & Mamun, 2013; Rong et al., 2015; McDowell, Cain & Brumley, 2019; Ha et al., 2019; Makama et al., 2021). In specific, while women with excessive GWG are more likely to have higher PPWR than those with adequate GWG, women with inadequate GWG tend to have lower PPWR. A meta-analysis of observational studies (Rong et al., 2015) reported that women with excessive gestational weight gain had higher risk of PPWR and that shorter or longer PPWR was determined by gestational weight gain rather than pre-pregnancy BMI. PPWR not only contributes to adverse maternal and neonatal outcomes (e.g. gestational diabetes mellitus, pre-eclampsia, caesarean delivery, stillbirth, large-for-gestational age newborn) but also predisposes women to obesity in later life and elevated risk of chronic diseases (e.g. Type 2 Diabetes Mellitus, cardiovascular disease and cancers) as well as intergenerational cycle of obesity for the women and offspring (Marchi et al., 2015; Farpor-Lambert et al., 2018; McKinley et al., 2018).

The interrelationship of pre-pregnancy BMI, GWG and PPWR that contributes to multiple complications in pregnancy and delivery and subsequent poor maternal and child health calls for effective nutrition and lifestyle intervention strategies for women of childbearing age before, during and after pregnancy. The Comprehensive Implementation Plan on Maternal, Infant and Young Child Nutrition (WHO, 2014) outlines five important action plans to be implemented by member countries and international partners as strategies to achieve the six global nutrition targets 2025 of stunting, anaemia, low birthweight, overweight, breastfeeding and wasting. Since then, several recommendations have been published as guidelines to promote optimum maternal health and nutrition which include the 'Recommendations on antenatal care for positive pregnancy experience' (WHO, 2016a); 'Good maternal nutrition – The best start in life' (WHO, 2016b); 'Nutritional interventions update – Multiple micronutrient supplements during pregnancy' (WHO, 2020a); 'Nutritional interventions update – Vitamin D supplements during pregnancy' (WHO, 2020b); and 'Recommendations on maternal and newborn care for a positive postnatal experience' (WHO, 2022b).

3.3 Scientific Basis

3.3.1 Pre-pregnancy weight

Obesity has detrimental impacts on almost all physiological functions of the body and significantly increases morbidity and mortality throughout the life course, including during pregnancy (Fruh, 2017; Chooi, Ding, & Magkos, 2019). Obese or overweight women are generally at higher risk for pregnancy complications. In a cross-sectional study of 3454 singleton pregnant Korean women, the risk of any adverse pregnancy outcomes was 1.5 times higher in overweight women and 2.5 times higher in obese women compared to women with normal pre-pregnancy BMI (Choi et al., 2022). In addition, obesity before pregnancy puts women at significantly higher risk of hypertensive disorders of pregnancy, gestational diabetes mellitus, postpartum depressive symptoms, a caesarean delivery, and birthing a large-for-gestational-age infant (Choi et al., 2022).

Obesity is also associated with infertility in reproductive-age women through complex and multifactorial interaction. Because of alterations in leptin, free fatty acid, and cytokine production from adipose tissue, obese women encounter disturbances of the hypothalamic-pituitary-ovarian (HPO) axis, which can affect oocyte maturation and endometrial epithelium receptivity (Gambineri et al., 2019). Moreover, insulin resistance and hyperinsulinemia in obese women cause hyperandrogenism and hyperestrogenism, which in turn cause anovulation, lower endometrial receptivity and, ultimately contribute to infertility (Gambineri et al., 2019). Obese women have been found to have a threefold higher risk of infertility than non-obese women, and they need a longer time for pregnancy (Silvestris et al., 2018). In addition, women with a higher pre-pregnancy BMI tend to gain excessive weight during pregnancy, which leads to postpartum weight retention. A longitudinal cohort study in China reported that women with higher pre-pregnancy waist circumference and gestational weight gain were at increased risk of retaining more weight at one month postpartum (Silvestris et al., 2018).

When energy intake exceeds energy expenditure, body fat mass increases. Weight gain and, ultimately, obesity are the results of a chronic positive energy balance. In contrast, weight loss typically occurs when energy intake is lower than energy expenditure. Therefore, modifying daily behaviours that have an impact on energy intake and expenditure can help in achieving and maintaining a healthy body weight. The American College of Cardiology/American Heart Association Task Force on Practice Guidelines and The Obesity Society in 2013 reported that any dietary strategies appear to be effective for losing weight if they can induce a chronic negative energy balance (Jensen et al., 2014). However, it is also important to consider a reasonable and realistic method to start and avoid radical changes that are neither healthy nor sustainable for long-term weight loss. Practicing a healthy diet, such as based on the Malaysian Dietary Guidelines 2020 (NCCFN, 2021), can provide the nutrients needed by the body while staying within the daily calorie goal for weight loss. A healthy diet typically consists of plant-based foods, fresh fruits and vegetables, antioxidants, soy, nuts, and sources of omega-3 fatty acids and is low in animal proteins, added/refined sugars, saturated and trans fats, and processed and ultra-processed foods (NCCFN, 2021).

Increasing energy expenditure through increased physical activity, including supervised or unsupervised exercise, occupational activity, household chores, transportation, and leisure-time activities, as well as limiting sedentary activities, is critical for weight loss (Petridou, Siopi & Mougios, 2019). Regular exercise is important for reducing body weight and fat, maintaining lower body weight and fat and improving metabolic fitness in obese people (Petridou, Siopi & Mougios, 2019). Evidence suggests that combined diet and exercise interventions were more effective than diet-only interventions in inducing weight loss at 6 months, with a typical weight loss of 8-11% (Chin, Kahathuduwa & Binks, 2016). Participation in sports and exercise activity before pregnancy may also favourably impact gestational weight gain. A study of 223 healthy Swedish women found that women with a high pre-pregnancy physical activity level gained 0.10 kg/week less weight during the third trimester than women with a medium physical activity level (Löf et al., 2008). Moreover, high-intensity interval training, resistance training, and endurance training are all effective options for reducing weight gain. Therefore, habitual exercise and an active lifestyle before pregnancy are important to contribute to weight loss and avoid excessive gestational weight gain.

A U-shaped curve exists between sleep duration and the risk of obesity. A systematic review and meta-analysis reported that short and long sleep durations increased the risk of obesity by 14% and 15%, respectively (Che et al., 2021). However, sleep deprivation is a more pervasive issue affecting people of all ages in modern societies (Chattu et al., 2018). Several meta-analysis studies have demonstrated that shorter sleep duration is associated with an increased incidence of obesity (Cappuccio et al., 2008; Sperry et al., 2015). Although there is a variety of definitions for short sleep duration, all (<7, <6, <5, <4 hours/night) were linked to an elevated risk of obesity in adults (Itani et al., 2017). Sleep plays a crucial role in energy metabolism. Increased food consumption is a physiological response to insufficient sleep because it provides the energy needed to maintain prolonged wakefulness (Markwald et al., 2013). Moreover, short sleep duration may be associated with obesity because of decreased energy expenditure and changes in the levels of appetite-regulating hormones, such as leptin and ghrelin (Bayon et al., 2014). Hence, ensuring sufficient sleep duration, even before getting pregnant, may be an important strategy in preventing obesity and weight gain.

Poor psychosocial status has also been implicated as a risk factor for being underweight and overweight/obese. A cross-sectional study comprising 2781 university students, found that those with higher ratings on psychosocial health were more likely to report having remained in the normal BMI category (Arnetz et al., 2020). In another study involving 1967 Swedish females aged 18-34 years, underweight women were more likely to have poorer psychological health compared with normal-weight women (Ali & Lindström, 2006). Similarly, overweight and obese women were more likely to have poor health-related behaviors and a lack of internal locus of control (Ali & Lindström, 2006). These findings imply that maintaining good psychosocial status is crucial for attaining a normal BMI.

3.3.2 Gestational weight gain

Inadequate GWG increases the risks of preterm birth and being born small for gestational age (Kapadia et al., 2015; Voerman et al., 2019). Women with normal pre-pregnancy weight but do not gain adequate weight as recommended (Table 3.1) are at risk of several pregnancy complications such as having low birth weight babies, preterm deliveries, and stillbirths (Yan, 2019). On the other hand, excessive GWG during pregnancy has been shown to cause adverse outcomes to both mother and child. Positive significant associations were found between excessive GWG with increased risk of gestational hypertension and diabetes, caesarean delivery, postpartum weight retention and maternal obesity post-delivery (Kapadia et al., 2015; Voerman et al., 2019). Meanwhile, infants born to mothers who experience excessive weight gain during pregnancy had increased risk of macrosomia, large for gestational age (LGA) and childhood obesity (Voerman et al., 2019).

Table 3.1: Weight gain recommendations during pregnancy

Pre-Pregnancy BMI (kg/m²)	Total Weight Gain in (kg) 1st Trimester	Mean and range of GWG in 2nd & 3rd Trimester	Total Weight Gain (kg)
Underweight < 18.5	0.5 – 2 kg	0.5 kg @ 0.44 – 0.58 kg/week	12.5 – 18.0
Normal 18.5 – 24.9		0.4 kg @ 0.35 – 0.50 kg/week	11.5 – 16.0
Overweight 25.0 – 29.9		0.3 kg @ 0.23 – 0.33 kg/week	7.0 – 11.5
Obese ≥ 30.0		0.2 kg @ 0.17 – 0.27 kg/week	5.0 – 9.0

Source: IOM (2009)

A balanced diet that includes a variety of nutrient-dense foods is important to achieve the recommended gestational weight gain. However, consuming too much high-calorie, high-fat, or sugary foods can lead to excessive GWG and increase the risk of gestational complications. Pregnant women adhering to dietary patterns characterized by high sugar and fast foods had higher odds of excessive GWG while pregnant women adhering to a balanced and healthy diet were protected from inadequate GWG and excessive GWG rate during pregnancy (Itani et al., 2020). Unhealthy dietary practices such as meal skipping, energy dense and low intake of fruits and vegetables are also important contributors to excessive GWG (Petrella et al., 2014; Chen, Loy & Chen, 2023).

Cultivating healthy dietary habits is just as important as consuming a healthy diet. With modern lifestyles, there is a shift in eating habits, leading to a rise in night eating. Night eating is described as consuming food mainly in the late evening or night and experiencing a strong urge to eat after waking up at night, resulting in late breakfast (Balieiro, 2022). While night eating is common during pregnancy due to foetal movements and cravings, studies have shown that maternal night eating habits were often characterized by diets that are high in carbohydrate and fat. Consequently, this is associated with increased risk of gestational diabetes and significant increase in weight gain, particularly in the third trimester (Loy et al., 2020; Chen, Loy & Chen, 2023). Eating meals at regular times during the day exhibited greater benefits than having heavier meals later in the day. In addition, a good diet quality with healthy food groups and minimal ultra-processed foods was associated with optimal gestational weight gain and subsequently, positive birth outcomes (Loy et al., 2020; Chen, Loy & Chen, 2023)

Physical activity is one of the modifiable factors for weight management. Weight management during pregnancy is essential to ensure healthy GWG is attained to reduce pregnancy complications. Klankhajhon & Sthien (2022) reported several benefits of physical activity for pregnant women such as reducing the risk of gestational diabetes mellitus (GDM), hypertensive disorders, discomfort during pregnancy and improving musculoskeletal fitness. Pregnant women are recommended to perform a moderate-intensity exercise for at least 20-30 minutes a day on most or all days of the week. However, vigorous, or high-risk activities are not recommended. Sedentary pregnant women were found to gain more weight during pregnancy as compared to active pregnant women among urban women in China (Jiang et al., 2012). In a systematic review of 22 studies including Asia showed that generally, pregnant women have low physical activity levels to achieve positive birth outcomes (Silva-Jose et al., 2022). Several studies among Malaysian pregnant women showed significant association between physical activity with GWG rates whereby those with low physical activity levels were more likely to have excessive GWG rates than active pregnant women (Farhana, Rohana & Alina, 2015; Yong et al., 2016; Woon, Yu & Chin, 2019a).

Besides physical activity, alterations in sleep patterns can also affect pregnancy health. Sleep quality generally deteriorates as pregnancy progresses mainly due to night-time wakefulness in the third trimester. The relationship is linked to the weight gain pattern and the change in sleep quality over the trimesters. Poor sleep quality and short sleep duration have been associated with excessive GWG and higher body fat compared with those having good sleep quality (Hill et al., 2021; Whitaker et al., 2021). Additionally, poor sleep quality has been linked to an increased risk of gestational diabetes and pre-eclampsia, which can affect weight gain during pregnancy. This may be due to hormonal changes that occur during pregnancy, which can disrupt normal sleep patterns and lead to increased appetite and cravings (Hill et al., 2021). Poor sleep quality can also lead to fatigue and decreased physical activity, which can contribute to weight gain. Although inadequate weight gain during pregnancy is multifactorial, poor sleep quality may also be a contributing factor. Pregnant women may have trouble sleeping due to physical discomfort, hormonal changes, or anxiety about the pregnancy. Lack of sleep can lead to fatigue and decreased appetite, which can contribute to inadequate weight gain (Balieiro et al., 2019).

Feelings of anxiety and depression are common during pregnancy due to uncertainties, hormonal changes, and stressful events. Psychosocial factors, such as stress, depression, and social support, have been shown to play a role in GWG. High levels of stress and depression can lead to unhealthy coping mechanisms such as overeating, while social support can positively influence healthy behaviours including behaviours related to weight management during pregnancy. Stress during pregnancy elicits a higher cortisol response leading to increased cravings and overeating, mainly related to emotional eating, leading to low nutrient intake (Braig et al., 2020; Athar et al., 2021). These cravings are often for foods that are unhealthy and energy dense. Moreover, stress can also reduce physical activity and lower metabolism, further promoting weight gain. Left unattended, stress could have severe complications for both mother and foetus.

3.3.3 Postpartum weight retention

Childbirth-related weight gain contributes to an increased risk of obesity in women (Langley-Evans, Pearce & Ellis, 2022). Excessive weight gain during pregnancy and postpartum weight retention (PPWR) are significant contributors to parity-related weight gain (Mazariegos et al., 2020). PPWR is defined as failing to return to pre-pregnancy weight one year after delivery (Kent-Marvick, 2022). However, weight retention could last for many years after the first year postpartum. PPWR can be calculated at 4 - 6 weeks as immediate PPWR, 3 to 6 months as short-term PPWR, and more than 1 year as long-term PPWR. Although the average retained weight at 6 months and 1 year postpartum is 1.6 to 4.1 kg and 0.5 to 1 kg, respectively, PPWR typically ranges from 0.5 to 3 kg within the first year after childbirth. However, it is possible for some women to retain more than 4 kg of weight during this period (Shao et al., 2018; Sha et al., 2019; Makama et al., 2021).

PPWR is associated with an increased risk of adverse outcomes in the following pregnancy including pre-eclampsia, gestational hypertension, gestational diabetes mellitus (GDM), caesarean delivery, and stillbirth with the risk appears to be linearly related to the amount of weight gain (McKinley et al., 2018). Besides, postpartum women who are overweight or obese have lower rates of breastfeeding intention, initiation and duration (Chang et al., 2020). PPWR is also linked to risk of chronic diseases such as type 2 diabetes, hypertension, and cardiovascular diseases. Women who retained more weight at one year postpartum had a higher risk of developing type 2 diabetes up to 15 years later (Gunderson et al., 2018). Mannan, Doi & Mamun (2016) reported that PPWR increased the risk of women developing hypertension in middle age. Additionally, PPWR was associated with higher levels of triglycerides which put women at risk of cardiovascular diseases (O'Reily et al., 2016).

Diet is one of the significant risk factors for PPWR, as poor dietary habits can lead to excessive calorie intake and weight gain during the postpartum period. Women who consumed a high-fat diet during pregnancy had a greater risk of PPWR than those who consumed a low-fat diet (Phelan et al., 2011). Additionally, women with higher intakes of sugar and processed foods during pregnancy were more likely to have higher levels of PPWR (Drehmer et al., 2016). It has been recommended that dietary interventions should be one of the key components of postpartum weight management (Ferguson, Paley & Panetti, 2019).

Physical activity plays an important role in the prevention of PPWR. Exercise interventions during the postpartum period could lead to significant reductions in weight retention, body mass index (BMI) and waist circumference (Davenport et al., 2018). Moreover, it has also been suggested that combining exercise with dietary interventions may have even greater benefits for preventing PPWR. A systematic review reported that targeting diet and physical activity as well as providing health professional support should be the core intervention components for postpartum weight management (Lim et al., 2019). Women should be encouraged to engage in regular physical activity, starting with low-intensity exercises in the early postpartum period and gradually increasing the intensity and duration as they recover.

In addition, breastfeeding has been linked to PPWR. Breastfeeding is expected to contribute to postpartum weight loss because of its increased energy demands (about 500 kcal/day), and possible fat mobilization (Lambrinou, Karaglani & Manios, 2019). Moreover, women who did not breastfeed had higher levels of visceral fat, which is associated with an increased risk of chronic diseases such as diabetes and cardiovascular disease (Giglia et al., 2014). However, breastfeeding is also often observed among women maintaining and gaining weight during postpartum. One possible explanation for this observation is the scepticism surrounding energy restriction during breastfeeding, as women fear it may negatively impact their milk production (Lambrinou, Karaglani & Manios, 2019).

Sleep has been identified as a potential modifiable risk factor for PPWR. Insufficient sleep during postpartum has been reported to be associated with higher levels of PPWR (Leclair et al., 2020). In addition, women who reported poor sleep quality during postpartum had significantly higher body fat at 6 months postpartum (Wolfson et al., 2003). Short sleep duration is reported to be associated with higher energy intake, mainly due to increased consumption of saturated fat, resulting in weight gain and an increased in BMI. The mechanisms underlying the relationship between sleep and PPWR are not fully understood, but it has been suggested that disrupted sleep or short sleep duration may affect hormonal regulation of appetite and caloric intake, increase hormones that stimulate hunger and reduce the saturating hormone leptin metabolism, leading to increased food intake to combat fatigue or stress (Papatriantafyllou et al., 2022).

Psychosocial factors have been found to be associated with PPWR. A systematic review reported that depression and stress are positively associated with PPWR (Bazzazian et al., 2021). Women with higher levels of depressive symptoms were more likely to have greater PPWR due to the adverse effect of depression on eating behaviours and physical activity (Herring et al., 2016). Kieffer et al. (2013) showed that women with high levels of stress were more likely to experience PPWR as stress could increase cortisol levels and consequently weight gain. Furthermore, body dissatisfaction was significantly associated with PPWR of more than 5 kg (Chen, Loy & Chen, 2023). Negative body image and poor self-esteem may influence the motivation of women to adopt healthy lifestyle behaviours.

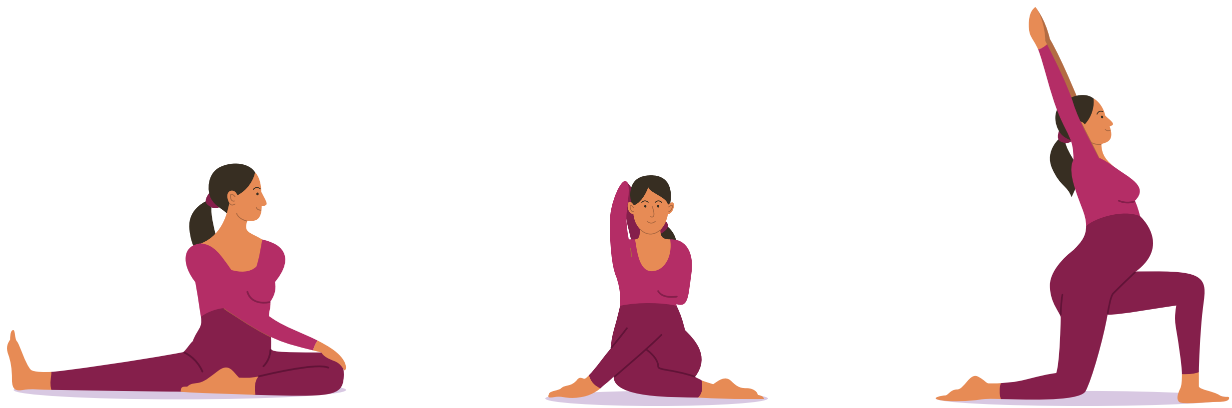
Social support has been recognized as an important factor in postpartum weight management. Women who reported higher levels of social support from their partners, family members, and friends were more likely to achieve and maintain recommended levels of physical activity and healthy eating behaviours during the postpartum period, which could lead to lower levels of PPWR. The social support may provide women with motivation and encouragement for adopting and maintaining healthy behaviours, as well as alleviate stress and fatigue associated with caring for a newborn (Faleschini et al., 2019). Support from health professionals, such as guidance on appropriate diet and physical activity, may also be beneficial in preventing PPWR (Makama et al., 2021). These findings suggest that social support from partners, family members, friends, and healthcare providers can play an important role in preventing PPWR by promoting healthy lifestyle behaviours and reducing stress and fatigue.

3.4 Current Status

The prevalence of adult overweight and obesity in Malaysia has gradually increased over the years with data from the National Health and Morbidity Survey show that the prevalence increased from 29.4% and 15.1% in 2011, 30.0% and 17.7% in 2015 to 30.4% and 19.7% in 2019. In all surveys (2011 – 2019), the prevalence of obesity was higher in women (17.6%, 20.6%, 24.7%) than men (12.7%, 15.0%, 15.3%), prevalence of overweight and obesity in women increased throughout the childbearing years and prevalence of underweight in women decreased from 9.1% (2011) to 5.7% (2019). The Maternal and Child Health Survey 2016 reported that maternal obesity increased from 5.9% in women aged 15-19 years to 27.6% in women aged 40-44 years. Several small-scale community and clinic studies reported that women with pre-pregnancy BMI as underweight, overweight and obesity ranged from 7-17%, 21-29% and 11-23%, respectively (Jan Mohamed et al., 2014; Yong et al., 2016; Kaur et al., 2019; Woon et al., 2019a; Shahrir et al., 2021; Yong et al., 2022b; Yong et al., 2022a). These data support the current global trend that women in childbearing age would enter pregnancy with higher pre-pregnancy BMI, predominantly as overweight or obese.

Yong et al. (2020) reported that 18.6% of pregnant women who attended health clinics in Seremban had excessive GWG and 45.5% had inadequate GWG. In Selangor, 23% of pregnant women had excessive GWG, and a larger proportion (47%) with inadequate GWG (Nurul Farehah et al., 2022). Similarly, in Kelantan, more than half (55.5%) of pregnant women had inadequate GWG (Farhana, Rohana & Alina, 2015). Among pregnant women in Johor, more experienced excessive (53%) than inadequate (29%) rate of GWG (Woon et al., 2019a). Both rural and urban pregnant women were found to have an excessive rate of GWG, although a higher percentage of excessive rate was seen among urban pregnant women (Kaur et al., 2019). When comparing pregnant women's pre-pregnancy BMI with their rate of GWG, underweight women had the highest prevalence of inadequate rate of GWG while the rate of GWG of overweight/obese women was comparable for inadequate (36.7%) and excessive (34.7%) rate of GWG (Ng et al., 2019). These data suggest that inappropriate total and rate of GWG are prevalent among pregnant women in Malaysia and that having less than optimal pre-pregnancy weight could further impact weight gain during pregnancy.

Despite PPWR being an important risk factor for obesity in later life and subsequent chronic diseases, there is lack of recent data on PPWR in Malaysia. Fadzil et al. (2018) examined the weight pattern of 420 mothers after 6 months after delivery and found that 33.8% of the mothers retained 5 kg or more weight during this period. The average postpartum weight retention was determined to be 3.12 ± 4.76 kg. A study involving 83 pregnant women attending a health clinic found that on average, these women retained 2.53 ± 4.01 kg of weight six months after giving birth (Yong et al., 2019). Additionally, about one-third of these women (32.5%) retained a weight of ≥ 4.55 kg during the same period. Zawahid, Muniandy & Shuhaimi (2022) reported that mothers who exclusively breastfed their infants experienced a decrease in weight after giving birth, with an average change of -1.04 ± 4.66 kg at 6 months postpartum. In contrast, mothers who did not exclusively breastfeed showed a slight increase in weight during the same period, with an average change of 0.32 ± 4.25 kg. However, there was no significant difference in postpartum weight and body mass index (BMI) changes between the two groups. As postpartum weight retention seems prevalent among Malaysian women, addressing PPWR is crucial to prevent the long-term health consequences associated with excessive weight retention and to promote maternal and child health in Malaysia.



3.5 Key Recommendations

Key Recommendation 1

Achieve and maintain a healthy body weight before planning a pregnancy.

How to Achieve

Weight Monitoring

1. Weigh yourself weekly in light clothing and without shoes, preferably before breakfast.
   i. Determine body mass index (BMI) based on your weight (kg) and height (m).
   ii. Achieve and maintain your BMI within a normal range (18.5 – 24.9kg/m^2) in a healthy way.
2. Avoid excessive weight gain or loss.
3. Seek health professional advice if you are underweight or overweight/obese.

Diet

1. Eat according to calorie recommendations based on age and physical activity level.
2. Practice a healthy diet based on the Malaysian Food Pyramid 2020.
3. Limit daily intake of high-fat, high-sugar, high-salt, processed and ultra-processed foods and beverages.

Lifestyle

1. Be active and exercise for at least 30 minutes daily.
2. Limit sedentary activities.
3. Aim for at least 7 hours of quality sleep every night.
4. Manage psychosocial health effectively.

Key Recommendation 2

Achieve recommended gestational weight gain.

How to Achieve

Weight Monitoring

1. Know your recommended gestational weight gain based on your pre-pregnancy BMI.
   a) Determine your pre-pregnancy BMI
   b) Identify your gestational weight gain recommendation (Table 3.1)
2. Weigh yourself weekly during gestation to monitor weight gain.
3. Attend your monthly antenatal visits to assess your weight gain pattern.
4. Seek health professional advice for inadequate or excessive gestational weight gain.

Diet

1. Achieve recommended energy intake based on trimester requirements through main meals and healthy snacks.
2. Plan your main meals using the Malaysian Healthy Plate concept. (Refer KM1, Figure 1.2).
3. Eat main meals on time and avoid meal skipping.
4. Limit daily intake of high-fat, high-sugar, high-salt, processed and ultra-processed foods and beverages.
5. If eating at night is required,
   a. Ensure eating 2 hours before bedtime.
   b. Choose healthy foods instead of high-calorie foods

Lifestyle

1. Perform physical activity/ exercise to ensure weight gain is within the recommended range (Table 3.1).

2. Ensure sufficient (6-8 hours) and good quality night sleep by limiting screen time and caffeine intake close to bedtime.

3. Be aware of your psychosocial state to avoid unhealthy eating patterns. Seek health professional help, if necessary.

Key Recommendation 3

Achieve optimal weight within six months to two years after delivery.

How to Achieve

Weight Monitoring

1. For women with underweight pre-pregnancy BMI, aim to achieve normal weight within two years of delivery.

2. For women with normal pre-pregnancy BMI, aim to achieve normal weight within two years of delivery

3. For women with overweight and obese pre-pregnancy BMI, aim to achieve at least pre-pregnancy weight within two years of delivery

4. If optimum weight is not achieved within the first six months, target to reduce 0.5 kg per week until the optimal weight achieved within two years of delivery.

5. Self-monitor your weight regularly at least once a month.

6. Seek health professional advice for effective post-partum weight management, if needed.

Diet

1. Eat according to the recommended calorie requirements.

2. Practice healthy dietary behavior by focusing on diet quality, healthy snacking, and regular mealtime.

3. Limit intake of high-fat, high-sugar, high-salt, processed and ultra-processed foods and beverages.

Lifestyle

1. Practice exclusive breastfeeding for the first six months and continue breastfeeding up to two years or more.

2. Do daily physical activities and gradually increase intensity according to individual ability (Refer KM4).

3. Aim for at least 6 hours of good quality sleep every day.

4. Get psychosocial support from family members and friends. If needed, seek health professional help.

3.6 References

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Key Message 4

Be physically active everyday

Key Message 4

Be physically active everyday

Dr. Vina Tan Phei Sean, Assoc. Prof. Dr. Rohana Abdul Jalil, Dr. Yong Heng Yaw, Ms. Lim Khee Li, Ms. Ng Wei Hui, Ms. Nur Shafawati Mohd Ghazali

4.1 Terminology

Physical activity

Any activity that involves moving the body voluntarily is considered as physical activity. Sub-components of physical activity can be occupational activity, exercise, recreational activity and sedentary activity.

Cardio exercises

An aerobic type of exercise which involves large muscle groups activation, such as those in arms and legs. Cardio exercise increases breathing and heart rate. Examples of cardio exercises include running, swimming, walking, zumba and biking. Over time, regular cardio exercises make heart and blood circulation stronger that contributes to improved fitness.

Cool-down

Cool-down session is conducted at the end of the exercise to gradually lower your heart rate and circulation. Cool-down also help to remove excess metabolites and waste generated during exercise. Cool-down is achieved by reducing the intensity of activity such as slowing down the run or walk if in relation to cardio activity or continuous movement without loads in resistance exercises. Stretching exercises are encouraged as part of cool-down session. A cool-down session may be 3 – 5 min or more, depending on peak intensity achieved and fitness of the individual. Individuals that achieve high peak intensities and/or have lower fitness may need longer cool-down sessions. Cool-down session is complete when you no longer feel breathlessness and heart rate is considered as light intensity level.

Exercise
Planned, structured physical activity that includes components of frequency, intensity, time and type of activity.

Exercise intensity
Exercise intensity is individualized and based on percentages of maximal ability of the person.

Resistance exercises
Resistance exercises or strength training are exercises that strengthens and builds muscles. Scientifically, it is known as an anaerobic type of activity. It can improve daily activities and movement, balance, and coordination. Common resistance exercises are push-ups, lunges, and bicep curls with loads and also includes weightlifting and core-strengthening programs like Pilates.

Flexibility exercises
Flexibility exercises are movements that stretch and lengthen muscles and connective tissues. It can help improve joint range-of-motion and treat muscle stiffness and shortening. Some examples are certain yoga and tai-chi movements as well as muscle-specific stretches.

Fitness
Body's ability to extract and use oxygen efficiently during higher demands of metabolic and energy demands from muscle, skeletal and cardiovascular systems in relation to physical movement. Examples of fitness-related situations are in sports or physical competitions, manual and pregnancy labour, and emergency situations that requires higher cardiovascular demands related to physical movement.

MET
An abbreviation for metabolic equivalent where 1 MET is equivalent to the amount of energy used at rest.

Sedentary activity
Activity that is classified as having low energy requirements, specifically <1.5 METs or in other words, 0.5 more times of energy used compared to resting energy expenditure (~1 MET). Sedentary activities are sitting or lying down but not sleeping. Examples of sedentary activities are sitting to work, lying down to read, sitting during transportation and sitting down for screen time.

Warm-up
Warm-up session is aimed to prepare the body for exercise such as the increased demand of blood circulation. This is in preparation for higher intensities of exercises compared to resting rates, regardless it is cardio, resistance or flexibility exercises. Warm-ups start at a slow-to-medium pace of cardio activities to give your body a chance to get ready for exercise. A warm-up session is around 3 – 5 min or longer depending on the type and intensity of the warm-up activities conducted. Warm-up session is complete when the body and face is warmer, and heart rate is at a higher and stable rate, like during light intensity activity. Examples of warm-up activities include purposeful walking, on-the-spot marching, arm swinging with high-steps and stepping (up-down on single stair step) with arm movements, all at a moderate pace.

Unstable chronic diseases
Chronic diseases imply non-communicable diseases that mostly lifestyle related and is a long-term issue. 'Unstable' implies the presence of signs and symptoms of the chronic diseases. Example of unstable chronic disease conditions are high blood pressure in hypertensive patients on medication; chest pain and discomfort in people treated for heart disease; or highly elevated blood glucose levels in treated type 2 diabetics.

4.2 Introduction

Regular physical activity is strongly encouraged for healthy women with uncomplicated pregnancies before, throughout, and after pregnancy. Physical activity (PA) and exercise during pregnancy are beneficial and safe for both mother and baby and are successfully conducted with modifications to exercise programs to suit the anatomical and physiological changes of pregnancy (ACOG, 2020). Studies have highlighted the benefits of PA and exercise during pregnancy, including a decreased risk of excessive gestational weight gain (GWG), gestational diabetes mellitus (GDM), and preeclampsia (Han, Middleton & Crowther, 2012; Domenjaz, Kayser & Boulvain, 2014; Elena R. Magro-Malossoet al., 2017; Di Biase et al., 2019; Sun & Chien, 2021). Evidence also indicates that PA during pregnancy was associated with lower adverse birth outcomes risk such as rates of preterm births, better sleep quality, reduce the risk of caesarean section procedures and postpartum recovery time, and reduces the length of labour and delivery complications (Elena Rita Magro-Malosso et al., 2017; Davenport, Meah, et al., 2018; ACOG, 2020). In addition, PA during pregnancy also reduces fatigue, stress, anxiety, reduced lower back pain, and improves wellbeing (Kołomańska, Zarawski & Mazur-Bialy, 2019). Thus, not only PA and exercise crucial for physical health and birth but also makes an impact on the psychosocial well-being of pregnant women.

The current PA and exercise recommendations for pregnant women stem from evidence and recommendations for healthy adults, in which healthy pregnant women should undertake regular physical activity throughout pregnancy and postpartum (ACOG, 2020). Pregnant women should aim for at least 150 min of moderate intensity exercises throughout the week for substantial health benefits. Furthermore, pregnant women should also incorporate a variety of cardio, muscle strengthening and flexibility type of activities in their exercise routine. Physically active women who habitually engage in vigorous intensity activity before pregnancy can safely continue these activities during pregnancy and the postpartum period with adjustments as the body changes (ACOG, 2020).

In addition, having a healthy diet and being physically active during pregnancy can help women gain weight within the guidelines, potentially reducing the risk of the adverse outcomes described for the mother including gestational diabetes and increased type 2 diabetes risk (Laredo-Aguilera et al., 2020; Hayes et al., 2021) and pre-eclampsia (Hayes et al., 2021; Ribeiro, Andrade & Nunes, 2022). Beyond any impact on GWG, diet and physical activity during pregnancy have been shown to affect maternal and infant outcomes such as macrosomia (Ferrari & Joisten, 2021) and pre-term births (Muhammad, Pramono & Rahman, 2021). With the current evidence on the benefits of being physically active during pregnancy and postpartum, comprehensive guides and recommendations are highlighted in this section.

4.3 Scientific Basis

Recently, evidence-based recommendations for PA and exercise throughout pregnancy were comprehensively conducted to present the benefits and potential harms in the form of twelve systematic reviews (Mottola et al., 2018) from this extensive and comprehensive work, the 2019 Canadian Guideline on Physical Activity Throughout Pregnancy was published. Further evidence-based tool, Get Active Questionnaire for Pregnancy (GAQ-P) in Figure 4.1a (Davenport et al., 2022) would also assist pregnant women and their physicians to incorporate PA and exercise for better health and pregnancy outcomes.

The GAQ-P is a self-administered questionnaire to screen out complicated pregnancies that have unstable chronic diseases, pregnancy-specific disorders and other disorders that are absolute contraindications to exercise during pregnancy (Davenport et al., 2022). If patients wish to consult physicians or other health professional providers, the Health Care Provider Consultation Form for Prenatal Physical Activity (Figure 4.1b) is available to guide and assist the health professionals.

While pregnancy and birth have been around since the beginning of time, findings on the physiological changes during pregnancy and birth were more recent due to advances in science and safety for pregnant women. Physiological changes during pregnancy from the cardiovascular and hemodynamic components were comparable to a person conducting PA. For instance, in the first trimester, heart rate and cardiac output increases by 3-5% and 5-10%, respectively. In the third trimester, heart rate is 15-20% higher than pre-pregnancy rates (ACOG, 2020). This would be similar to the cardiovascular demands when engaging in light intensity of physical activity or exercise. Thus, the heart is working harder throughout pregnancy and being physically active and exercising helps train and further strengthen the cardiovascular system. Regular PA and exercises will also prepare the cardiovascular system for labour where the cardiac output will increase to 30-50% from pre-pregnancy rates and heart rate goes up by 40-50% during uterine contractions (ACOG, 2020). This increase in heart rate is equivalent to conducting moderate intensity level exercises. Ribeiro, Andrade & Nunes (2022) in their review found moderate-to-high evidence level where physically active pregnant women are more likely to have vaginal deliveries instead of Caesarean or instrumental deliveries. A systematic review (Davenport et al., 2018) of twenty randomised-controlled trials (n=3,819) showed that there was a significant 24% reduction of instrumental deliveries in pregnant women who exercised versus those who did not. It could be possible that when pregnant women are fitter, they are able to endure physiological demands better during labour.

4.3.1 Readiness - GAQ-P

Safety and harm of PA and exercise on pregnant women and baby may be the key misconception and can be advised based on current scientific evidence. While there are certain rare conditions that recommend bedrest during pregnancy, bedrest or sedentary behaviour may be more harmful than engaging in physical activity for pregnant women (Jones et al., 2022). More studies on sedentary behaviour during pregnancy revealed that sitting and lying down for long periods increases inflammation, have higher incidences of macrosomic infants, increased gestational diabetes (GDM) risk and negative effects on foetal development (Acosta-Manzano et al., 2020; Fazzi et al., 2017; Jones et al., 2022; Yong et al., 2020). To minimise adverse outcomes of exercise during pregnancy, guidelines on exercise are provided from comprehensive scientific reviews (ACOG, 2020; Mottola et al., 2018). In Malaysian setting, environmental heat and humidity levels should be noted and avoid exercising when surrounding temperature are high and being directly under the sun. Proper hydration should also be maintained along with avoiding exercise in the heat as constant high body temperatures are detrimental for foetal growth and development. Fatigue will also be higher when exercising in the heat.

4.3.2 Type of exercises

During pregnancy, the three main types of exercises such as cardio, resistance or strength training and flexibility activities are all beneficial to the pregnant women and baby. Each type of exercise will provide specific benefits as outlined in Table 4.1 (see Appendices). More importantly, exercises need to follow the exercise principles like having adequate warm-up & cool-downs, targeted intensity (Table 4.2, see Appendices) or load that is suitable for the individual's ability and fitness (Kenney, Wilmore & Costill, 2015). There are multiple aspects on following exercise principles but mainly it is to 1) prevent injuries and pain related to exercise and 2) effectiveness of exercise to provide the required benefits. For instance, active cool-down sessions may have the ability to return body pH levels to normal faster and have a positive effect on blood lactate levels (Nahon et al., 2021). Other beneficial exercises during pregnancy are like Pilates that helps with pregnancy-related lower back pain through core stabilization movements (Sonmezer, Ozkoslu & Yosmaoglu, 2021).

From an extensive review of studies, Ribeiro et al. (2021) indicated that many pregnant women achieved the health benefits by achieving the current exercise guidelines of performing moderate level exercises for 150 min a week with 20 – 30 min sessions on most days of the week. Resistance and aerobic type of exercises are both conducted and there were no harm or issues towards pregnant women or foetus (Ribeiro, Andrade & Nunes, 2022). However, it is vital that pregnant women know of the signs and symptoms for stopping exercise such as when there is vaginal bleeding, abdominal pain, regular uterine contractions, amniotic fluid leakage, persistent excessive shortness of breath, dizziness, headache, severe chest pain, muscle weakness, calf pain or swelling (ACOG, 2020; Mottola et al., 2018). Exercises in hot weather, high altitudes or exercises that may increase the risk of falling or injury should be avoided (ACOG, 2020; Mottola et al., 2018). More common-sense advice when being physically active during pregnancy is to wear and use proper sports attire. Suitable sports attire for the pregnant woman's size and shape when engaging in exercise would prevent chaffing, provide support for changes in body shapes and comfort to conduct exercises.

4.3.3 Posture during pregnancy

Pregnant women are going through biomechanical and hormonal changes that could influence posture, balance and the walking pattern (Conder, Zamani & Akrami, 2019). These can cause changes in the center of pressure of the pregnancy body, risk of low back pain, reduce gait velocity and increase risk of fall (Bertuit et al., 2017). All of these could seriously impact the quality of life (QOL) of pregnant women. Studies also showed that lordosis angle is significantly more in pregnant women than non-pregnant women (Yoo, Shin & Song, 2015). Increase of lordosis and kyphosis can cause deviation of the center of gravity and hence increase the risk of musculoskeletal pain such as back pain or hip pain (Yoo, Shin & Song, 2015). Hence, maintaining a good lying, sitting and standing posture during pregnancy are crucial to prevent unnecessary stress or strain toward the muscle and ligament.

According to the WHO guideline, it is recommended to exclusively feed infants with breast milk until they reach six months of age. Hence, postnatal mothers are required to spend most of their time in carrying and breastfeeding their infants. Mothers are unconsciously bending their trunk during breastfeeding and this can cause unwanted stress towards the neck and back (Gumasing, Villapando & Siggoat, 2019). Aburub et al. (2022) indicated that improper posture during breastfeeding could cause musculoskeletal pain such as low and upper back, neck, shoulder and hand. This is affecting their daily activities and QOL (Aburub et al., 2022). Hence, breastfeeding posture education is important in nursing mothers to reduce the risk of musculoskeletal pain so that mother and infants could enjoy the breastfeeding journey positively (Aburub et al., 2022).

4.3.4 Sedentary lifestyle

A sedentary lifestyle refers to a state in which pregnant women engage in minimal physical activity, resulting in very low energy expenditure. Sedentary activities are defined as activities that involve energy expenditure between 1.0 and 1.5 metabolic equivalent task units (METs). Examples of sedentary lifestyles include engaging in the following activities while in a seated or lying position (excluding sleeping): working at a desk or computer, browsing social media/using a phone, watching television, reading/writing/drawing/using a computer, and traveling by car, bus, train, or airplane (WHO, 2018).

Previous studies have shown that approximately 55 - 60% of waking hours (7-9 hours per day) in the adult population is spent on sedentary activities (Park et al., 2020). Similar to the general population, pregnant women also spend half or more of their day engaged in sedentary activities (Badon et al., 2018). Overall, a sedentary lifestyle can slow down body metabolism, blood circulation, fat breakdown, as well as affect blood sugar and blood pressure control. Sedentary behaviour also has negative effects on mental well-being, including the risk of depression. Additionally, it can impact the health of pregnant women and their foetus, such as gestational diabetes, high blood pressure, and macrosomic (large) baby at birth. Studies have indicated that pregnant women with high levels of sedentary behaviour and low levels of physical activity have a higher risk of gestational diabetes compared to pregnant women with low levels of both sedentary behaviour and physical activity (Jones et al., 2020). Therefore, reducing sedentary behavior during pregnancy and increasing regular physical activity is crucial to maintaining the health of both mother and baby during pregnancy. The World Health Organization advises pregnant women to engage in regular physical activity throughout pregnancy and limit the amount of inactive or sitting time, replacing it with either low, moderate, or high-intensity physical activities (WHO, 2018).

4.3.5 Sleeping

Implementing simple lifestyle modifications can potentially enhance sleep quality among pregnant women (Gooley, Mohapatra & Twan, 2018). Pregnant women are advised to limit fluid intake before bedtime to minimize the need to wake up during the night to urinate. Taking a pre-sleep bath has been linked to improved sleep for pregnant women. Regular physical activity during the day, such as walking, can facilitate easier sleep onset at night, promote better blood circulation, and reduce the occurrence of leg cramps during pregnancy. Wearing socks while sleeping is beneficial for maintaining foot warmth, optimizing toe circulation, and preventing leg cramps. The World Health Organization recommends pregnant women to avoid large fatty meals and alcohol, quit smoking and elevate the head of the bed while sleeping to prevent and relieve heartburn symptoms. Incorporating these adjustments into daily routine can enhance pregnant women's sleep quality, ensuring a more restful and uninterrupted night's sleep.

4.4 Current Status

From evidence-based studies, proper and regular physical activity during pregnancy is beneficial and safe for both women and foetuses (Ribeiro, Andrade & Nunes, 2022). For example, a higher level of physical activity throughout pregnancy was associated with a reduced risk of emergency Caesarean section and lowered the excessive gestational weight gain (Ribeiro, Andrade & Nunes, 2022). Yet only a minority of pregnant women achieved the recommended level of physical activity, and that higher physical activity and reduced sedentary time were associated with improved health outcomes (Meander et al., 2021).

Several Malaysian-based studies on maternity and physical activity provide insights on the local current scenario. Physical inactivity is one of the contributing factors to maternal obesity. Data from the Malaysian National Obstetric Registry revealed that 28.6% and 22.6% of pregnant women from Klang Valley were overweight and obese, respectively (Nurul-Farehah & Rohana, 2020). In a prospective study based in Seremban, Yong and colleagues (2020) reported that highly sedentary pregnant women with excessive gestational weight gain had a significantly greater risk of having gestational diabetes mellitus (Yong et al., 2020).

Predictors of physical inactivity among antenatal women include pre-pregnancy physical activity status, socioeconomic status, smoking, unhealthy eating and other health related factors (Mohamad Yusof et al., 2020). In the urban city of Kuala Lumpur, prevalence of physical inactivity among pregnant women in their first trimester was 38.3% (Nor, Idris & Isa, 2022). The highest energy expenditure was seen in the household category, followed by the occupational category, whereas the sports/exercise category had the lowest energy expenditure, most likely due to these respondents spending their time isolating at home. Unfortunately, there were little to no participation observed in the vigorous intensity category (Nor, Idris & Isa, 2022). According to Nurul-Farehah and colleagues (2022), with regards to physical activity (PA), light intensity PA had the highest volume, while household and caregiving activities were the most prevalent types of PA performed during pregnancy in Selangor. However, there was no vigorous activity was reported (Nurul-Farehah et al., 2022).

Barriers to being physically active during pregnancy and postpartum are not uncommon. A study among 168 pregnant women in Malaysia showed that barriers towards conducting PA includes complaints of leg cramps, backaches, fatigue and epigastric discomfort in pregnant women (Aslam et al., 2021). Other barriers include the lack of knowledge about physical activity and exercise prescriptions among doctors and nurses to motivate the patients to exercise and be physically active (Matthews et al., 2017). Hence, programs such as Health Professional Education Program (HPEP) for doctors and nurses should incorporate formal education on physical activity and exercise to assist pregnant women engage in physical activity. In addition, pregnant women may not know of the recommended guidelines for physical activity during pregnancy. Factors such as health support systems, informational support and the benefits associated with physical activity may play an important role to influence women's physical activity behaviour. Fears and myths of physical activity being harmful to foetus, causes injuries and physical discomforts to pregnant women and family commitments needs to be addressed and managed, hence the need for improved physical activity and health education, professional and social support are required during pregnancy (Shum, Ang & Shorey, 2022). Currently, there is lack of nationwide data regarding physical activity levels in pregnant Malaysian women. However, the prevalence of physical inactivity is expected to be much higher in pregnancy compared with non-pregnant women because women generally tend to reduce their physical activity once they become pregnant and as the pregnancy progresses (Garland, 2017; Santos et al., 2016; Shum, Ang & Shorey, 2022). To achieve optimum health before, during and after pregnancy, guidelines for Malaysian pregnant women should be developed for the future interventions in both clinical and community setting.

In general, physical inactivity among Malaysian women is remains prevalence as in other higher income countries (Nor, Idris & Isa, 2022). Although much effort has been made in term of Malaysia's capacity for physical activity promotion such as physical activity surveillance, physical activity plan and policies including physical activity research (Khoo et al., 2020), limited data are available on pregnant Malaysian women's physical activity levels (Nor, Idris & Isa, 2022). Health seeking behaviours of pregnant women are usually prominent in antenatal care services (Gopalakrishnan, Eashwar & Muthulakshmi, 2019). More needs to be done to support and advise on physical activity by health care professionals positioned at these health facilities. A multidisciplinary approach needs to be implemented in promoting physical activity among pregnant women that involves medical officers, nurses, physiotherapists, exercise professionals (university graduates in the field of exercise science) and nutritionists.

4.5 Key Recommendations

Key Recommendation 1

Be ready for exercise during pregnancy and postpartum.

How to achieve:

1. Identify conditions that may hinder physical activity during pregnancy like the presence of unstable chronic diseases, i.e. mild-to-severe cardiovascular and respiratory diseases, uncontrolled diabetes, epilepsy, thyroid dysfunction, and anemia. Repeat assessment if health condition changes.

2. Be aware of certain pregnancy status that may be contraindicated for exercise like previous miscarriages, multiple pregnancies after 28 weeks, incompetent cervix and abnormal bleeding.

3. Refer to healthcare providers that may help determine readiness for exercise during pregnancy using the Get Active Questionnaire for Pregnancy (GAQ-P) and Health Care Provider Consultation Form for Prenatal Physical Activity in Figure 4.1.

Key Recommendation 2

Be physically active daily and conduct cardio, resistance and flexibility exercises regularly.

How to Achieve

1. Achieve physical activity goals in two domains –
   i) Daily movement like walking at work or as transport, shopping, and using stairs and
   ii) Planned regular exercises like brisk walking, strength training, yoga and swimming.

2. Use the exercise guidelines on cardio, resistance and flexibility exercises in Table 4.1 (refer Appendices) that are suited for most pregnant women including those with GDM, underweight/ overweight/ obese, and excessive gestational weight gain (EGWG) as weight management strategies.

3. Exercise base on own comfort and ability to achieve targeted exercise intensity and duration.

4. Conduct exercises that are enjoyable to you and plan to do it solo or with family and friends.

5. Use the 10-point Borg scale (Table 4.2, see Appendices) to aim for a 3 to 5 level in each exercise session based on own perceived intensity.

Key Recommendation 3

Maintain good posture during pregnancy and while breastfeeding.

1. Stand tall with open the chest, relax the shoulders, straighten the back, keep head up and look forward. Ensure weight is equal between both feet and avoid standing in the same position for a long time.

2. Sit in a chair with the spine and thighs fully supported, ensure both feet are on the floor and if necessary, use a pillow to support the lower back.

3. Lie on the side with a pillow between both knees and thighs, and lie more on the side after 24 weeks and more of pregnancy. While getting up from lying, advice to roll onto the side, drop the legs off the side of the bed and use the arms to push up into a sitting position. Sit upright with both feet on the floor before coming up into standing.

4. While breastfeed, sit up straight, lean back in the chair and place a pillow to provide back support. The baby's head rests on the mother's arm and support the baby's back with the other arm.

5. While carrying a baby, think about the comfort of mother and child, and make sure the mother's back is straight while carrying the baby. If necessary, use a baby carrier.

6. Consult physiotherapist during antenatal classes to help reduce the risk of strain and discomfort in shoulders, back and pelvic joints during pregnancy.

Key Recommendation 4

Maintain good posture for a healthy pregnancy.

How to achieve

(Figures of different postures mentioned here are in Appendices).

1. Consult physiotherapist during antenatal classes to help reduce the risk of strain and discomfort in shoulders, back and pelvic joints during pregnancy.
2. Stand with proper posture that focuses on even weight distribution on both feet and no protruded back.
3. Sit upright with back supported using a small pillow. Both feet planted on the ground or small stool.
4. Lie on the side with pillow support for head, arms, legs to ensure neck, shoulders and hips are aligned for a straight spine. If lying supine, pillow under knees may help support weight distribution.
5. Maintain good posture during breastfeeding without hunching and baby supported at correct height for feeding, regardless of feeding while seated or lying on the side.

Key Recommendation 5

Limit sedentary behaviour.

How to Achieve

1. Incorporate more physical movement to daily routine i.e. getting up to turn electric switches on and off on your own (Table 4.3, see Appendices).
2. Insert 1 to 3 min movement breaks for every 30 min of sitting or lying down i.e. screentime.
3. Limit total recreational screentime to 2 hour/day by setting reminders on electronic devices i.e. social media, online shopping, YouTube, television and movies.
4. Do these activities whenever possible:
    i. Use stairs often.
    ii. Walk instead of driving to nearby destination.
    iii. Do active hobbies such as cooking, baking and having nature walks.

Key Recommendation 6

Achieve quality sleep during pregnancy.

How to Achieve

1. Consume small-moderate sized meals that is easy to digest food about two hours before sleep.
2. Avoid caffeinated beverages and alcohol especially in the evenings and before sleep.
3. Drink less water at night.
4. Bathing before bed.
5. Exercise regularly and be physically active during the day.
6. Practice a good sleep routine by putting away gadgets and mobile phones 30 minutes before going to sleep.
7. Make sure your bedroom is tidy and comfortable, preferably dark, quiet and with good temperature.
8. If necessary, wear socks.
9. Prop upper body higher with pillows when experiencing heartburn.
10. Sleep on your side instead of staying in a supine position for prolonged periods of time.

4.5.1 Additional Information

4.5.1.1 Pelvic floor muscle exercises (PFME) or Kegel exercises

Systematic reviews on the benefits of pelvic floor muscle exercises or Kegel exercises showed that PFME could benefit pregnant women in many ways, however, the studies on PFME were of low to moderate quality and more higher quality research is required in this area (Davenport & Nagpal et al., 2018; Sobhgol et al., 2019). Highlights from the significant findings from the systematic reviews are as listed:

- PFME reduced odds of having prenatal urinary incontinence (UI) by 50% in pregnant women, with or without cardio exercise
- PFME reduced odds of having postnatal UI by 37% in postpartum women, with or without cardio exercise
- PFME reduced second stage of labour on average by 20.9 min for all and by 21.0 min in primiparous women
- PFME reduced severity of perineal trauma (third- and fourth-degree perineal tear)

PFME can be done daily by squeezing the muscles that would be used to stop urinary mid-stream (Romeikienė & Bartkevičienė, 2021). PFME can be conducted by squeezing the muscles for 6 second, relax for 6 second and repeat the sequence 8 times for a total of three sets with 2-minutes rest between sets (El-Shamy & El-Fatah, 2019). Breathe as normal when holding the squeeze or contraction. When pelvic floor muscles get stronger, the exercises can be increased in repetitions or duration of holding the contraction/ squeeze. This will increase the strength and endurance of the pelvic floor muscles. PFME can be done in any position like lying down, sitting or standing and it is not noticeable to others.

4.5.1.2 Postnatal exercise

Postpartum period in Malaysia is mostly dictated by cultural norms and taboos especially for the first 30 days after delivery. However, ACOG recognizes this time as an opportune period for health and medical professionals to recommend and instil healthy lifestyle practices for women (ACOG, 2020). Postpartum exercise was noted to start as early as a few days post-delivery, but this varies on the nature of delivery (vaginal, caesarean) and the related medical or health complications (ACOG, 2020). There is no hard and fast rule on when to start exercise postpartum. In Asian-based studies, postpartum exercise interventions started 72 hours post-delivery (Ashrafinia et al., 2014), one month or six weeks post-delivery (Heh et al., 2008; Osman et al., 2016) and as late as 6 month to a year post-delivery (Tripette et al., 2014; Youngwanichsetha, Phumdoung & Ingkathawornwong, 2013). However, PFME were recommended to start during the immediate postpartum period.

Benefits of exercise (that were not PFME) during postpartum includes postpartum weight loss, decreased risk of postnatal depression, less experience of fatigue, improved fitness, better glycemic control and sleep quality (Mullins, Sharma & McGregor, 2021). More importantly, postpartum exercise did not affect lactation or infant growth (Mullins, Sharma & McGregor, 2021). However, there were limited studies in the systematic review to strongly link those benefits to postpartum exercises and more robust research are required in this area.

Exercise prescription for postpartum exercises can be based on the recommendations in KR2, Table 4.1 (see Appendices), with a reassessment of exercise intensity levels and not based on the exercise intensities (e.g., speed, distance and strength) prior to delivery. During the postpartum period, the body is still in recovery mode and exercise intensity, duration and ability are all very individualised as well. For women with Caesarean sections, be as active as you can without aggravating the wound and healing process. Exercise have been known to expedite wound healing in older adults (Emery et al., 2005) and molecular mechanisms from exercise on wound healing are being better understood (Chen et al., 2022). Thus, postpartum exercise may provide numerous benefits when it can be conducted without pain or exacerbating prior wounds or injuries.

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Appendices

Figure 4.1a: Get Active for Physical Activity Questionnaire during Pregnancy (GAQ-P)

GET ACTIVE QUESTIONNAIRE FOR PREGNANCY

CSEP | SCPE

NAME (+ NAME OF PARENT/GUARDIAN IF APPLICABLE) [PLEASE PRINT]:			
TODAY'S DATE (DD/MM/YYYY):	YOUR DUE DATE (DD/MM/YYYY):	NO. OF WEEKS PREGNANT:	AGE:

Physical activity during pregnancy has many health benefits and is generally not risky for you and your baby. But for some conditions, physical activity is not recommended. This questionnaire is to help decide whether you should speak to your Obstetric Health Care Provider (e.g., your physician or midwife) before you begin or continue to be physically active.

Please answer YES or NO to each question to the best of your ability. **If your health changes as your pregnancy progresses you should fill in this questionnaire again.**

1.	In this pregnancy, do you have:		
	a. Mild, moderate or severe respiratory or cardiovascular diseases (e.g., chronic bronchitis)?	Y	N
	b. Epilepsy that is not stable?	Y	N
	c. Type 1 diabetes that is not stable or your blood sugar is outside of target ranges?	Y	N
	d. Thyroid disease that is not stable or your thyroid function is outside of target ranges?	Y	N
	e. An eating disorder(s) or malnutrition?	Y	N
	f. Twins (28 weeks pregnant or later)? Or are you expecting triplets or higher multiple births?	Y	N
	g. Low red blood cell number (anemia) with high levels of fatigue and/or light-headedness?	Y	N
	h. High blood pressure (preeclampsia, gestational hypertension, or chronic hypertension that is not stable)?	Y	N
	i. A baby that is growing slowly (intrauterine growth restriction)?	Y	N
	j. Unexplained bleeding, ruptured membranes or labour before 37 weeks?	Y	N
	k. A placenta that is partially or completely covering the cervix (placenta previa)?	Y	N
	l. Weak cervical tissue (incompetent cervix)?	Y	N
	m. A stitch or tape to reinforce your cervix (cerclage)?	Y	N
2.	In previous pregnancies, have you had:		
	a. Recurrent miscarriages (loss of your baby before 20 weeks gestation two or more times)?	Y	N
	b. Early delivery (before 37 weeks gestation)?	Y	N
3.	Do you have any other medical condition that may affect your ability to be physically active during pregnancy? What is the condition? Specify:	Y	N
4.	Is there any other reason you are concerned about physical activity during pregnancy?		

Go to Page 2 *Describe Your Physical Activity Level*

Page 1

KM4

Be physically active everyday

Describe Your Physical Activity Level

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During a typical week, what types of physical activities do you take part in (e.g., swimming, walking, resistance training, yoga)?

____

____

____

During the same week, please describe ON AVERAGE how often and for how long you engage in physical activity of a light, moderate or vigorous intensity. See definitions for intensity below the box.

ON AVERAGE	FREQUENCY (times per week)	INTENSITY (see below for definitions)	DURATION (minutes per session)
How physically active were you in the **six months before pregnancy**?	☐ 0 ☐ 3-4 ☐ 1-2 ☐ 5-7	☐ light ☐ moderate ☐ vigorous	☐ <20 ☐ 31-60 ☐ 20-30 ☐ >60
How physically active have you been **during this pregnancy**?	☐ 0 ☐ 3-4 ☐ 1-2 ☐ 5-7	☐ light ☐ moderate ☐ vigorous	☐ <20 ☐ 31-60 ☐ 20-30 ☐ >60
What are your physical activity goals for the **rest of your pregnancy**?	☐ 0 ☐ 3-4 ☐ 1-2 ☐ 5-7	☐ light ☐ moderate ☐ vigorous	☐ <20 ☐ 31-60 ☐ 20-30 ☐ >60

Light intensity physical activity: You are moving, but you do not sweat or breathe hard, such as walking to get the mail or light gardening.

Moderate intensity physical activity: Your heart rate goes up and you may sweat or breathe hard. You can talk, but could not sing. Examples include brisk walking.

Vigorous intensity physical activity: Your heart rate goes up substantially, you feel hot and sweaty, and you cannot say more than a few words without pausing to breathe. Examples include fast stationary cycling and running.

General Advice for Being Physically Active During Pregnancy

Follow the advice in the 2019 Canadian Guidelines for Physical Activity throughout Pregnancy: **csepguidelines.ca/pregnancy**

It recommends that pregnant women get at least 150 minutes of moderate-intensity physical activity (resistance training, brisk walking, swimming, gardening), spread over three or more days of the week. **If you are planning to take part in vigorous-intensity physical activity, or be physically active at elevations above 2500 m (8200 feet), then consult with your health care provider.** If you have any questions about physical activity during pregnancy, consult a Qualified Exercise Professional or your health care provider beforehand. This can help ensure that your physical activity is safe and suitable for you.

Declaration

To the best of my knowledge, all of the information I have supplied on this questionnaire is correct. **If my health changes, I will complete this questionnaire again.**

☐ **I answered NO to all questions on Page 1.**
Sign and date the declaration below.
Physical activity is recommended.

I answered YES to one or more questions on Page 1 and I will speak with my health care provider before beginning or continuing physical activity.
*The Health Care Provider Consultation Form for Prenatal Physical Activity can be used to start the conversation (**www.csep.ca/getactivequestionnaire-pregnancy**).*

☐ **I have spoken with my health care provider who has recommended that I take part in physical activity during my pregnancy.**
Sign and date the declaration below.

NAME (+ NAME OF PARENT/GUARDIAN IF APPLICABLE) [PLEASE PRINT]:		SIGNATURE (OR SIGNATURE OF PARENT/GUARDIAN IF APPLICABLE):
TODAY'S DATE (DD/MM/YYYY):	TELEPHONE (OPTIONAL):	EMAIL (OPTIONAL):

Figure 4.1b: Health Care Provider Consultation Form for Prenatal Physical Activity

HEALTH CARE PROVIDER CONSULTATION FORM FOR PRENATAL PHYSICAL ACTIVITY

CSEP | SCPE

PATIENT NAME:	DUE DATE (DD/MM/YYYY):	TODAY'S DATE (DD/MM/YYYY):

Your patient wishes to begin or continue to be physically active during pregnancy. Your patient answered "Yes" to one or more questions on the Get Active Questionnaire for Pregnancy and has been asked to seek your advice (**www.csep.ca/getactivequestionnaire-pregnancy**).

Physical activity is safe for **most** pregnant individuals and has many health benefits. However, a **small number of patients** may need a thorough evaluation before taking part in physical activity during pregnancy.

The Society of Obstetricians and Gynaecologists of Canada/Canadian Society for Exercise Physiology *2019 Canadian Guideline for Physical Activity throughout Pregnancy* recommends that pregnant women get at least 150 minutes of moderate intensity physical activity each week (see next page or **csepguidelines.ca/pregnancy**). But there are contraindications to this goal for some conditions (see right).

Specific concern from your patient and/or from a Qualified Exercise Professional:

To ensure that your patient proceeds in the safest way possible, they were advised to consult with you about becoming or continuing to be physically active during pregnancy. Please discuss potential concerns you may have about physical activity with your patient and indicate in the box below any modifications you might recommend:

- ☐ Unrestricted physical activity based on the *SOGC/CSEP 2019 Canadian Guidelines for Physical Activity throughout Pregnancy*.
- ☐ Progressive physical activity
  - ☐ Recommend avoiding:
  - ☐ Recommend including:
- ☐ Recommend supervision by a Qualified Exercise Professional, if possible.
- ☐ Refer to a physiotherapist for pain, impairment and/or a pelvic floor assessment.
- ☐ Other comments:

Absolute contraindications

Pregnant women with these conditions should continue activities of daily living, but not take part in moderate or vigorous physical activity:

- ☐ ruptured membranes,
- ☐ premature labour,
- ☐ unexplained persistent vaginal bleeding,
- ☐ placenta previa after 28 weeks gestation,
- ☐ preeclampsia,
- ☐ incompetent cervix,
- ☐ intrauterine growth restriction,
- ☐ high-order multiple pregnancy (e.g. triplets),
- ☐ uncontrolled Type I diabetes,
- ☐ uncontrolled hypertension,
- ☐ uncontrolled thyroid disease,
- ☐ other serious cardiovascular, respiratory or systemic disorder.

Relative contraindications

Pregnant women with these conditions should discuss advantages and disadvantages of physical activity with you. They should continue physical activity, but modify exercises to reduce intensity and/or duration.

- ☐ recurrent pregnancy loss,
- ☐ gestational hypertension,
- ☐ a history of spontaneous preterm birth,
- ☐ mild/moderate cardiovascular or respiratory disease,
- ☐ symptomatic anemia,
- ☐ malnutrition,
- ☐ eating disorder,
- ☐ twin pregnancy after the 28th week,
- ☐ other significant medical conditions.

CSEP | SCPE

SOGC/CSEP 2019 CANADIAN GUIDELINE FOR PHYSICAL ACTIVITY THROUGHOUT PREGNANCY

The evidence-based guideline outlines the right amount of physical activity women should get throughout pregnancy to promote maternal, fetal, and neonatal health.

Research shows the health benefits and safety of being active throughout pregnancy for both mother and baby. Physical activity is now seen as a critical part of a healthy pregnancy. Following the guideline can reduce the risk of pregnancy-related illnesses such as depression, by at least 25%, and of developing gestational diabetes, high blood pressure and preeclampsia by 40%.

Pregnant women should get at least 150 minutes of moderate-intensity physical activity each week over at least three days per week. But even if they do not meet that goal, they are encouraged to be active in a variety of ways every day. Please visit **csepguidelines.ca/pregnancy** for more information. The guideline makes six recommendations:

1

All women without contraindication should be physically active throughout pregnancy. Specific subgroups were examined:

- Women who were previously inactive.
- Women diagnosed with gestational diabetes mellitus.
- Women categorized as overweight or obese (pre-pregnancy body mass index ≥25kg/m^2).

2

Pregnant women should accumulate at least 150 minutes of moderate-intensity physical activity each week to achieve clinically meaningful health benefits and reductions in pregnancy complications.

3

Physical activity should be accumulated over a minimum of three days per week; however, being active every day is encouraged.

4

Pregnant women should incorporate a variety of aerobic and resistance training activities to achieve greater benefits. Adding yoga and/or gentle stretching may also be beneficial.

5

Pelvic floor muscle training (e.g., Kegel exercises) may be performed on a daily basis to reduce the risk of urinary incontinence. Instruction in proper technique is recommended to obtain optimal benefits.

6

Pregnant women who experience light-headedness, nausea or feel unwell when they exercise flat on their back should modify their exercise position to avoid the supine position.

No. 367-2019 Canadian Guideline for Physical Activity throughout Pregnancy
JOINT SOGC/CSEP CLINICAL PRACTICE GUIDELINE | Volume 40, ISSUE 11, P1528-1537, November 01, 2018

Table 4.1: Description, purpose, prescription and benefits of cardio, resistance and flexibility exercises during pregnancy based on compiled scientific evidence

	Cardio
Frequency	At least 3x/ week, daily if possible.
Intensity	At least moderate intensity where talking is choppy due to noticeable change in breathing rate; or monitor by heartrate by age calculation*, 60 – 80% HRmax; or by perceived exertion of 3 – 5 out of 10-Borg scale.
Time	Anywhere from 20 – 30 min per time or longer.
Type	Whole body movements such as brisk walking, jogging, swimming, zumba, dancing, step-aerobics, yoga, tai-chi.
Progression	Increase speed, time, distance after 4 – 6 weeks.
Benefits	Controls blood pressure and glucose levels, prevents and treats pre-eclampsia and gestational diabetes mellitus.

*HRmax by age calculation = (220 - Age in years) x % intensity

	Strength
Frequency	At least twice a week, exercise same muscle groups on alternate days.
Intensity	Use own body weight to start. If using weights, use the weights that can be lifted for 15 times without tiring. Based on repetitions and sets. Range of 8 – 15 reps depending on number of sets to complete, 1 – 3 sets per muscle group.
Time	Depending on repetitions, sets and number of exercises conducted.
Type (examples)	Upper body exercises – seated rows, triceps extensions, shoulder chops Core body exercises – alternate leg-hand lifts, side-twists Lower body exercises – side-lying leg abduction, lunges, squats.
Progression	After the first 6 weeks, can increase either the reps, sets or weights. Change/ increase either reps, sets or weights about every 4 weeks or based on goals.
Benefits	Maintain strength and posture to last throughout the pregnancy when body is heavier.

*reps = repetitions

	Flexibility
Frequency	Daily or as frequent as needed. Stiff areas can be stretched multiple times a day.
Intensity	Stretch based on ability untill muscles feel slightly strained. Stretches can be passive (assisted by others) or active (done on your own). Stretch dynamically (“bounce”) or statically (hold stretch) to lengthen and stretch the muscle and joints.
Time	Generally, maintain a stretch for 15 – 30 seconds each time, rest and repeat.
Type	Upper body stretches – wrist/ lower arm, triceps, shoulder and neck stretches Core body stretches – child’s pose, cat-and-cow stretch Lower body stretches – pelvic tilts, calf, inner thigh, glutes stretches
Progression	Hold stretches longer or repeat more often when the muscle feels stiff.
Benefits	Stretch tight muscles due to postural changes brought by shifts body size and centre of gravity.

Sources: Ruchat et al. (2018); Mottola et al. (2018); ACOG (2020).

Table 4.2: Rate of perceived exertion - The 10-point Borg Scale

When rating exertion, give a number that corresponds to how hard and strenuous you perceive the work to be. The perception of exertion is mainly felt as strain and fatigue in your muscles, breathlessness or any aches.

0 “Nothing at all”, means that you do not feel any exertion, no muscle fatigue, no breathlessness or difficulties in breathing.

1 “Very weak” means a very light exertion such as taking a shorter walk at your own pace.

3 “Moderate” is somewhat easy but not too hard. It feels good and not difficult to go on.

5 “Strong”. The work is hard and tiring, but it is not difficult to proceed. The effort and exertion is about half as intense as “Maximal”.

7 “Very strong” is very strenuous. You can still go on, but you really have to push yourself and you are very tired.

10 “Extremely strong – Maximal” is an extremely strenuous level. For most people, this is the most strenuous exertion they have ever experienced.

• “Absolute maximum” is the highest possible, for example “12” or somewhat more.

Note: For moderate to high exercise intensity, aim for a 3 to 5 rating in the Borg 10-point CR scale (R) (CR10) (© Gunnar Borg, 1982, 1998,2004).

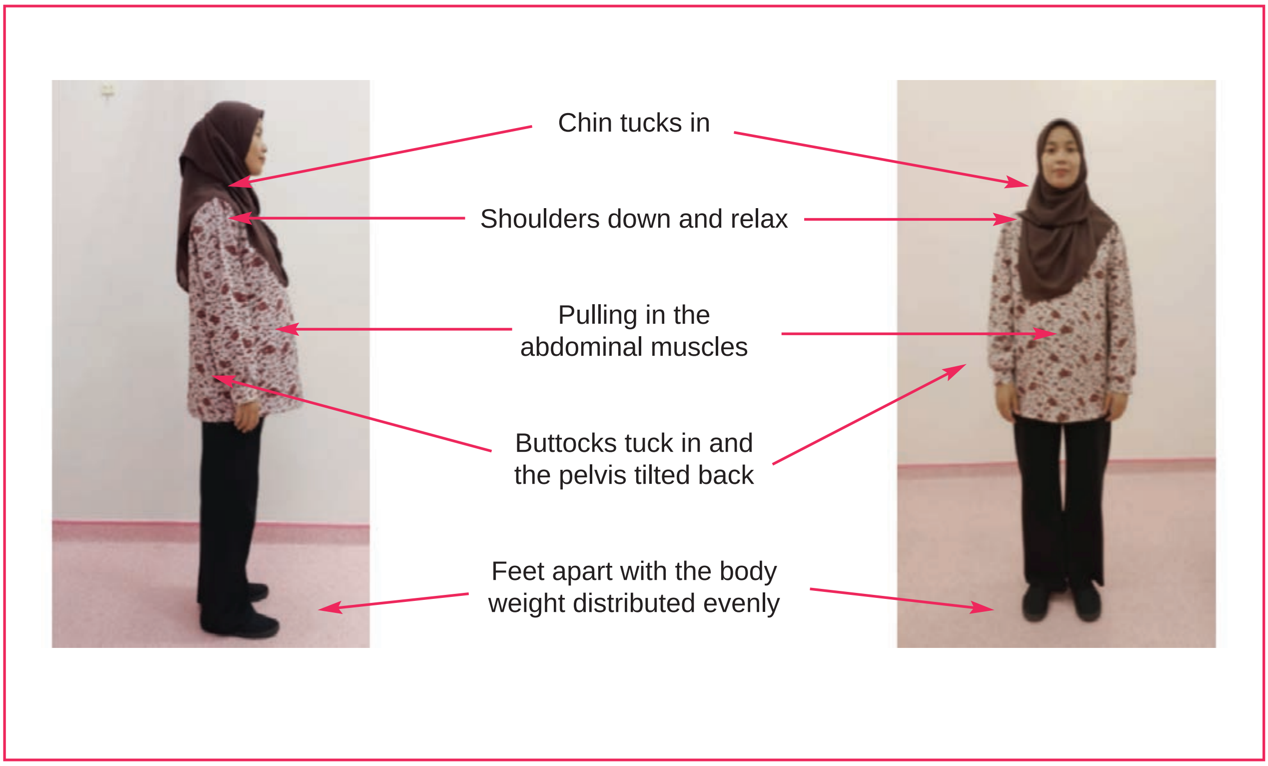


Figure 4.2: Posture when standing

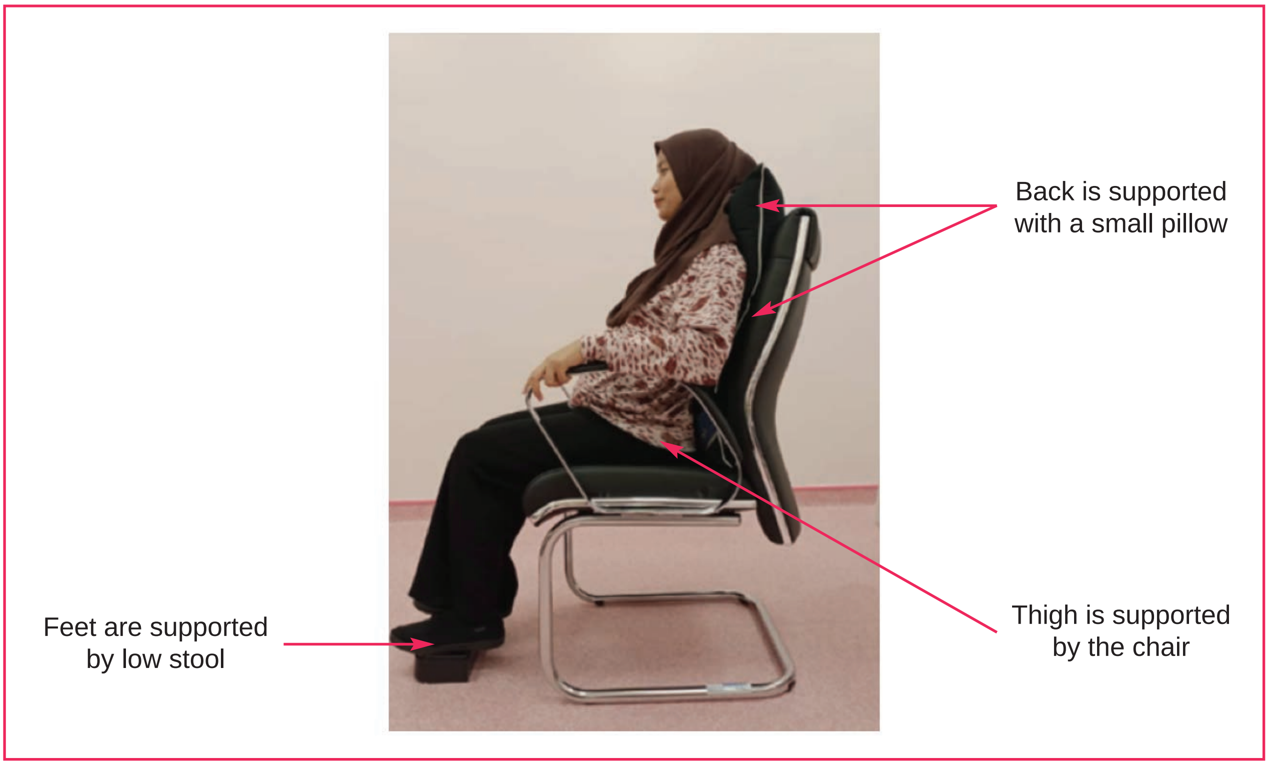


Figure 4.3: Posture when sitting

Put a pillow to support the head and knees to prevent stretch on the lower back and knees.

Put a pillow to support the head, forearm and in between the knees to prevent strain on the sacroiliac joint.

Figure: 4.4: Posture when lying

Sit all the way back in the chair. Put a small pillow behind for support. Keep your shoulders relaxed and feet flat on the floor. Avoid looking down at your baby the whole time.

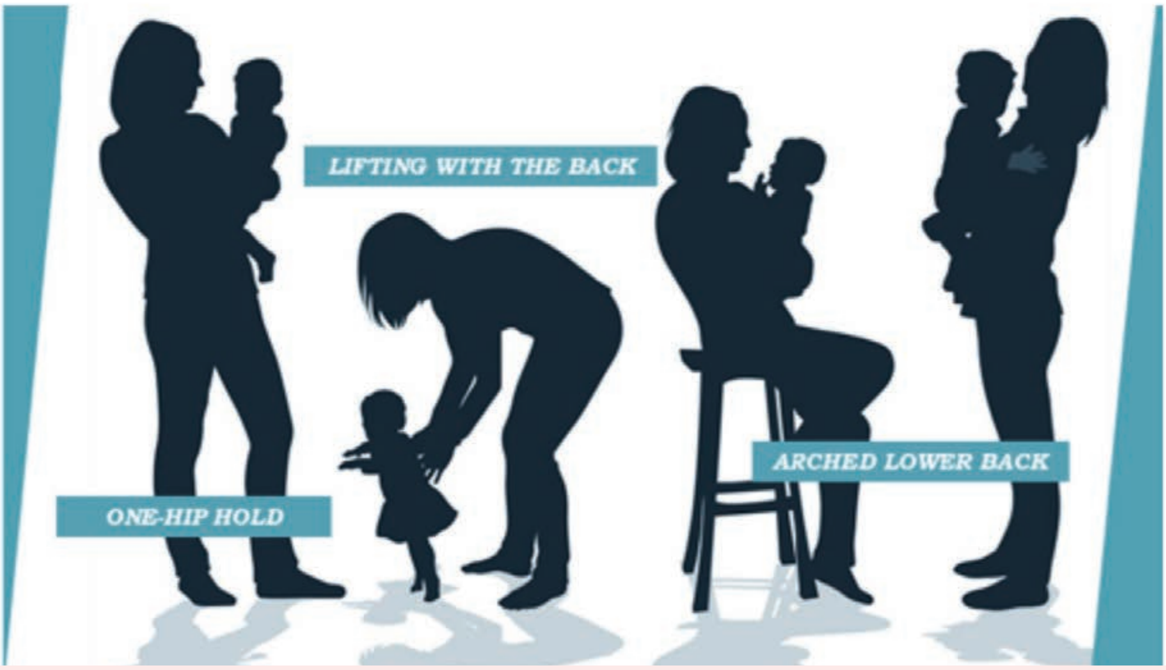
Figure 4.5: Posture during breastfeeding

Carrying/ holding a baby in comfort and correct position.

Baby is supported by the bottom and upper back.

Mother stands with back straight and neutral pelvic tilt – imagining a line straight up from the head that is pulling the body up and straight.

Figure 4.6: Posture when carrying/ holding baby



Avoid holding a baby in these positions
- One-hip hold. Instead, stand with weight evenly distributed on both feet.
- Lifting with lower back. Instead, bend at hips and knees, keep back straight
- Arched lower back. Instead, pull-in tummy and straighten posture.

Figure 4.7: Posture to avoid when carrying/ holding baby

Table 4.3: Suggestions to disrupt sedentary behaviour and limit bedrest

General	Work/ Outdoors	Screentime
• Use stairs often • Walk with purpose/ fast • Choose active transport – walk, cycle to destination or park vehicles further away from destination • Select active hobbies – gardening, house-cleaning, baking, cooking, caring for pets, play a musical instrument. • Bedrest only if medically required.	• Change passive sitting to active desk work – try standing desks, have frequent movement breaks. • Choose active communication – stand when on phone or texting, walk to desk or department instead of using phone. • Organise walking or standing meetings.	• Insert 1 to 3 minute movement breaks for every 30 minute of screentime. • Limit total recreational screentime to 2 hours/day by setting reminders on devices – social media, online shopping, YouTube, television, movies. • Stand, pace or move while having screentime.

Table 4.4: Suggestions to achieve quality sleep during pregnancy

Food intake	Environment	Daily routine
• Avoid large meals, high-fat food, cruciferous vegetables, large amount of fluids 2 hours before bedtime. • No caffeine and alcohol intake in the evenings.	• Clean and comfortable bed Bedroom is dark, quiet and at a comfortable temperature. • Sleep on the side, supported with pillows. • Prop upper body if having heartburn symptoms.	• Exercise daily, light stretching before bedtime. • Avoid screentime 30 minute before bedtime – read a book, moisturize skin with massaging movements, breathing exercises. • Wear socks if feet are cold.

Choose clean and safe food

Key Message 5

Choose clean and safe food

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5.1 Terminology

Cross-contamination
Cross-contamination is the transfer of dangerous microorganisms from other types of food, or through non-food surfaces such as human hands, kitchen utensils and equipment. It could also be a direct transfer of microorganisms from raw food to cooked food.

Food and water-borne diseases
Food and water-borne diseases are any diseases caused by the ingestion of food and water contaminated by pathogenic bacteria, viruses and parasites. Diseases such as food poisoning, typhoid fever, cholera, hepatitis A and dysentery (diarrhoea) are examples of food and water-borne diseases. These diseases, if not treated, could be fatal.

Food poisoning
Food poisoning usually starts with acute (intense and temporary) vomiting and/or diarrhoea and/or other effects on the bodily system due to ingestion of food contaminated with toxic microorganisms or other toxic components.

Fresh food
Fresh food is raw food which does not have off-colour, without an undesirable smell, not wilted and with desirable texture (unchanged).

Heavy metal contamination (Lead, Mercury, and Cadmium)
Environmental pollution is contamination caused by the presence of lead, mercury and cadmium heavy metals in the food supply chain. The presence of lead and mercury is widespread in the environment and is usually through water and food supply chains. Cadmium is a heavy metal that can contaminate food such as potatoes and leafy vegetables.

Listeria monocytogenes
Listeria monocytogenes is the bacteria that causes listeriosis infection. It is a type of facultative anaerobic bacteria, which is able to live in the presence of oxygen, grow and multiply in host cells, and is one of the most virulent food-borne pathogens, with 20 to 30 percent of the clinical infection causing death.

Natural mineral water
Natural mineral water shall be ground water which is obtained for human consumption from subterranean water-bearing strata through a spring, well, bore, or other exit, with or without the addition of carbon dioxide. The natural mineral water must be subjected to one or more of the following treatments, i.e. filtration, chlorination (followed by dechlorination), aeration.

Potable water
Water shall be clean and free from contamination, objectionable taste and odour.

Packaged drinking water
Packaged drinking water shall be potable water or treated potable water, other than natural mineral water, that is hermetically sealed in bottles or other packages and is intended for human consumption.

Water from water vending machine
Water from a water vending machine is water that is dispensed by a vending machine and is not bottled in a tight bottle by bottlers, for human consumption. The water vending machine means any self-service machine that upon insertion of a coin, token or by any other means automatically dispenses unit volume of water for drinking or other purposes involving a likelihood of the water being consumed by humans.

Water from water filtration machine
Water from a water filtration machine is water that has been treated through a filtration treatment system, in which impurities will be removed through a barrier, either fine physical barriers, chemical processes or biological processes, depending on consumers' preference.

5.2 Introduction

Foodborne diseases have been defined by the World Health Organization (WHO, 2022) as diseases that are infectious or naturally toxic, caused by agents that enter human bodies through the consumption of food. Most of the foodborne disease cases are caused by cross-contamination of microbiological agents and improper handling at the food chain level. Food can be contaminated due to the usage of contaminated raw ingredients, improper defrosting of food, food not cooked properly, food being exposed and served more than 4 hours, improper storing of food, preparing and serving food in an unhygienic way and using dirty and contaminated utensils.

Food poisoning is caused by eating contaminated food. Contamination can happen due to bacterial, virus and parasite infection. Toxins are produced by plants or animals like mushrooms, algae, and certain fishes (i.e., puffer fish). It is also produced by bacteria in food before it is eaten. The serious health effect of food-borne diseases is gastroenteritis and can result in other complications like cancer, birth, nerve and liver defects and kidney syndromes.

Food-borne diseases are caused by the presence of dangerous substances in food or drinks defined by Codex as biological, chemical, or physical agents or condition of food which has the potential to affect the health seriously. Most of these dangerous substances have been identified and handled by food safety control steps (WHO, 2007). Children, pregnant women and the elderly are the vulnerable populations, which have the highest risks of being infected with food-borne diseases and experiencing serious and bad health effects.

Handling safe and hygienic food involves 5 levels which are choosing, storing, preparing, cooking, and serving. Contamination of food can happen at any level of food handling. Many of these practices and prevention can be done to reduce food poisoning on any level. Guidebook on safe Food Handling has been published by the Food Safety and Food Quality Department, Ministry of Health. It can be used as a guide for pregnant and lactating mothers.

Pregnant and lactating women need to drink clean water during pregnancy to prevent the risk of water-borne diseases (for example, cholera). Water from uncontrolled sources has to be avoided (river and private tanks) unless the water has been treated or boiled before drinking. Mineral water, packaged drinking water, water from a water vending machine or water from a water filtration machine can be used as alternatives. Pregnant women are advised to carefully select vegetables that can be eaten raw such as cucumber, cabbage and carrot which may be contaminated with *Listeria monocytogenes,* which causes listeriosis. Pregnant women are also advised to drink processed milk to avoid similar contamination.

Recently, awareness of contamination of heavy metals in seafood especially fish, is increasing because fish is an important protein source. In Malaysia, fish is an important protein source which contributes to 23% of animal protein eaten. The Food and Agriculture Organization (FAO) and World Health Organization (WHO) estimated that between 15 to 20% of animal protein sources are from aquatic animals, especially fish (Tukimat et al., 2006). The United States Food and Drug Administration (FDA) and Environmental Protection Agency (EPA) suggested that pregnant women, lactating women and women planning to become pregnant should choose fish with low mercury levels. Among fish which have low mercury levels are salmon, tilapia, tuna, catfish, codfish and white codfish. Big size fish which may contain high mercury levels such as sharks, swordfish and king mackerel should be limited. In the human body, mercury will act as a kind of neurotoxin, in fact, it can disturb the brain and nervous system. Mercury exposure may be very dangerous to pregnant women and children. This is because at the early development stage, children's brains are still developing and easily absorb nutrients. Although in low doses, mercury may affect children's development, slowing their ability to walk and talk, shortening their focus and contributing to learning problems.

Food safety is the responsibility of a few agencies, especially the Food Safety and Quality Division (FSQD), the Ministry of Health (MOH). Food Act 1983 and Food Regulations 1985 were enforced in Malaysia to protect the public against health hazards and fraud in the preparation, sale and use of food, and for matters incidental thereto or connected therewith. The Food Hygiene Regulations 2009 which was gazetted in 2009 provide the infrastructure to control the hygiene and safety of food including preparation, handling, distribution, selling and food consumption which is sold in the country to protect the health of the public.

5.3 Scientific Basis

With Malaysia's total population of 32.66 million in 2020, the population's demand for health is increasing. In 2020, food poisoning is the most important factor in food and water-borne diseases contributing to more than 29 incidence rates per 100,000 Malaysian population. Other disease incidence rates reported in 2020 are cholera (0.35), hepatitis A (0.14), typhoid (0.20), and dysentery (0.48) (Ministry of Health Malaysia, 2021).

Health hazards from food can exist from any part of the food chain, from raw materials to handling, and through all stages involved in the processing, transport, storage, sales, and consumption of food. Food-borne disease outbreaks can be reduced if consumers and food handlers understand the importance of proper food hygiene practices. A study on food handlers trained at a food handler training school recognized by the Malaysian Ministry of Health showed that hygiene practices were in line with the requirements of the Food Act 1983 and Food Hygiene Regulations 2009. Food handlers have played a role in implementing good hygiene practices in the food service industry. Work experience and consistent exposure to food-handling activities have added value to food-handling practices (Mazni, Toh & Mohamed Azam, 2013).

Improper food handling and storage can cause food spoilage and in turn, cause severe food poisoning if consumed. Temperature plays an important role in food safety, as proper temperature treatment and storage on food ensure microbial growth could be controlled to avoid spoilage. Food should be held at a safe temperature (less than 5°C for cold foods, or more than 60°C for hot foods) to slow or stop the growth of microorganisms (WHO, 2008). On the other hand, maintaining food in the danger zone from 5°C to 60°C can cause microorganisms to multiply very quickly (FAO, 2008). Foods, especially potentially hazardous food such as minced meats, rolled roasts, large joints of meat and whole poultry need to be properly cooked to an internal temperature of at least 70°C to kill almost all dangerous microorganisms and should be consumed within 4 hours to ensure microorganism levels remain below the dangerous threshold (WHO, 2008). In the case of reheating food, the internal temperature of the food should achieve at least

74°C, and it to be consumed within 2 hours to prevent microorganisms growth from reaching levels that can cause food-borne illnesses (NCCFN, 2021).

Listeria monocytogenes is a facultatively anaerobic, non-spore-forming foodborne pathogen that can cause severe listeriosis. Although listeriosis is relatively rare, it accounts for a high mortality rate (30%), especially in the YOPI (young, old, pregnant, immunodeficient) group. Previous studies have reported that *L. monocytogenes* has been found in poultry, seafood, and vegetables in Malaysia in the past 15 years (Kuan et al., 2013). *L. monocytogenes* are widely distributed in the environment and although it does not usually cause symptoms in healthy people, they can cause disease (listeriosis) in pregnant women and new-born babies. Listeriosis occurs as a result of *L. monocytogenes* infection from food. If a pregnant woman has listeriosis, it causes influenza-like symptoms and can cause premature birth and reduced foetal movement. If not treated immediately, listeriosis poses a dangerous threat to the foetus through infection of the placenta, membranes, and amniotic fluid, causing uterine sepsis and foetal loss (Hof, 2003). Individuals at high risk for the disease are pregnant women, neonates, the elderly, immunocompromised individuals and adults. Listeriosis can be a dangerous disease with a mortality of about 20% and the case fatality rate can increase in the highest risk group (Ranjbar & Halaji, 2018).

The presence of lead and mercury is widespread in the environment and can be found in water and the food chain. Food is usually the main source of maternal exposure to these two metals. Heavy metals such as mercury and lead are naturally toxic and cause a variety of health problems including neurobehavioral and developmental disorders in infants and children. Considering the importance of developmental processes in the early stages of life, infants and children are the most susceptible groups to the contamination of these metals. Breast milk is often the only source of nutrition during the first months of life. Breast milk can be contaminated with these metals and through breastfeeding, babies can be exposed to these heavy metals (Park et al., 2018). Heavy metal levels in breast milk are generally lower than levels found in infant formula (Dorea, 2004). Cadmium is a heavy metal that has many industrial uses including batteries, pigments and metal coatings. Although many organisms can regulate the concentration of cadmium in their tissues, research shows some forms of aquatic life such as oysters and clams do not perform this activity well. Pregnant women should limit their intake of Bluff oysters and queen scallops due to high concentrations of cadmium (Taylor et al., 2018).

The practice of consuming traditional medicine, remedies and herbal products is common among pregnant and breastfeeding women. These practices are usually based on the advice and experience of the older generation. However, caution should be practiced while consuming such products, as scientific studies on these practices in terms of their health effects are very limited (Jun et al., 2021). Aside from that, the wide variety of these products as well as their interaction with other products makes the risk assessments of such practices complicated. In a correlation study by Yusof et al. (2016), a positive correlation was found between the consumption of these products and obstetric outcomes. It was found that the usage of *Kacip Fatimah* was associated with preterm labour, while *Tongkat Ali* herbal coffee had a significant association with hypertensive disorders in pregnancy and foetal distress. Meanwhile, the usage of *Jamu* was significantly associated with low birth weight in grandmultiparae (woman who has given birth 5 or more times). Therefore, if pregnant and breastfeeding women decide to consume such products, awareness of the type of ingredients present in the products as well as its suitable dosage should be given due consideration. If in doubt, it is advisable to refrain from consuming such products and get advice from medical practitioners and certified traditional medicine practitioners.

5.4 Current Status

A report from the Malaysian Ministry of Health (2018) showed that food and water-borne diseases continue to be public health problems in Malaysia and other developed countries. These include diseases such as cholera, typhoid and paratyphoid fever, hepatitis A, diarrhoea and food poisoning. In Malaysia, notification of infectious diseases including foodborne diseases such as cholera, typhoid, food poisoning, hepatitis A and dysentery is mandatory as required under the provisions of Prevention and Control (Act 342) of Infectious Diseases. Cases reported to the nearest District Health Offices are fully investigated and appropriate controls and preventive measures can be taken to prevent the spread and repeat episodes. The surveillance system has been strengthened through the introduction of an electronic mode of disease notification called the Communicable Disease Control Information System (CDCIS) harmonizing case definitions and providing guidelines on disease management.

Food and water-borne diseases namely cholera, typhoid, dysentery, viral hepatitis A and food poisoning are caused by consuming contaminated food and water. It can lead to permanent health problems and disabilities. Food poisoning can cause death if not treated promptly. For example, typhoid fever is a disease caused by the bacteria *Salmonella typhi.* This infection occurs through the consumption of food or drinks contaminated by the faeces or urine of an infected person. These symptoms show effects after 1-3 weeks of exposure.

Apart from the danger of food poisoning, food contamination can also be caused by the presence of heavy metals in food. In Malaysia, fish is one of the main sources of protein that contributes to 23% of animal protein in food (Tukimat et al., 2006). A total of 33% of 2675 people in Peninsular Malaysia consume fish in their daily diet. The ten types of fish that are highly consumed in Peninsular Malaysia in descending order are indian mackerel, anchovy, yellowtail and yellow-stripe scads, tuna, sardines, torpedo scad, Indian and short-fin scads, pomfret, red snapper and king mackerel (Nurul Izzah et al., 2016). Indian mackerel or its scientific name *Rastrelliger kanagurta* is one of the sea fish that is highly consumed in Malaysia (Goh et al., 2021). A study by Azmi et al. (2019) showed that the concentration of heavy metals in Indian mackerels and other fish commonly consumed in Malaysia are still below the maximum limit based on the estimated weekly intake by Malaysians (Azmi et al., 2019), and is within the limit allowed by the Malaysian Food Act (1983) and Food Regulations (1985) (As: 1 mg/kg; Cd: 1 mg/kg, Pb: 1 mg/kg and Hg: 1mg/kg). However, there are several studies showed that the commonly consumed fish in Malaysia has varied heavy metal contents, and there are instances that the heavy metal contents are above the allowable limits (Alina et al., 2012; Mohd Yusoff et al,. 2018; Octavianti & Jaswir, 2017; Rosli et al., 2018; Salam et al., 2021). A study by Tukimat et al. (2006) showed that the consumption rate of lead by the people in Tanjung Karang was 0.008 mg/day which is below the maximum limit allowed by several organizations such as WHO (0.430 mg/day), USFDA and FAO. The total intake of cadmium metal in a day was in the range of 0.006 mg/day to 0.096 mg/day with an average of 0.031 mg/day. The limit allowed in nutrition by WHO is 0.064 mg/day (Tukimat et al., 2006; WHO, 2006).

In addition, contaminated food results in 600 million cases of foodborne diseases and 420,000 deaths worldwide every year, which are involved 7.69% and 7.5% of people, respectively. A majority of such patients are affected by norovirus and *Campylobacter. Listeria monocytogenes* is the most fatal (Lee & Yoon, 2021). In addition, *Listeria monocytogenes* is responsible for approximately 2,500 illnesses and 500 deaths in the United States (US) per year, in which listeriosis is the leading cause of death among foodborne bacterial pathogens, with a death rate higher than that of *Salmonella* and *Clostridium botulinum* (Ramaswamy et al., 2007). In Malaysia, a study on the assessment of *Listeria spp.* and *Listeria monocytogenes* in selected vegetable gardens showed that *Listeria spp.* detected in cabbage (30%), cucumber (20%), long beans (10%) and carrots (10%) while *Listeria monocytogenes* was detected in cabbage (10%), long beans (10%) and carrots (10%). This study provided basic data on the contamination of *Listeria spp.* at the farm level (Jeyaletchumi et al., 2011).

5.5 Key Recommendations

Key Recommendation 1

Prevent food and water borne diseases.

How to achieve

1. Choose and buy fresh food. Food should be prepared using clean utensils, cooked at the correct temperature and kept in suitable, clean and safe containers to ensure food safety.
2. Limit the intake of raw or half-cooked food during pregnancy to reduce the risk of foodborne diseases.
3. Acquire information on the food safety steps and the type of food recommended during pregnancy and breastfeeding to reduce the risk of food and water-borne diseases.
4. Drink clean potable water from controlled and treated water sources during pregnancy and breastfeeding to avoid the risk of water-borne diseases (e.g. cholera).
5. Water from uncontrolled and untreated sources should be avoided (e.g. rivers, lake, and individual water tanks), unless it has been boiled or treated before drinking.
6. Natural mineral water, packaged drinking water and water from water vending machines or filtration systems can be consumed as alternatives.

Key Recommendation 2

Prepare safe and clean food for pregnant and breastfeeding women.

How to achieve

Food preparation

1. Wash fruits and vegetables that are to be eaten raw with safe and clean water.
2. Wash all surfaces of crockery, cutlery, cooking utensils and other equipment used for food preparation with soap and rinse thoroughly with water to avoid cross-contamination.
3. Separate cutting boards for handling raw foods and cooked foods.
4. Eat cooked food immediately after cooking and do not keep/ serve food for more than 4 hours at room temperature.
5. Do not touch cooked food with dirty hands. Wash hands well with soap and water. Use a spoon to handle food.
6. Heat food properly and appropriately according to the time, temperature and type of food when using the oven/ microwave oven.
7. Any food that needs reheating must be reheated to reach an internal temperature of 74°C. Do not reheat food more than once.

Food storage

1. Store cooked food separately from raw food to avoid cross-contamination.
2. Store raw food such as chicken, fish and meat in the freezer compartment.
3. Store food in a clean and airtight container.
4. Read and understand storage labels when buying packaged food.
5. Do not keep cooked foods for more than 4 hours at room temperature because bacteria have started contaminating the food.
6. Do not leave drinking water in a hot environment (between 37°C to 60°C), such as in a car parked under a hot sun. Change water if it is dirty, cloudy, smelly or if there is mould growth. Check and clean water containers regularly.

Key Recommendation 3

Make sure that the food of pregnant women is free of bacteria such as *Listeria monocytogenes* and *Salmonella spp.* that are harmful to the foetus.

How to Achieve

1. Make sure the food that you drink or eat, such as milk, cheese, honey and eggs are properly processed (not raw).

2. Ensure vegetables, including salads *(ulam)*, that can be eaten raw, such as *ulam raja*, cucumber and carrot are washed and cleaned properly before consumption.

3. Make sure the frozen and raw meat is cooked properly at the right temperature so that the existing germs can be completely killed, and the food should be eaten within 4 hours after cooking.

4. Food that does not require heating, such as canned tuna/sardines, should be eaten immediately after opening and if there is excess, it should be removed from the can and put in another closed container to be stored in the refrigerator.

5. Food stored in the refrigerator should not exceed the recommended number of days as found on the product label.

6. Wash hands thoroughly and dry them after handling raw food (eggs, meat, fish, and poultry) and vegetables.

7. Clean kitchen surfaces, cooking utensils, chopping boards/ knives after handling raw food and vegetables.

Key Recommendation 4

Ensure that the food of pregnant and breastfeeding women is safe from environmental pollutant substances such as lead, mercury and cadmium.

How to achieve

1. Ensure that the consumption of fish, shellfish and marine products are safe and free from contaminated water and environmental pollution.

2. Drink only treated water as heavy metals could possibly accumulate in the groundwater.

3. Wash all fresh fruits and vegetables to remove pesticides and fungicides residues.

4. Limit food or drink consumption of canned products as the inner coating of certain cans are made from metals including lead.

Key Recommendation 5

Select safe food from restaurants/ food outlets/ stalls.

How to achieve

1. Choose restaurants/ food outlets/ stalls that are clean in terms of environment, food handlers and have good basic facilities such as sinks and toilets.

2. Avoid consuming high-risk raw foods and foods with raw ingredients such as sushi, sashimi, sandwiches or salads from restaurants/ food outlets/ stalls.

3. Choose food that is served hot, ideally at 60°C.

4. For buffet food, it is advisable to take the food immediately after they are being served.

5. Limit consumption of street food that is sold openly.

5.5.1 Special Consideration

Be cautious in practicing traditional medicine, remedies and herbal products for pregnant and breastfeeding women. Traditional medicine, remedies and herbal products are practiced by many pregnant and breastfeeding women, mainly based on practices by the older generation. Although these practices may bring beneficial effects to the mothers, foetus and infants, the risks involved should be taken into consideration. Many traditional medicine, remedies and herbal products are consumed based on traditional practices without proper scientific studies. The source of the products may also be questionable and may be contaminated with other unknown products. Another issue is the dosage of such products, as the consumption amount is based on experience rather than proper scientific studies. Traditional medicine, remedies and herbal products, including traditional tinctures, concoctions and herbal liquor may contain steroids, alcohol and other substances which may bring adverse effects to the pregnant and breastfeeding women as well as the foetus and infant. Pregnant and breastfeeding women should practice caution to information from hearsay and the internet, as not all are correct and accurate. When in doubt, seek advice and consultations from medical practitioners and certified traditional medicine practitioners.

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Maternal Dietary Guidelines for Malaysia (MDGM) 2023
Focus Group Discussion
on the
Key Messages, Key Recommendations &
How To Achieve
(21st February 2023)

Focus Group Discussion (FGD) on the Key Messages, Key Recommendations, and How to Achieve of the Maternal Dietary Guidelines for Malaysia (21st February 2023)

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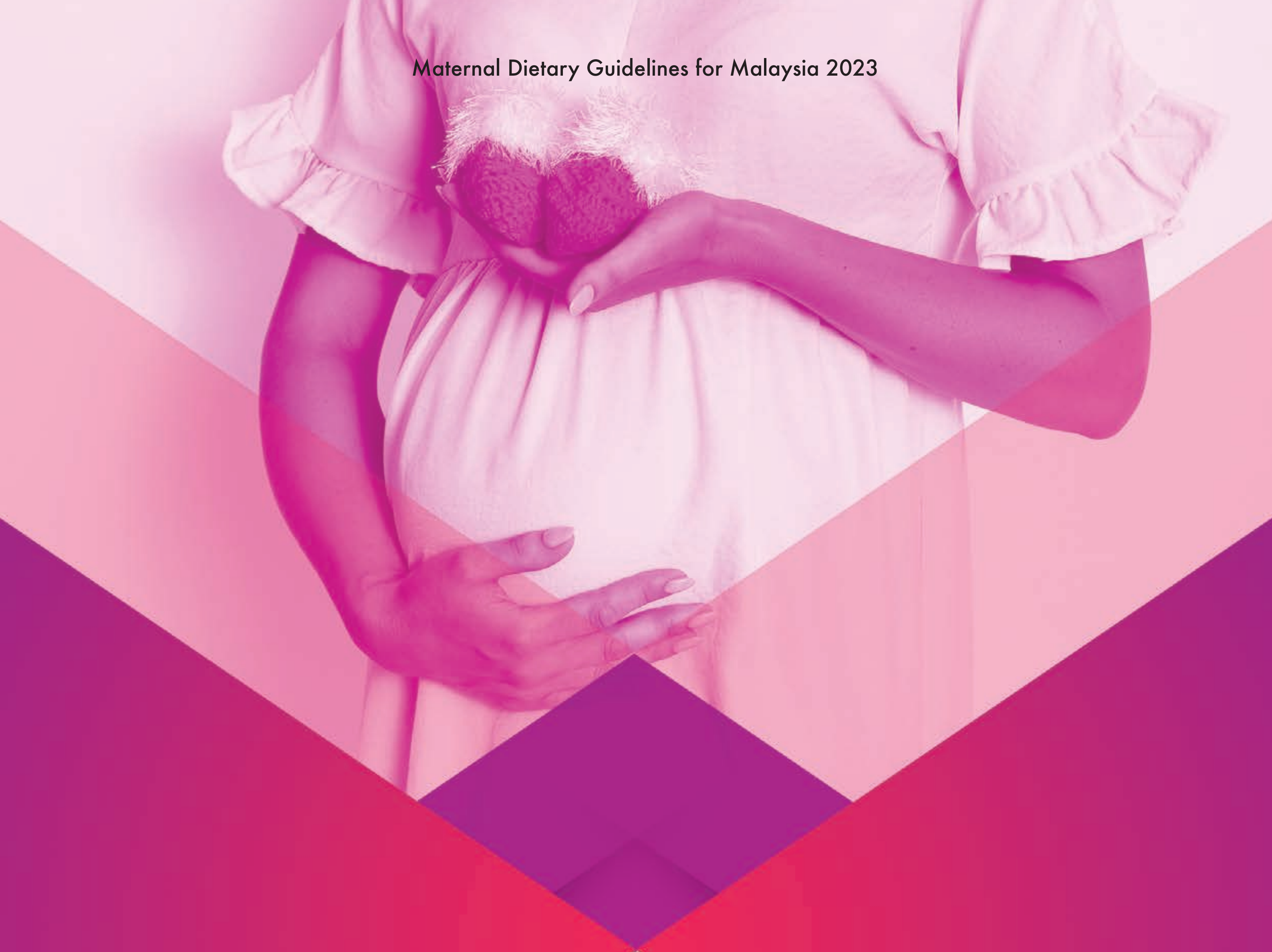

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